

Lecture Notes:
Commutative Algebra
&
Introduction to Algebraic Geometry

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(Based on the notes taken by Friedrich Günther and Niklas Müller in Summer Term 2019)

Contents

I	Commutative Algebra	1
1	Rings and Ideals	2
1.1	Ideals: Examples and Operations	4
1.2	Prime Ideals and Maximal Ideals	6
1.3	Nilradical, Jacobson Radical and Local Rings	8
1.4	Chinese Remainder Theorem	10
1.5	Hilbert's Basis Theorem	11
2	Modules	13
2.1	Submodules and Quotient Modules	14
2.2	Operations on Submodules	15
2.3	Direct Sums and Direct Products	16
2.4	Finitely Generated Modules	18
2.5	Exact Sequences	23
2.6	Tensor Products of Modules	26
2.7	Algebras, Restriction and Extension of Scalars	31
2.8	Tensor Product of Algebras	33
2.9	Exactness of Tensor Products	35
3	Localizations	39
3.1	Localization of a Module	41
3.1.1	Submodules and Canonical Isomorphisms	43
3.2	Local Properties of Rings and Modules	45
3.3	Ideals in Ring of Fractions	46
4	Integral Dependence	49
4.1	Going-Up Theorem	50
4.2	Integral Closure	52
4.3	Going-Down Theorem	55
5	Noetherian Modules and Rings	57
5.1	Modules of Finite Length	63

6	Dimension Theory	66
6.1	Noether's Normalization Lemma	67
7	Valuation Rings	70
7.1	Discrete Valuation Rings	73
II	Introduction to Algebraic Geometry	76
8	Affine Varieties	77
8.1	Hilbert's Nullstellensatz	80
8.2	Irreducible Components	86
9	Projective Varieties	91
10	Morphisms	97

Part I

Commutative Algebra

Chapter 1

Rings and Ideals

In this chapter, we introduce and study some basic notions and results concerning rings and ideals. Note that some of them were already presented in the courses “Lineare Algebra I (WS 2024/25)” and “Algebra (WS 2025/26)”, but are included here for completeness and clarity.

Throughout these notes, all rings are assumed to be *commutative with identity*, unless explicitly stated otherwise. In particular, any nontrivial ring $R \neq \{0\}$ contains at least two different elements: 0 and 1.

Definition 1.1. Let R and S be rings. A *ring homomorphism* $\varphi: R \rightarrow S$ is a map fulfilling the following properties:

- (a) $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in R$,
- (b) $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in R$,
- (c) $\varphi(1_R) = 1_S$.

Note that property (c) does not hold automatically and hence must be required.

Example 1.2. Let R and S be nontrivial rings. The following maps between rings fulfill (a) and (b), but not (c).

- (1) The zero map $R \rightarrow S$, $x \mapsto 0$.
- (2) The map $R \rightarrow R \times R$, $x \mapsto (x, 0)$.
- (3) The map $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$, $\bar{n} \mapsto \bar{3} \cdot \bar{n}$.

Definition 1.3. Let R be a ring. A subset $S \subseteq R$ is a *subring* of R if it is a ring with respect to the operations induced by those in R and moreover $1_R \in S$.

Remark 1.4. Let R be a ring and let $S \subseteq R$ be a subset.

- (a) The subset S is a subring if and only if it is closed under subtraction and under multiplication and $1_R \in S$.

- (b) If S is a subring, then the inclusion map $S \rightarrow R$, $x \mapsto x$, is a ring homomorphism.

Remark 1.5. In the definition of a subring $S \subseteq R$, the condition $1_R \in S$ is not automatically satisfied and must be imposed. Indeed, a subset of a ring may itself form a ring under the induced operations without containing the identity element of the larger ring. For example, let R be a nontrivial ring and consider

$$S := \{(x, 0) \mid x \in R\} \subseteq R \times R.$$

The subset S is a ring under the operations inherited from $R \times R$: indeed, S is nonempty, closed under subtraction and multiplication, and $(1, 0)$ is the multiplicative identity of S . However, the identity element of $R \times R$ is $(1, 1)$ and since R is nontrivial we have $(1, 0) \neq (1, 1)$. Since $(1, 1) \notin S$, the ring S is not a subring of $R \times R$.

Definition 1.6. Let R be a ring and let $a, b \in R$. We say that b *divides* a , or that b is a *divisor of* a , and write $b \mid a$, if there exists an element $c \in R$ such that $a = bc$.

Definition 1.7. Let R be a ring and let $r \in R$.

- (a) The element r is called a *unit* if it is invertible with respect to multiplication, i.e., there exists $s \in R$ such that $rs = 1$. We denote by R^* the set of all units in R . Notice that $(R^*, \cdot)$ is an abelian group.
- (b) The element r is called a *zero divisor* if there exists $s \in R \setminus \{0\}$ such that $rs = 0$. A nontrivial ring whose only zero divisor is 0 is called an *integral domain*.
- (c) The element r is called *nilpotent* if there exists $n \in \mathbb{N}_{>0}$ such that $r^n = 0$.
- (d) Assume now that $r \neq 0$ and $r \notin R^*$.

- (i) The element r is called *prime* if for all $a, b \in R$

$$r \mid ab \implies r \mid a \quad \text{or} \quad r \mid b.$$

- (ii) The element r is called *irreducible* if for all $a, b \in R$

$$r = ab \implies a \in R^* \quad \text{or} \quad b \in R^*.$$

Remark 1.8.

- (a) In a nontrivial ring any nilpotent element is a zero divisor. However, the converse is not true in general. For instance, in the ring $\mathbb{Z}/6\mathbb{Z}$ the element $\bar{2}$ is a zero divisor, but it is not nilpotent.

(b) In general, the notions of prime and irreducible elements are independent: a prime element need not be irreducible, and an irreducible element need not be prime. For instance, in the ring $\mathbb{Z} \times \mathbb{Z}$ the element $(1, 0)$ is prime, but not irreducible since $(1, 0) = (1, 0) \cdot (1, 0)$. Also, in the ring $\mathbb{Z}[i\sqrt{5}] := \{a + bi\sqrt{5} \mid a, b \in \mathbb{Z}\}$ the element 2 is irreducible, but not prime. Thus, without additional assumptions on the ring, there is no implication between the two concepts. However, in special classes of rings the situation improves:

- (i) In an integral domain, every prime element is irreducible.
- (ii) In a unique factorization domain (UFD), the two notions coincide.

1.1 Ideals: Examples and Operations

Definition 1.9. Let $(R, +, \cdot)$ be a ring. A subset $I \subseteq R$ is called an *ideal* if:

- (a) $(I, +)$ is a subgroup of $(R, +)$, and
- (b) for all $r \in R$, for all $a \in I$, it holds $ra \in I$.

We give now some examples of ideals and define operations on them.

Example 1.10. Let R be a ring.

- (a) The subsets $\{0\}$ and R are ideal and are called *trivial ideals*.
- (b) Given two ideals $I, J \subseteq R$, we define

$$\begin{aligned} I + J &:= \{i + j \mid i \in I, j \in J\}, \\ IJ &:= \left\{ \sum_{k=1}^n i_k j_k \mid n \in \mathbb{N}, i_k \in I, j_k \in J \right\}, \\ I \cap J &:= \{a \mid a \in I \text{ and } a \in J\}. \end{aligned}$$

All three of them are ideals and it holds $IJ \subseteq I \cap J$. Notice that

$$I \cup J := \{a \mid a \in I \text{ or } a \in J\}$$

is an ideal if and only if $I \subseteq J$ or $J \subseteq I$.

- (c) Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a family of ideals in R . We define

$$\begin{aligned} \sum_{\alpha \in \mathcal{A}} I_\alpha &:= \left\{ \sum_{\alpha \in \mathcal{A}} i_\alpha \mid i_\alpha \in I_\alpha \text{ for all } \alpha \in \mathcal{A} \text{ and } i_\alpha \neq 0 \text{ only for finitely many } \alpha \right\}, \\ \bigcap_{\alpha \in \mathcal{A}} I_\alpha &:= \{a \mid a \in I_\alpha \text{ for all } \alpha \in \mathcal{A}\}. \end{aligned}$$

Both of them are ideals. Now, assume that $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ is a chain of ideals, i.e., for any $\alpha, \beta \in \mathcal{A}$, either $I_\alpha \subseteq I_\beta$ or $I_\beta \subseteq I_\alpha$. Then,

$$\bigcup_{\alpha \in \mathcal{A}} I_\alpha := \left\{ a \mid a \in I_\alpha \text{ for some } \alpha \in \mathcal{A} \right\}$$

is also an ideal.

(d) Let $a \in R$. We define

$$(a) := \{\lambda a \mid \lambda \in R\}.$$

This is an ideal called *the ideal generated by a* ; we denote it by aR as well.

(e) Let $I \subseteq R$ be an ideal. We say that I is *principal* if there exists $a \in R$ such that $I = (a)$.

(f) Let $a_1, \dots, a_n \in R$. We define

$$(a_1, \dots, a_n) := a_1R + \dots + a_nR = \left\{ \sum_{i=1}^n \lambda_i a_i \mid \lambda_i \in R \right\}.$$

This is called *the ideal generated by $a_1, \dots, a_n$* .

(g) Let $\{a_i\}_{i \in \mathcal{A}}$ be a subset of R . We define

$$(a_i)_{i \in \mathcal{A}} := \sum_{i \in \mathcal{A}} a_i R_i.$$

This is called *the ideal generated by the subset $\{a_i\}_{i \in \mathcal{A}}$* .

(h) Let $I \subseteq R$ be an ideal. We say that the subset $\{a_i \mid i \in \mathcal{A}\} \subseteq I$ is a *system of generators for I* if $I = (a_i)_{i \in \mathcal{A}}$. The ideal I is said to be *finitely generated* if there exists a finite system of generators for I .

Remark 1.11.

- (a) Let R be a ring and let $I \subseteq R$ be an ideal. If $1_R \in I$, then $I = R$. In particular, a proper ideal $I \subsetneq R$ is not a subring of R .
- (b) A proper ideal $I \subsetneq R$ of a ring R might be itself a ring. In this case, $1_I \neq 1_R$ is an idempotent element of R . For instance, the ring $S \subseteq R \times R$ given in Remark 1.5 is an ideal in $R \times R$. A natural question then arises: does there exist an example of a subset of a ring that is itself a ring but neither a subring nor an ideal of the ambient ring? The answer is positive. Indeed, the same ring S can be regarded as a subset of the polynomial ring $(R \times R)[x]$, but it is neither a subring nor an ideal of $(R \times R)[x]$.

1.2 Prime Ideals and Maximal Ideals

Definition 1.12. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

- (a) The ideal I is called *prime* if for any $a, b \in I$

$$ab \in I \implies a \in I \text{ or } b \in I.$$

- (b) The ideal I is called *maximal* if for any ideal J

$$I \subseteq J \subseteq R \implies J = I \text{ or } J = R.$$

Proposition 1.13. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

- (i) The ideal I is prime if and only if R/I is an integral domain.
(ii) The ideal I is maximal if and only if R/I is a field.

Corollary 1.14. In a ring any maximal ideal is prime.

The converse does not hold true in general.

Example 1.15. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

- (a) If R is an integral domain that is not a field, then $I = (0)$ is prime, but not maximal.
(b) In $R = \mathbb{Z}[x, y]$, the ideal $I = (x)$ is prime, but not maximal since

$$(x) \subsetneq (x, y) \subsetneq \mathbb{Z}[x, y].$$

Proposition 1.16. Let R be a ring and let $I \subseteq R$ be an ideal. There is a one-to-one inclusion-preserving correspondence between the ideals of R containing I and the ideals of R/I . Under this correspondence, prime (resp. maximal) ideals correspond to prime (resp. maximal) ideals.

Hint: Let $\pi: R \rightarrow R/I$ be the canonical projection. If J is an ideal in R/I , then $\pi^{-1}(J)$ is an ideal in R containing I .

Proof. Exercise. □

Proposition 1.17. (i) Every nontrivial ring $R \neq 0$ has a maximal ideal.

- (ii) Let R be a ring and let $I \subsetneq R$ be a proper ideal. Then there exists in R a maximal ideal containing I .

Proof. (i) Let $\mathcal{I} := \{I' \subsetneq R \mid I' \text{ is an ideal of } R\}$. Since $(0) \in \mathcal{I}$, the set $\mathcal{I}$ ordered by inclusion is a nonempty partially ordered set (poset for short). Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a chain in $\mathcal{I}$, i.e., a totally ordered family of proper ideals in R . Since $\bigcup_{\alpha \in \mathcal{A}} I_\alpha$ is a proper ideal, every totally ordered subset of $\mathcal{I}$ has an upper bound in $\mathcal{I}$. Hence, by Zorn's Lemma, there exists a maximal element for the poset $(\mathcal{I}, \subseteq)$, i.e., a maximal ideal in R .

(ii) Apply (i) to the quotient ring R/I and then the one-to-one inclusion-preserving correspondence given by Proposition 1.16. Alternatively, argue as in (i) setting

$$\mathcal{I} := \{I \subseteq I' \subsetneq R \mid I' \text{ is an ideal of } R\}.$$

□

Corollary 1.18. *Every non-unit of a ring R is contained in a maximal ideal.*

Proof. Let $r \in R \setminus R^*$. Then the principal ideal (r) is a proper ideal of R . Now we apply Proposition 1.17 (ii) and we are done. □

Proposition 1.19. *Let R be a ring.*

(i) *Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n \subsetneq R$ be prime ideals and let $I \subseteq R$ be an ideal such that $I \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Then $I \subseteq \mathfrak{p}_i$ for some $i \in \{1, \dots, n\}$.*

(ii) *Let $I_1, \dots, I_n \subseteq R$ be ideals and let $\mathfrak{p} \subsetneq R$ be a prime ideal such that $\mathfrak{p} \supseteq \bigcap_{j=1}^n I_j$. Then $\mathfrak{p} \supseteq I_j$ for some $j \in \{1, \dots, n\}$. Moreover, if $\mathfrak{p} = \bigcap_{j=1}^n I_j$, then $\mathfrak{p} = I_j$ for some $j \in \{1, \dots, n\}$.*

Proof. (i) The statement will be shown by induction on n in the form: If $I \not\subseteq \mathfrak{p}_i$ for $1 \leq i \leq n$, then $I \not\subseteq \bigcup_{i=1}^n \mathfrak{p}_i$.

For $n = 1$ the statement is trivial. Let now $n \geq 2$ and assume the statement is true for $n - 1$. Then for each $1 \leq i \leq n$ by the inductive hypothesis there exists $x_i \in I$ such that $x_i \notin \mathfrak{p}_j$ for all $j \neq i$. If for some $1 \leq i \leq n$ it holds $x_i \notin \mathfrak{p}_i$, then $x_i \notin \mathfrak{p}_j$ for all $1 \leq j \leq n$, thus we are done. Otherwise, it holds $x_i \in \mathfrak{p}_i$ for all $1 \leq i \leq n$. Consider the element

$$y := \sum_{i=1}^n x_1 x_2 \dots x_{i-1} x_{i+1} \dots x_n.$$

Note that $y \in I$ and $y \notin \mathfrak{p}_i$ for all $1 \leq i \leq n$. Indeed, assume by contradiction that $y \in \mathfrak{p}_i$ for some i . Then $x_1 x_2 \dots x_{i-1} x_{i+1} \dots x_n \in \mathfrak{p}_i$ since each of the other summands is a multiple of $x_i \in \mathfrak{p}_i$ and hence in $\mathfrak{p}_i$. But then, since $\mathfrak{p}_i$ is prime, we would have $x_j \in \mathfrak{p}_i$ for some $j \neq i$, a contradiction. Hence $y \notin \bigcup_{i=1}^n \mathfrak{p}_i$ and we are done.

- (ii) Assume that $\mathfrak{p} \not\supseteq I_j$ for all $1 \leq j \leq n$. Then there exists $x_j \in I_j$ but $x_j \notin \mathfrak{p}$ for each $1 \leq j \leq n$. Thus

$$\prod_{j=1}^n x_j \in \prod_{j=1}^n I_j \subseteq \bigcap_{j=1}^n I_j.$$

But, since $\mathfrak{p}$ is prime, $\prod_{j=1}^n x_j \notin \mathfrak{p}$ and thus $\mathfrak{p} \not\supseteq \bigcap_{j=1}^n I_j$.

Now, if $\mathfrak{p} = \bigcap_{j=1}^n I_j$, then in particular we have $\mathfrak{p} \supseteq \bigcap_{j=1}^n I_j$ and therefore there is some $j_0 \in \{1, \dots, n\}$ such that $I_{j_0} \subseteq \mathfrak{p}$. On the other hand, $\mathfrak{p} \subseteq I_j$ for $1 \leq j \leq n$, hence $\mathfrak{p} = I_{j_0}$. □

1.3 Nilradical, Jacobson Radical and Local Rings

Proposition/Definition 1.20. *Let R be a ring and let $I \subseteq R$ be an ideal. The set*

$$\sqrt{I} := \{a \in R \mid a^n \in I \text{ for some } n \in \mathbb{N}_{>0}\}$$

is an ideal called the radical of I .

Proof. Let $r \in R$ and let $a \in \sqrt{I}$. Then $a^n \in I$ for some $n \in \mathbb{N}_{>0}$ and so $(ra)^n = r^n a^n \in I$ since I is an ideal. Now let $x, y \in \sqrt{I}$. Then there exist $n, m \in \mathbb{N}_{>0}$ such that $x^n, y^m \in I$. By the binomial formula we get

$$(x + y)^{m+n-1} = \sum_{k=0}^{m+n-1} \binom{m+n-1}{k} \underbrace{x^k y^{m+n-1-k}}_{\in I},$$

hence, $(x + y)^{m+n-1} \in I$, which implies $x + y \in \sqrt{I}$. Thus, we have shown that $\sqrt{I}$ is indeed an ideal. □

Proposition 1.21. *Let R be a ring. Let I and J be two ideals in R . Then:*

- (i) $\sqrt{I} \supseteq I$,
- (ii) $\sqrt{\sqrt{I}} = \sqrt{I}$,
- (iii) $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.
- (iv) $\sqrt{I} = R \iff I = R$.

Definition 1.22. Let R be a ring. The *nilradical* of R is defined as

$$N_R := \sqrt{(0)} = \{a \in R \mid a^n = 0 \text{ for some } n \in \mathbb{N}_{>0}\}.$$

In other words, the set of nilpotent elements in R is indeed an ideal called the nilradical of R .

Proposition 1.23. *Let R be a ring. Then in the quotient ring R/N_R the only nilpotent element is 0.*

Proof. Let $[x] \in R/N_R$ be nilpotent. This means there is $k \in \mathbb{N}$ such that $[x]^k = [x^k] = 0$ in R/N_R , hence $x^k \in N_R$. Because $x^k \in N_R$, there is $l \in \mathbb{N}$ such that $(x^k)^l = x^{kl} = 0$ in R , and thus $x \in N_R$. Because $x \in N_R$, $[x] = 0$ in R/N_R . $\square$

The following theorem describes the nilradical of a ring and, more in general, the radical of any ideal.

Theorem 1.24. *Let R be a ring and let $I \subseteq R$ be an ideal. Then:*

(i) *The nilradical of R is the intersection of all prime ideals of R , i.e.,*

$$N_R = \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P.$$

(ii) *The radical of I is the intersection of all prime ideals containing I , i.e.,*

$$\sqrt{I} = \bigcap_{\substack{I \subseteq P \subsetneq R \\ \text{prime}}} P.$$

Proof. (i) “ $\subseteq$ ”: Let $P' \subsetneq R$ be a prime ideal in R . If $x \in N_R$, there is $k \in \mathbb{N}$ such that $x^k = 0 \in P'$. Since P' is prime, it holds $x \in P'$, and thus $N \subseteq \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

“ $\supseteq$ ”: Let now $x \in R \setminus N_R$ and let $\mathcal{S}_x$ be the set of proper ideals $I \subsetneq R$ with the property that if $n \in \mathbb{N}_{>0}$, then $x^n \notin I$. Since $(0) \in \mathcal{S}_x$, the set $\mathcal{S}_x$ is a nonempty poset (with respect to inclusion). As in the proof of Proposition 1.17 (i), by Zorn’s Lemma $\mathcal{S}_x$ has a maximal element, say P' .

We show now that P' is a prime ideal in R . To see this, let $a, b \in R \setminus P'$. Then the ideals $(a) + P'$ and $(b) + P'$ are strictly larger than P' (with respect to inclusion) and thus are not in $\mathcal{S}_x$. Since $(a) + P', (b) + P' \notin \mathcal{S}_x$, there is $m \in \mathbb{N}_{>0}$ such that $x^m \in (a) + P'$ and there is $k \in \mathbb{N}_{>0}$ such that $x^k \in (b) + P'$. Thus $x^m x^k = x^{m+k} \in [(a) + P'][(b) + P'] \subseteq (ab) + P'$. Since $x^{m+k} \in (ab) + P'$, the ideal $(ab) + P' \notin \mathcal{S}_x$, and thus $(ab) + P' \supsetneq P'$, which implies $ab \notin P'$. Finally, we have found a prime ideal P' such that $P' \in \mathcal{S}_x$. By definition of $\mathcal{S}_x$, the ideal P' does not contain any power of x . In particular, $x \notin P'$ and hence $x \notin \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

(ii) Let $\pi: R \rightarrow R/I$ be the canonical projection. Note that $\pi^{-1}(N_{R/I}) = \sqrt{I}$. Hence, we first apply (i) to the ring R/I and then use the correspondence between prime ideals in R containing I and prime ideals in R/I given by Proposition 1.16.

$\square$

Definition 1.25. Let R be a ring. The *Jacobson radical* J_R of R is defined as the intersection of all maximal ideals, i.e.,

$$J_R := \bigcap_{\substack{\mathfrak{m} \subseteq R \\ \text{maximal}}} \mathfrak{m}.$$

Proposition 1.26. Let R be a ring. The Jacobson radical J_R admits the following description

$$J_R = \{x \in R \mid 1 - xy \in R^* \text{ for all } y \in R\}.$$

Proof. “ $\subseteq$ ”: Let $x \in J_R$ and assume that $1 - xy$ is not a unit for some $y \in R$. Then by Proposition 1.17 (ii) the element $1 - xy$ is contained in some maximal ideal $\mathfrak{m}$ of R and by definition of Jacobson radical, $x \in \mathfrak{m}$ as well. But then $1 \in \mathfrak{m}$, which is a contradiction.

“ $\supseteq$ ”: If $x \notin J_R$, then $x \notin \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ in R . Hence, $(x) + \mathfrak{m} \supsetneq \mathfrak{m}$ and thus $(x) + \mathfrak{m} = (1)$, i.e., there are elements $y \in R$ and $z \in \mathfrak{m}$ such that $xy + z = 1$. In other words, the element $1 - xy = z \in \mathfrak{m}$ and therefore $1 - xy$ is not a unit. $\square$

Definition 1.27. A ring R is called *local* if it contains only one maximal ideal $\mathfrak{m}$. Then the quotient $R/\mathfrak{m}$ is a field, called the *residue field* of R . In literature, a local ring is often denoted by $(R, \mathfrak{m})$ or $(R, \mathfrak{m}, k)$, where $k := R/\mathfrak{m}$.

Proposition 1.28. (i) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be an ideal such that every element in $R \setminus \mathfrak{m}$ is a unit. Then R is a local ring and $\mathfrak{m}$ is its unique maximal ideal.

(ii) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be a maximal ideal such that every element in the set $1 + \mathfrak{m} := \{1 + x \mid x \in \mathfrak{m}\}$ is a unit. Then R is local and $\mathfrak{m}$ is its unique maximal ideal.

Proof. Exercise. $\square$

1.4 Chinese Remainder Theorem

Definition 1.29. Let R be a ring and let $I, J \subseteq R$ be ideals. Then I and J are called *coprime* (or *comaximal*) if $I + J = R$.

Remark 1.30. Let R be a ring and I, J be ideals in R .

- (i) If the ideals I, J are coprime, it holds $IJ = I \cap J$.
- (ii) Two ideals $I, J \subseteq R$ are coprime if and only if there exist $i \in I, j \in J$ such that $i + j = 1$.

Proposition 1.31 (Chinese Remainder Theorem). *Let R be a ring and $I_1, \dots, I_n \subseteq R$ be ideals. We define a ring homomorphism by*

$$\varphi: R \longrightarrow \prod_{j=1}^n (R/I_j), \quad x \longmapsto (x + I_1, \dots, x + I_n).$$

- (i) *If I_k and I_l are coprime for $k \neq l$, then $\prod_{j=1}^n I_j = \bigcap_{j=1}^n I_j$.*
- (ii) *φ is surjective if and only if I_k and I_l are coprime for $k \neq l$.*
- (iii) *φ is injective if and only if $\bigcap_{j=1}^n I_j = (0)$.*

1.5 Hilbert's Basis Theorem

Definition 1.32. A ring R is called *Noetherian* if every ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq \dots \subseteq I_k \subseteq \dots$$

is stationary, i.e., there exists $N \in \mathbb{N}_{>0}$ such that $I_k = I_N$ for $k \geq N$.

Example 1.33. Any principal ideal domain (PID) is a Noetherian ring, as shown in the course “Algebra (WS 2025/26)”.

The following well-known theorem allows us to construct new Noetherian rings from a given one.

Theorem 1.34 (Hilbert's Basis Theorem). *Let R be a Noetherian ring. Then $R[x_1, \dots, x_n]$ is also Noetherian.*

Proof. First of all, it suffices to show the theorem when $n = 1$ since

$$R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n].$$

By contradiction, assume $R[x]$ is not Noetherian. Let $I_1 \subsetneq I_2 \subsetneq \dots$ be a strictly ascending chain of ideals. Set $I := \bigcup_{n \in \mathbb{N}} I_n$. We construct a sequence $\{f_1, f_2, \dots\}$ of polynomials in $R[x]$ as follows:

- (a) We choose $f_1 \in I - \{0\}$ of minimal degree.
- (b) Assume we found $f_1, f_2, \dots, f_{n-1}$. We pick f_n in $I \setminus (f_1, \dots, f_{n-1})$ of minimal degree.

Note that this is a well-defined construction: indeed, $\mathbb{N}$ is well-ordered and step (b) is repeated infinitely many times since the given chain of ideals is strictly ascending.

Claim 1. $\deg(f_n) \geq \deg(f_{n-1})$ for $n \geq 2$.

Proof of Claim 1. First of all, $\deg(f_2) \geq \deg(f_1)$ by (a). Assume by contradiction $\deg(f_n) < \deg(f_{n-1})$ for some $n \geq 3$. Since by construction $f_n \in I \setminus (f_1, \dots, f_{n-1}) \subseteq I \setminus (f_1, \dots, f_{n-2})$, we found a polynomial in $I \setminus (f_1, \dots, f_{n-2})$ whose degree is lower than $\deg(f_{n-1})$, against the choice of f_{n-1} as a polynomial in $I \setminus (f_1, \dots, f_{n-2})$ of minimal degree. $\square$

For $i \in \mathbb{N}$ define $a_i \in R$ to be the leading coefficient of f_i and define the ideal $J_i := (a_1, \dots, a_i)$. We have then an ascending chain of ideals of R , namely $J_1 \subseteq J_2 \subseteq J_3 \subseteq \dots$. As R is noetherian, there is $N \in \mathbb{N}$ such that $J_k = J_N$ for all $k \geq N$. Consider the element $a_{N+1} \in J_{N+1} = J_N = (a_1, \dots, a_N)$. Then for $1 \leq i \leq N$ there are $u_i \in R$ such that $a_{N+1} = \sum_{i=1}^N u_i a_i$. Now set

$$f'_{N+1} := u_1 x^{\deg(f_{N+1}) - \deg(f_1)} f_1 + \dots + u_N x^{\deg(f_{N+1}) - \deg(f_N)} f_N.$$

The leading term of f'_{N+1} is $a_{N+1} x^{\deg(f_{N+1})}$ and $f'_{N+1} \in (f_1, \dots, f_N)$. Consider the polynomial

$$g := f'_{N+1} - f_{N+1}.$$

It holds $g \in I \setminus (f_1, \dots, f_N)$ since otherwise $f_{N+1} = f'_{N+1} - g \in (f_1, \dots, f_N)$. Moreover, by construction $\deg(g) < \deg(f_{N+1})$ and this contradicts the choice of f_{N+1} as a polynomial in $I \setminus (f_1, \dots, f_N)$ of minimal degree. $\square$

Chapter 2

Modules

Definition 2.1. Let A be a ring. An A -module is an abelian group $(M, +)$ together with a map $\mu: A \times M \rightarrow M$, $(a, m) \mapsto a \cdot m := \mu(a, m)$, which satisfies the following properties for all $a, b \in A$ and for all $x, y \in M$:

- (a) $a(x + y) = ax + ay$,
- (b) $(a + b)x = ax + bx$,
- (c) $(ab)x = a(bx)$,
- (d) $1_A x = x$.

Example 2.2. Let A be a ring. The following are some examples of modules:

- (a) An ideal $I \subseteq A$ is an A -module. In particular, A is itself an A -module.
- (b) If A is a field, then an A -module is an A -vector space.
- (c) Let $(G, +)$ be an abelian group. Then G is a $\mathbb{Z}$ -module with respect to the scalar multiplication

$$n \cdot x := \begin{cases} \underbrace{x + \dots + x}_{n\text{-times}} & \text{if } n \geq 0 \\ \underbrace{(-x) + \dots + (-x)}_{(-n)\text{-times}} & \text{if } n < 0 \end{cases}$$

- (d) Assume $A = k[X]$, where k is a field. Let V be a k -vector space and let $T \in \text{End}(V)$ be a linear map. Then we can regard V as an A -module via the following construction: for $f = \sum_{i=0}^n a_i X^i \in k[X]$ set $f(T) := \sum_{i=0}^n a_i T^i$ and define $\mu: k[X] \times V \rightarrow V$ for $f \in k[X]$ and $v \in V$ by

$$f \cdot v := f(T)(v).$$

Definition 2.3. Let A be a ring and M, N be A -modules. A map $f: M \rightarrow N$ is called an *A -module homomorphism* (or *A -linear*) if the following properties are satisfied:

- (i) $f(x + y) = f(x) + f(y)$ for all $x, y \in M$,
- (ii) $f(ax) = af(x)$ for all $x \in M$, for all $a \in A$.

Example 2.4. If A is a field, then an A -module homomorphism is a linear map.

Remark 2.5.

- (a) Let M, N be A -modules. The set $\text{Hom}_A(M, N)$ of all A -module homomorphisms from M to N is an A -module in the following way. We define the sum of $f, g \in \text{Hom}_A(M, N)$ via

$$(f + g)(x) := f(x) + g(x) \quad \text{for all } x \in M,$$

and the scalar multiplication of $a \in A$ and $f \in \text{Hom}_A(M, N)$ via

$$(af)(x) := af(x) \quad \text{for all } x \in M.$$

These operations make $\text{Hom}_A(M, N)$ into an A -module. When the ring A is clear from the context, we simply write $\text{Hom}(M, N)$.

- (b) The composition of A -module homomorphisms is again an A -module homomorphism.
- (c) Let M, M', N, N'' be A -modules and let $U: M' \rightarrow M$ and $V: N \rightarrow N''$ be A -module homomorphisms. Then we have the following induced A -module homomorphisms:

$$\begin{aligned} \bar{U}: \text{Hom}_A(M, N) &\longrightarrow \text{Hom}_A(M', N), & \bar{U}(f) &:= f \circ U, \\ \bar{V}: \text{Hom}_A(M, N) &\longrightarrow \text{Hom}_A(M, N''), & \bar{V}(f) &:= V \circ f. \end{aligned}$$

- (d) For any A -module M we have a natural isomorphism

$$\text{Hom}_A(A, M) \cong M, \quad f \longmapsto f(1).$$

2.1 Submodules and Quotient Modules

Definition 2.6. Let A be a ring and M be an A -module. A *submodule* $N \subseteq M$ is a subgroup of M which is closed under scalar multiplication, i.e., $a \cdot x \in N$ for $a \in A$ and $x \in N$.

Definition 2.7. Let M be an A -module and let $N \subseteq M$ be a submodule. The abelian group M/N inherits an A -module structure from M via

$$a \cdot (x + N) := ax + N \quad \text{for } a \in A, x \in M.$$

The A -module M/N is called the *quotient of M by N* . The canonical projection $\pi: M \rightarrow M/N$ is an A -module homomorphism.

Proposition 2.8. Let M be an A -module and let $N \subseteq M$ be a submodule. There is a one-to-one inclusion-preserving correspondence between the submodules of M which contain N and submodules of the quotient M/N .

Definition 2.9. Let A be a ring and let M, N be A -modules. Let $f: M \rightarrow N$ be an A -module homomorphism.

- (a) The *kernel of f* is defined as $\ker(f) := \{x \in M \mid f(x) = 0\}$. This is indeed a submodule of M .
- (b) The *image of f* is defined as $\operatorname{im}(f) := \{f(x) \mid x \in M\} = f(M)$. This is indeed a submodule of N .
- (c) The *cokernel of f* is defined as the quotient $\operatorname{coker}(f) := N/\operatorname{im}(f)$.

Theorem 2.10 (Homomorphism Theorem/First Isomorphism Theorem). Let A be a ring, let M, N be A -modules and let $f: M \rightarrow N$ be an A -module homomorphism. Then

$$M/\ker(f) \cong \operatorname{im}(f).$$

2.2 Operations on Submodules

Definition 2.11. Let M be an A -module and let $\{M_i\}_{i \in I}$ be a family of submodules of M . We define the *intersection of the submodules M_i*

$$\bigcap_{i \in I} M_i := \left\{ m \mid m \in M_i \text{ for all } i \in I \right\}$$

and the *sum of the submodules M_i*

$$\sum_{i \in I} M_i := \left\{ \sum_{i \in I} m_i \mid m_i \in M_i \text{ for all } i \in I \text{ and } m_i \neq 0 \text{ for only finitely many } i \right\}.$$

Both are submodules of M . Moreover, we say that the sum $\sum_{i \in I} M_i$ is *direct* and denote it by $\bigoplus_{i \in I} M_i$ if every element $m \in \sum_{i \in I} M_i$ can be written uniquely as $\sum_{i \in I} m_i$ where $m_i \in M_i$ for each $i \in I$.

Remark 2.12. Let M_1 and M_2 be submodules of M . The sum $M_1 + M_2$ is direct if and only if $M_1 \cap M_2 = \{0\}$.

Now, assume $\{M_i\}_{i \in I}$ is a chain of submodules of M , i.e., for any $i, j \in I$, either $M_i \subseteq M_j$ or $M_j \subseteq M_i$. Then

$$\bigcup_{i \in I} M_i := \left\{ m \mid m \in M_i \text{ for some } i \in I \right\}$$

is also a submodule of M .

Since we can't multiply two elements of a module, there is no product of submodules. However, we define the product of an ideal $I \subseteq A$ with the A -module M as

$$IM := \left\{ \sum_{i=1}^n a_i m_i \mid a_i \in I, m_i \in M, n \in \mathbb{N} \right\},$$

which is a submodule of M .

Theorem 2.13 (Second Isomorphism Theorem). *Let A be a ring, let M be an A -module and let M_1 and M_2 be submodules of M . Then*

$$(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2).$$

Proof. Consider

$$M_2 \longrightarrow M_1 + M_2 \longrightarrow (M_1 + M_2)/M_1, \quad x \longmapsto 0 + x \longmapsto x + M_1.$$

The composition $\varphi: M_2 \rightarrow (M_1 + M_2)/M_1$ is surjective since for $m_i \in M_i$ it holds $(m_1 + m_2) + M_1 = m_2 + M_1$. Moreover, we have $\ker(\varphi) = M_1 \cap M_2$. Now the first isomorphism theorem gives the thesis. $\square$

Theorem 2.14 (Third Isomorphism Theorem). *Let A be a ring. Let $L \supseteq M \supseteq N$ be A -modules. Then*

$$(L/N)/(M/N) \cong L/M.$$

Proof. Consider

$$\psi: L/N \longrightarrow L/M, \quad x + N \longmapsto x + M.$$

This is a well-defined A -module homomorphism since $N \subseteq M$. It is clearly surjective and moreover $\ker(\psi) = M/N$. Therefore, the first isomorphism theorem shows the assertion. $\square$

2.3 Direct Sums and Direct Products

Definition 2.15. Let A be a ring and let $\{M_i\}_{i \in I}$ be a family of A -modules. We define:

(i) The *direct product* of the M_i

$$\prod_{i \in I} M_i := \left\{ f: I \rightarrow \bigsqcup_{i \in I} M_i \mid f(i) \in M_i \text{ for } i \in I \right\}.$$

This is an A -module whose elements are usually written as sequences $(m_i)_{i \in I}$ such that $m_i \in M_i$ for $i \in I$. The operations in $\prod_{i \in I} M_i$ are componentwise.

(ii) The *direct sum* of the M_i

$$\bigoplus_{i \in I} M_i := \left\{ f \in \prod_{i \in I} M_i \mid f(i) \neq 0 \text{ for only finitely many } i \right\}.$$

This is a submodule of the direct product $\prod_{i \in I} M_i$.

Remark 2.16. In general, direct sum and direct product of a family $\{M_i\}_{i \in I}$ of A -modules are not the same. They coincide if the index set I is finite.

Remark 2.17 (Internal and External Direct Sum). Let M be an A -module. For a family $\{M_i\}_{i \in I}$ of submodules of M we have at this stage two notions of their direct sum: namely, the direct sum as submodules in the sense of Definition 2.11 and the direct sum as a family of modules in the sense of Definition 2.15. For the time being, we refer to them, respectively, as *internal direct sum* and *external direct sum*. Indeed, the two concepts coincide up to isomorphism and this is the reason why one refers to both of them simply as direct sum. For the benefit of the reader, we explain in detail the case of two submodules M_1 and M_2 . Suppose that the sum $M_1 + M_2$ is direct in the sense of Definition 2.11 and denote it by $M_1 \oplus_i M_2$. On the other hand, we consider the external direct sum of M_1 and M_2 , which we denote here by $M_1 \oplus_e M_2$. Consider

$$f: M_1 \oplus_i M_2 \rightarrow M_1 \oplus_e M_2, \quad m_1 + m_2 \mapsto (m_1, m_2).$$

This map is well-defined since the sum $M_1 + M_2$ is direct in the sense of Definition 2.11. It is easy to see that f is indeed an isomorphism of A -modules.

Let now M_1 and M_2 be A -modules. Consider the external direct sum $M_1 \oplus_e M_2$. We define:

$$\widetilde{M}_1 := \{(m_1, 0) \mid m_1 \in M_1\} \quad \text{and} \quad \widetilde{M}_2 := \{(0, m_2) \mid m_2 \in M_2\}.$$

Both are submodules of $M_1 \oplus_e M_2$ and clearly $\widetilde{M}_i \cong M_i$. It is easy to see that the sum $\widetilde{M}_1 + \widetilde{M}_2$ is direct in the sense of Definition 2.11. Moreover, we have

$$\widetilde{M}_1 \oplus_i \widetilde{M}_2 = M_1 \oplus_e M_2.$$

Remark 2.18 (Direct Sum of Rings and Direct Sum of Ideals).

Let A be a nontrivial ring.

- (i) Assume that A is a direct sum of rings $A = \bigoplus_{j=1}^n A_j = A_1 \times \dots \times A_n$. Then the ring

$$I_j := \{(0, \dots, 0, a_j, 0, \dots, 0) \mid a_j \in A_j\} \cong A_j$$

is an ideal of A , but not a subring since $1_A = (1, \dots, 1) \notin I_j$. Hence, the ring A , considered as an A -module, is the direct sum of the ideals I_j , i.e., $A = \bigoplus_{j=1}^n I_j$.

- (ii) Assume now that A , considered as an A -module, is a direct sum of ideals $A = \bigoplus_{j=1}^n I_j$. Set $B_j := \bigoplus_{k \neq j} I_k$ and note that $A/B_j \cong I_j$, which shows that I_j is indeed a ring, and hence there is a ring isomorphism

$$A \cong \bigoplus_{j=1}^n (A/B_j).$$

Since $A = \bigoplus_{j=1}^n I_j$, we may write $1_A = (y_1, \dots, y_n)$ with $y_j \in I_j$. Then we have that

$$1_{A/B_j} = 1_A + B_j = (y_1, \dots, y_n) + B_j = (0, \dots, 0, y_j, 0, \dots, 0) + B_j$$

and by means of the isomorphism $A/B_j \cong I_j$ we get $1_{I_j} = y_j$. Note that $1_{I_j} = y_j$ is idempotent and $I_j = (y_j)$ as an ideal in A .

2.4 Finitely Generated Modules

We now introduce some terminology and key concepts needed for what follows.

Definition 2.19. Let A be a ring and M be an A -module.

- (a) Let $x \in M$. The set $Ax := \{ax \mid a \in A\} \subseteq M$ is a submodule of M which is called *the submodule generated by x* .
- (b) The A -module M is called *principal* if there exists $x \in M$ such that $M = Ax$.
- (c) Let $\{x_i\}_{i \in I}$ be a subset of M . The sum $\sum_{i \in I} Ax_i$ is called *the submodule generated by the subset $\{x_i\}_{i \in I}$* .
- (d) If $M = \sum_{i \in I} Ax_i$, then the set $\{x_i\}_{i \in I}$ is called a *set of generators of M* .
- (e) The A -module M is called *finitely generated* if there exists a finite set of generators of M , i.e., $M = Ax_1 + \dots + Ax_n$ for some $x_1, \dots, x_n \in M$.
- (f) The A -module M is called *free* if $M \cong \bigoplus_{i \in I} M_i$, where $M_i \cong A$ for each $i \in I$.

(g) For $n \in \mathbb{N}$, we denote by A^n the direct sum of A with itself n times:

$$A^n := \bigoplus_{i=1}^n A = \underbrace{A \oplus \dots \oplus A}_{n\text{-times}}.$$

For $i \in \{1, \dots, n\}$, we define $e_i := (0, \dots, 0, \overset{i}{1}, 0, \dots, 0)$. The set $\{e_i\}_{i=1}^n$ is called the *canonical basis* (or *canonical set of generators*) of A^n . By convention, we set $A^0 := \{0\}$.

(h) Let X be a set. We define

$$A^X := \left\{ f: X \rightarrow A \mid f(x) \neq 0 \text{ for only finitely many } x \right\}.$$

For $f, g \in A^X$ and for $a \in A$ we define for each $x \in X$

$$(f + g)(x) := f(x) + g(x) \quad \text{and} \quad (a \cdot f)(x) := af(x).$$

The operations above make A^X into an A -module. Indeed, there is an A -linear isomorphism $A^X \cong \bigoplus_{x \in X} M_x$, where $M_x = A$ for all $x \in X$. Thus, A^X is a free module called *the free module over X* . For each $x \in X$ define $e_x \in A^X$ by

$$e_x(y) = \delta_{xy} = \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{if } x \neq y. \end{cases}$$

The set $\{e_x\}_{x \in X}$ is called the *canonical basis* (or *canonical set of generators*) of A^X .

Remark 2.20. (a) Note that the free module A^X can be described as the A -module S of finite formal linear combinations

$$\sum_{i=1}^n a_i x_i,$$

where $n \in \mathbb{N}_{>0}$, $a_i \in A$ and $x_i \in X$. Indeed, there is an A -linear isomorphism

$$A^X \rightarrow S, \quad f \mapsto \sum_{x \in X} f(x) \cdot x,$$

where the sum is finite since $f(x) \neq 0$ for only finitely many $x \in X$. We will adopt the description of A^X as S , since it is more convenient for computations and constructions. Note that in this description of A^X the canonical basis is given by the set X itself: each $x \in X$ can be regarded as the formal linear combination $1 \cdot x$.

(b) Note that the trivial module $M = 0$ is finitely generated: the unique set of generators is the empty-set.

We will mostly focus on finitely generated modules. The following result provides a useful characterization of these modules.

Proposition 2.21. *Let A be a ring and let M be an A -module. Then M is a finitely generated A -module if and only if it is isomorphic to a quotient of a finitely generated free A -module. More precisely, $M \cong A^n/B$ if and only if M has n generators.*

Proof. “ $\Rightarrow$ ”: Let $x_1, \dots, x_n$ be generators of M . Define

$$\varphi: A^n \longrightarrow M, \quad (a_1, \dots, a_n) \longmapsto \sum_{i=1}^n a_i x_i.$$

The map φ is a surjective A -module homomorphism, and thus $A^n/\ker(\varphi) \cong M$ by the homomorphism theorem.

“ $\Leftarrow$ ”: If $M \cong A^n/B$, we have a surjective A -module homomorphism $\varphi: A^n \rightarrow M$. Then the images $\varphi(e_i)$ of the canonical generators e_i generate M . $\square$

Proposition 2.22 (Nakayama’s Lemma). *Let A be a ring and let M be a finitely generated A module. Let I be an ideal of A contained in the Jacobson radical J_A . Then $IM = M$ implies $M = \{0\}$.*

Proof. We argue by contradiction. Suppose $M \neq \{0\}$ and let $u_1, \dots, u_n$ be a minimal set of generators of M . Then $u_n \in M = IM$, thus we may write $u_n = a_1 u_1 + \dots + a_n u_n$ for some $a_i \in I$. But then

$$(1 - a_n)u_n = a_1 u_1 + \dots + a_{n-1} u_{n-1} \tag{2.1}$$

and since $a_n \in I \subseteq J_A$, it follows from Proposition 1.26 that $1 - a_n$ is a unit in A . Hence, by multiplying (2.1) with $(1 - a_n)^{-1}$ we have that $u_n \in Au_1 + \dots + Au_{n-1}$, a contradiction since the set of u_i is a minimal set of generators of M . $\square$

Corollary 2.23. *Let A be a ring and let M be a finitely generated A -module. Let $N \subseteq M$ be a submodule and let I be an ideal contained in the Jacobson radical J_A . Then $M = IM + N$ implies $N = M$.*

Proof. Observe that

$$I(M/N) = (IM + N)/N.$$

Since $M = IM + N$, we have

$$I(M/N) = M/N.$$

Moreover, because M is finitely generated, the quotient module M/N is finitely generated as well. Therefore, we apply Nakayama’s Lemma to M/N and get $M/N = 0$, equivalently, $N = M$. $\square$

Definition 2.24. Let A be a ring and let M be an A -module. The *annihilator* of M is defined as

$$\text{Ann}(M) := \{a \in A \mid am = 0 \text{ for all } m \in M\}$$

This is an ideal in A .

Definition 2.25. An A -module M is called *faithful* if $\text{Ann}(M) = 0$.

Remark 2.26. (a) Let M be an A -module and let $I \subseteq \text{Ann}(M)$ be an ideal. Then we may regard M as an (A/I) -module as follows: for $a \in A$ and for $x \in M$, we define

$$(a + I) \cdot x := ax.$$

This scalar multiplication is well-defined since $IM = 0$.

(b) Let M be a finitely generated A -module and let $\{x_1, \dots, x_n\}$ be a minimal set of generators of M . Let $I \subseteq J_A$ be an ideal. Since $I \subseteq \text{Ann}(M/IM)$, the quotient M/IM has a natural structure of (A/I) -module by (a). Denote by $[x_1], \dots, [x_n]$ the residue classes of $x_1, \dots, x_n$ in M/IM ; these classes generate M/IM as an A -module as well as an (A/I) -module. Since the x_i 's form a minimal set of generators, by using the same argument as in the proof of Nakayama's lemma we have that $[x_j] \neq 0$ for $1 \leq j \leq n$ and $[x_i] \neq [x_j]$ for any $i \neq j$.

Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A -module. Since $\mathfrak{m} \subseteq \text{Ann}(M/\mathfrak{m}M)$, by Remark 2.26(a) the quotient module $M/\mathfrak{m}M$ can be regarded as an $(A/\mathfrak{m})$ -module, i.e., a k -vector space. Since M is a finitely generated A -module, then $M/\mathfrak{m}M$ is a finitely generated A -module, hence a finite dimensional k -vector space. Let $\{x_1, \dots, x_n\}$ be a minimal set of generators of M and denote by $[x_1], \dots, [x_n]$ the residue classes of $x_1, \dots, x_n$ in $M/\mathfrak{m}M$. By Remark 2.26(b) the classes $[x_1], \dots, [x_n]$ are pairwise distinct. Indeed, we can say more.

Proposition 2.27. Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A -module. Let $\{x_1, \dots, x_n\}$ be a minimal set of generators of M . Then the classes $[x_1], \dots, [x_n]$ form a basis of $M/\mathfrak{m}M$ as a k -vector space. In particular, any minimal set of generators of M contains exactly $\dim_k(M/\mathfrak{m}M)$ elements.

Proof. It is enough to show that $[x_1], \dots, [x_n]$ are linearly independent over k .

Let

$$\overline{a}_1[x_1] + \dots + \overline{a}_n[x_n] = [0], \quad \overline{a}_i \in A/\mathfrak{m} = k.$$

Hence, for $1 \leq i \leq n$ there exists $b_i \in \mathfrak{m}$ such that

$$a_1x_1 + \dots + a_nx_n = b_1x_1 + \dots + b_nx_n \in \mathfrak{m}M.$$

Assume by contradiction that $\overline{a_i} \neq \overline{0}$, i.e., $a_i \notin \mathfrak{m}$, for some i . Then $a_i - b_i \notin \mathfrak{m}$ since $b_i \in \mathfrak{m}$, and hence $a_i - b_i$ is invertible since $\mathfrak{m}$ is the unique maximal ideal of the local ring A . It follows that

$$x_i = \sum_{j \neq i} (a_i - b_i)^{-1} (b_j - a_j) x_j,$$

against the minimality of $\{x_1, \dots, x_n\}$. $\square$

Hence, if $\{x_1, \dots, x_n\}$ is a minimal set of generators of M , then $\{[x_1], \dots, [x_n]\}$ is a basis of the k -vector space $M/\mathfrak{m}M$. The following result shows that the converse holds true as well.

Proposition 2.28 (Nakayama's Lemma for Local Rings). *Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A -module. Let $x_1, \dots, x_n$ be elements of M whose residue classes $[x_1], \dots, [x_n] \in M/\mathfrak{m}M$ form a basis of this k -vector space. Then $x_1, \dots, x_n$ generate M .*

Proof. If $M/\mathfrak{m}M = 0$, then the unique subset of M fulfilling the assumptions is the empty-set. The thesis is then true since $M = 0$ by Nakayama's lemma. Now assume $M/\mathfrak{m}M \neq 0$. Then there are elements $x_1, \dots, x_n \in M$ fulfilling the assumptions. Let $N := \sum_{i=1}^n Ax_i$. Observe that the map A -linear map

$$\varphi: N \rightarrow M/\mathfrak{m}M, \quad x \mapsto [x]$$

is surjective since $\{[x_i] \mid 1 \leq i \leq n\}$ is a basis of $M/\mathfrak{m}M$.

Since M is finitely generated and $J_A = \mathfrak{m}$, if we show that $N + \mathfrak{m}M = M$, then $N = M$ by Corollary 2.23 and we are done. Hence, let us show that $N + \mathfrak{m}M = M$.

" $\subseteq$ " is clear. For " $\supseteq$ ", let $x \in M$ and consider $[x] \in M/\mathfrak{m}M$. Since φ is surjective, there is $n \in N$ such that $[n] = [x]$, and thus $x - n \in \mathfrak{m}M$. Therefore, there exists $y \in \mathfrak{m}M$ such that $x = n + y \in N + \mathfrak{m}M$. $\square$

Remark 2.29. Let $(A, \mathfrak{m}, k)$ be a local ring. Then the following statements are equivalent:

- (a) Let M be a finitely generated A -module. If $M = \mathfrak{m}M$, then $M = 0$.
- (b) Let M be a finitely generated A -module. Let $x_1, \dots, x_n$ be elements of M whose residue classes $[x_1], \dots, [x_n] \in M/\mathfrak{m}M$ form a basis of this k -vector space. Then $x_1, \dots, x_n$ generate M .

Hence, for a local ring Nakayama's Lemma (Proposition 2.22) and Proposition 2.28 are indeed equivalent and this motivates the name "Nakayama's lemma for local rings" given to Proposition 2.28.

2.5 Exact Sequences

Definition 2.30. Let A be a ring. A sequence of A -modules and A -module homomorphisms

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots \quad (2.2)$$

is called

- (i) *exact at M_i* if $\ker(f_{i+1}) = \operatorname{im}(f_i)$,
- (ii) *exact* if it is exact at every M_i .

In particular, a sequence $0 \rightarrow M' \xrightarrow{f} M$ is exact if and only if f is injective, a sequence $M \xrightarrow{g} M'' \rightarrow 0$ is exact if and only if g is surjective and

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

is exact if and only if f is injective, g is surjective and $\ker(g) = \operatorname{im}(f)$. A sequence of this form is called a *short exact sequence*.

Remark 2.31. Any long exact sequence of the form 2.2 can be split into short exact sequences. Indeed, for each i set $N_i := \operatorname{im}(f_i) = \ker(f_{i+1})$. Then we have a short exact sequence

$$0 \longrightarrow N_i \xrightarrow{j} M_i \xrightarrow{f_{i+1}} N_{i+1} \longrightarrow 0,$$

where $j: N_i \rightarrow M_i$ is the inclusion map.

Proposition 2.32. (i) *Let*

$$M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0 \quad (2.3)$$

be a sequence of A -modules and A -module homomorphisms. Then (2.3) is exact if and only if for all A -modules N the sequence

$$\begin{array}{ccccccc} 0 \longrightarrow \operatorname{Hom}_A(M'', N) & \xrightarrow{\bar{v}} & \operatorname{Hom}_A(M, N) & \xrightarrow{\bar{u}} & \operatorname{Hom}_A(M', N) & & \\ & & f \longmapsto f \circ v & & & & \\ & & & & g \longmapsto g \circ u & & \end{array}$$

is exact.

(ii) *Let*

$$0 \longrightarrow N' \xrightarrow{u} N \xrightarrow{v} N'' \quad (2.4)$$

be a sequence of A -modules and A -module homomorphisms. Then (2.4) is exact if and only if for all A -modules M the sequence

$$\begin{array}{ccccccc} 0 \longrightarrow \operatorname{Hom}_A(M, N') & \xrightarrow{\bar{u}} & \operatorname{Hom}_A(M, N) & \xrightarrow{\bar{v}} & \operatorname{Hom}_A(M, N'') & & \\ & & f \longmapsto u \circ f & & & & \\ & & & & g \longmapsto v \circ g & & \end{array}$$

is exact.

Proposition 2.33 (Snake Lemma). *Let*

$$\begin{array}{ccccccc} M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' \end{array}$$

be a commutative diagram of A -modules and A -module homomorphisms with exact rows. Then there is an exact sequence

$$\begin{array}{ccccccc} \ker(f') & \xrightarrow{\bar{u}} & \ker(f) & \xrightarrow{\bar{v}} & \ker(f'') & & \\ & & & & \searrow d & & \\ & & \text{coker}(f') & \xrightarrow{\bar{u}'} & \text{coker}(f) & \xrightarrow{\bar{v}'} & \text{coker}(f'') \end{array} \quad (2.5)$$

where $\bar{u}, \bar{v}$ are restrictions of u, v and $\bar{u}', \bar{v}'$ are induced by u', v' . The map d is called the connecting homomorphism (or boundary homomorphism). Moreover, if u is injective, then so is the first map in (2.5), and if v' is surjective, then so is the last map in (2.5).

Proof. We have a diagram as follows:

$$\begin{array}{ccccccc} & 0 & & 0 & & 0 & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \cdots \cdots \cdots \rightarrow & \ker(f') & \xrightarrow{\bar{u}} & \ker(f) & \xrightarrow{\bar{v}} & \ker(f'') \cdots \cdots \cdots \rightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \cdots \cdots \cdots \rightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' \cdots \cdots \cdots \rightarrow 0 \\ & \downarrow f' & & \downarrow f & & \downarrow f'' & \\ 0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' \cdots \cdots \cdots \rightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ & \text{coker}(f') & \xrightarrow{\bar{u}'} & \text{coker}(f) & \xrightarrow{\bar{v}'} & \text{coker}(f'') & \longrightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ & 0 & & 0 & & 0 & \end{array}$$

Define $\bar{u} := u|_{\ker(f')}: \ker(f') \rightarrow \ker(f)$. We have to check just that $f(u(x)) = 0$ for $x \in \ker(f')$. Since the diagram above is commutative, we have $f(u(x)) = u'(f'(x)) = u'(0) = 0$. Similarly, one can check that $\bar{v} := v|_{\ker(f)}: \ker(f) \rightarrow \ker(f'')$ is well-defined. Note that $\bar{u}'$ (resp. $\bar{v}'$) is well-defined if $u'(\text{im}(f')) \subseteq \text{im}(f)$ (resp. $v'(\text{im}(f)) \subseteq \text{im}(f'')$). Let $y = f'(x) \in \text{im}(f')$. Then $u'(y) = u'(f'(x)) = f(u(x))$, hence $u'(y) \in \text{im}(f)$. Similarly for $v'(\text{im}(f)) \subseteq \text{im}(f'')$.

Define now $d: \ker(f'') \rightarrow \text{coker}(f')$ as follows. Let $x \in \ker(f'') \subseteq M''$. Since v is surjective, there exists $t \in M$ such that $v(t) = x$. Hence, $0 = f''(x) = f''(v(t)) = v'(f(t))$, and thus $f(t) \in \ker(v') = \text{im}(u')$, i.e., there exists $y \in N'$ such that

$u'(y) = f(t)$ and we can define $d(x) := [y] = y + \text{im}(f')$. For the time being, it seems that our definition of d depends on two choices: the choice of $t \in M$ such that $v(t) = x$ and the choice of $y \in N'$ such that $u'(y) = f(t)$. Hence, let $t' \in M$ such that $v(t') = x = v(t)$ and let $y' \in N'$ such that $u'(y') = f(t')$. We have to show that $y + \text{im}(f') = y' + \text{im}(f')$, i.e., $y - y' \in \text{im}(f')$. Observe that by definition of t and t' we have $t - t' \in \ker(v) = \text{im}(u)$, i.e., there exists $z \in M'$ such that $u(z) = t - t'$. On the other hand, by definition of y and y' we have

$$u'(y - y') = f(t - t') = f(u(z)) = u'(f'(z)),$$

hence

$$y - y' - f'(z) \in \ker(u') = 0 \implies y - y' = f'(z).$$

Finally, $y - y' \in \text{im}(f')$ and this shows that the map d is well-defined. It is now easy to check that d is an A -module homomorphism and this is left to the reader. As for the exactness of (2.5), we observe for example that

$$\text{im}(\bar{u}) = \text{im}(u)|_{\ker(f)} = \ker(v)|_{\ker(f)} = \ker(\bar{v}),$$

where the first equality holds because $f \circ u = u' \circ f'$ and u' is injective, the second equality is true because $\text{im}(u) = \ker(v)$ and the third equality is immediate since $\bar{v}$ is the restriction of v to $\ker(f)$. By diagram chasing the reader can fully check the exactness of (2.5). $\square$

Definition 2.34. Let $\mathcal{C}$ be a class of A -modules. A function $\lambda: \mathcal{C} \rightarrow \mathbb{Z}$ is called *additive* if for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ it holds that $\lambda(M') - \lambda(M) + \lambda(M'') = 0$.

Remark 2.35. Let $\mathcal{C}$ be a class of A -modules such that $M = 0 \in \mathcal{C}$. Let $\lambda: \mathcal{C} \rightarrow \mathbb{Z}$ be an additive function. Since $0 \rightarrow M \rightarrow M \rightarrow M \rightarrow 0$ is a short exact sequence, we have by additivity $\lambda(M) = \lambda(M) + \lambda(M)$, hence $\lambda(M) = 0$.

Example 2.36. Let $A = k$ be a field and $\mathcal{C}$ be the class of finite-dimensional k -vector spaces. Then $\lambda: \mathcal{C} \rightarrow \mathbb{Z}$, $V \rightarrow \dim_k V$, is additive.

Proposition 2.37. *Let*

$$0 \longrightarrow M_0 \longrightarrow M_1 \longrightarrow \cdots \longrightarrow M_n \longrightarrow 0$$

be an exact sequence of A -modules where all the modules M_i and the kernels of all homomorphisms belong to $\mathcal{C}$. Then for any additive function $\lambda: \mathcal{C} \rightarrow \mathbb{Z}$ on $\mathcal{C}$ we have

$$\sum_{i=0}^n (-1)^i \lambda(M_i) = 0.$$

Proof. Like in Remark 2.31, split the exact sequence into short exact sequences $0 \rightarrow N_i \rightarrow M_i \rightarrow N_{i+1} \rightarrow 0$, where $N_0 = N_{n+1} = 0$. Then for each $0 \leq i \leq n$ we have $\lambda(M_i) = \lambda(N_i) + \lambda(N_{i+1})$, namely:

$$\begin{aligned}\lambda(M_0) &= \lambda(N_0) + \lambda(N_1) \\ \lambda(M_1) &= \lambda(N_1) + \lambda(N_2) \\ &\dots \\ \lambda(M_{n-1}) &= \lambda(N_{n-1}) + \lambda(N_n) \\ \lambda(M_n) &= \lambda(N_n) + \lambda(N_{n+1})\end{aligned}$$

We sum these $n+1$ equality with alternating signs, and since $\lambda(N_0) = \lambda(N_{n+1}) = 0$, on the right-hand side all terms cancel pairwise, yielding

$$\sum_{i=0}^n (-1)^i \lambda(M_i) = 0.$$

□

2.6 Tensor Products of Modules

Definition 2.38. Let A be a ring and let M, N, P be A -modules. A map $f: M \times N \rightarrow P$ is called *A-bilinear* if for each $x \in M$, the map $N \rightarrow P$, $y \mapsto f(x, y)$, is A -linear, and for each $y \in N$ the map $M \rightarrow P$, $x \mapsto f(x, y)$, is A -linear.

Given A -modules M, N , we construct an A -module T , called the *tensor product of M and N* , characterized by the property that, for any A -module P , there is a natural one-to-one correspondence between A -bilinear maps $M \times N \rightarrow P$ and A -linear maps $T \rightarrow P$. The following proposition makes this precise.

Proposition 2.39. *Let A be a ring and let M, N be A -modules. Then there exists a pair (T, g) , where T is an A -module and $g: M \times N \rightarrow T$ is an A -bilinear map with the following property: For any A -module P and any A -bilinear map $f: M \times N \rightarrow P$, there exists a unique A -linear map $f': T \rightarrow P$ such that $f = f' \circ g$, i.e., the diagram*

$$\begin{array}{ccc} & & T \\ & \nearrow g & \downarrow f' \\ M \times N & \xrightarrow{f} & P \end{array} \quad (2.6)$$

commutes. Moreover, if (T, g) and (T', g') are two pairs with the above property, then there is a unique isomorphism $h: T \rightarrow T'$ such that $h \circ g = g'$.

Proof. (i) *Uniqueness:* Assume we have two pairs (T, g) and (T', g') with this property. Applying the property of (T, g) with $P = T'$ and $g': M \times N \rightarrow T'$,

and similarly applying the property of (T', g') with $P = T$ and $g: M \times N \rightarrow T$, we obtain the following commutative diagrams

$$\begin{array}{ccc} & T & \\ g \nearrow & \downarrow h & \\ M \times N & \xrightarrow{g'} & T' \end{array} \quad \begin{array}{ccc} & T' & \\ g' \nearrow & \downarrow h' & \\ M \times N & \xrightarrow{g} & T \end{array}$$

By combining the two diagrams above, we have the following commutative diagram

$$\begin{array}{ccc} & T & \\ g \nearrow & \downarrow h & \\ M \times N & \xrightarrow{g'} & T' \\ g \searrow & \downarrow h' & \\ & T & \end{array} \quad \text{id}_T$$

and applying the property of (T, g) with $P = T$ and $g: M \times N \rightarrow T$, we obtain $h' \circ h = \text{id}_T$. Similarly we show that $h \circ h' = \text{id}_{T'}$.

- (ii) *Existence:* Let C be the free A -module $A^{M \times N}$, whose elements are finite formal linear combinations of elements of $M \times N$ with coefficients in A , i.e., they are expressions of the form $\sum_i a_i(x_i, y_i)$.

Let D be the submodule of C generated by the elements of C of the form

- $(x + x', y) - (x, y) - (x', y),$
- $(x, y + y') - (x, y) - (x, y'),$
- $(ax, y) - a(x, y),$
- $(x, ay) - a(x, y),$

where $x, x' \in M$, $y, y' \in N$, $a \in A$. Let $T = C/D$. For each canonical generator (x, y) of C denote by $x \otimes y$ its image in T . Then T is generated by elements of the form $x \otimes y$, and from the definitions we have for $x \in M$, $y \in N$ and $a \in A$:

$$\begin{aligned} (x + x') \otimes y &= x \otimes y + x' \otimes y, & x \otimes (y + y') &= x \otimes y + x \otimes y', \\ (ax) \otimes y &= a(x \otimes y) = x \otimes (ay). \end{aligned}$$

Hence, the map $g: M \times N \rightarrow T$, $(x, y) \mapsto x \otimes y$ is A -bilinear. Let P be an A -module and let $f: M \times N \rightarrow P$ be an A -bilinear map. Then

$$\bar{f}: C \rightarrow P, \quad \sum_{i=1}^k a_i(x_i, y_i) \mapsto \sum_{i=1}^k a_i f(x_i, y_i)$$

is an A -linear map that vanishes on all generators of D , hence $D \subseteq \ker(\bar{f})$. Thus, $\bar{f}$ induces a well-defined A -linear map $f': T \rightarrow P$ such that $f'(x \otimes y) = f(x, y)$. Since the elements of the form $x \otimes y$ generate T , the condition $x \otimes y \mapsto f(x, y)$ uniquely determines f' . Thus, f' is the unique A -linear map such that $f' \circ g = f$. In other words, the pair (T, g) satisfies the required property. $\square$

Definition 2.40. Let A be a ring and let M, N be A -modules. A pair (T, g) fulfilling the property of Proposition 2.39 is called *the tensor product of M and N as A -modules* and is denoted by $(M \otimes_A N, \otimes)$. When the ring A is clear from the context, we just write $M \otimes N$ instead of $M \otimes_A N$.

Remark 2.41. Let M, N be A -modules.

- (a) It is clear from the construction that $M \otimes_A N$ is generated as an A -module by the elements of the form $x \otimes y$, where $x \in M$ and $y \in N$. These elements are called *pure tensors*. If $(x_i)_{i \in I}$ respectively $(y_j)_{j \in J}$ are families of generators of M respectively N , then the set $\{x_i \otimes y_j \mid i \in I, j \in J\}$ generates $M \otimes_A N$. In particular, if M and N are finitely generated A -modules, then so is $M \otimes N$.
- (b) Every element in $M \otimes_A N$ can be written as a finite sum of pure tensors. Indeed, let $z \in M \otimes_A N$. Then

$$z = \sum_{i=1}^n a_i(x_i \otimes y_i) = \sum_{i=1}^n (a_i x_i) \otimes y_i = \sum_{i=1}^n x_i \otimes (a_i y_i).$$

- (c) The notation $x \otimes y$ is inherently ambiguous unless we specify the tensor product to which it belongs. Indeed, let M' respectively N' be submodules of M respectively N , and let $x \in M'$, $y \in N'$. It can happen that $x \otimes y$ is zero as an element of $M \otimes N$, while $x \otimes y$ is nonzero as an element of $M' \otimes N'$. For example, take $A = \mathbb{Z}$, $M = \mathbb{Z}/4\mathbb{Z}$, $M' = 2\mathbb{Z}/4\mathbb{Z}$, $N = N' = \mathbb{Z}/2\mathbb{Z}$, $x = \bar{2}$ and $y = \bar{1}$. Then $x \otimes y = 0$ as an element of $M \otimes N = (\mathbb{Z}/4\mathbb{Z}) \otimes \mathbb{Z}/2\mathbb{Z}$ since

$$x \otimes y = \bar{2} \otimes \bar{1} = 2(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes \bar{0} = 0.$$

On the other hand, $M' \otimes N' = (2\mathbb{Z}/4\mathbb{Z}) \otimes \mathbb{Z}/2\mathbb{Z}$ is generated by $x \otimes y = \bar{2} \otimes \bar{1}$. Moreover, $M' \otimes N'$ is a nontrivial $\mathbb{Z}$ -module since

$$(2\mathbb{Z}/4\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}, \quad (\bar{a}, \bar{b}) \mapsto \overline{a/2} \cdot \bar{b},$$

is a nonzero $\mathbb{Z}$ -bilinear map. Therefore, it must be $x \otimes y \neq 0$.

Proposition 2.42. Let M and N be A -modules. Let $x_i \in M$ and $y_i \in N$ be elements such that $\sum_i x_i \otimes y_i = 0$ as an element of $M \otimes N$. Then there exist finitely generated submodules $M_0 \subseteq M$ and $N_0 \subseteq N$ such that $\sum_i x_i \otimes y_i = 0$ as an element of $M_0 \otimes N_0$.

Proof. If $\sum_i x_i \otimes y_i = 0$ in $M \otimes N$, then $\sum_i (x_i, y_i) \in D$. Therefore, we can write

$$\sum_i (x_i, y_i) = \sum_j a_j z_j,$$

where $a_j \in A$ and each $z_j \in D$ is one of the (maybe infinitely many!) generators of D . Let M_0 be the submodule of M generated by the x_i 's and all the elements of M that appear as first coordinates in each of the z_j . Similarly, let N_0 be the submodule of N generated by the y_i 's and all the elements of N that appear as second coordinates in each of the z_j . Then, by construction of $M_0 \otimes N_0$, it is clear that $\sum_i x_i \otimes y_i = 0$ as an element of $M_0 \otimes N_0$. $\square$

One can also define the tensor product of finitely many A -modules.

Definition 2.43. Let A be a ring. Let $M_1, \dots, M_n, P$ be A -modules. A map $f: M_1 \times \dots \times M_n \rightarrow P$ is called *n -linear over A* if for each $i \in \{1, \dots, n\}$ and for all fixed $\bar{x}_j \in M_j$ with $j \neq i$, the map $M_i \rightarrow P$, $x_i \mapsto f(\bar{x}_1, \dots, x_i, \dots, \bar{x}_n)$, is A -linear.

Proposition 2.44. Let $M_1, \dots, M_n$ be A -modules. Then there exists a pair (T, g) , consisting of an A -module T and an n -linear map $g: \prod_{i=1}^n M_i \rightarrow T$, with the following property: For any A -module P and any n -linear map $f: \prod_{i=1}^n M_i \rightarrow P$, there exists a unique linear map $f': T \rightarrow P$ such that $f = f' \circ g$. Moreover, if (T, g) and (T', g') are two pairs with this property, then there is a unique isomorphism $h: T \rightarrow T'$ such that $h \circ g = g'$.

Definition 2.45. Let A be a ring. Let $M_1, \dots, M_n$ be A -modules. A pair (T, g) fulfilling the property of Proposition 2.44 is called *the tensor product of $M_1, \dots, M_n$ as A -modules* and denoted by $(M_1 \otimes_A \dots \otimes_A M_n, \otimes)$. When the ring A is clear from the context, we just write $M_1 \otimes \dots \otimes M_n$ instead of $M_1 \otimes_A \dots \otimes_A M_n$.

Proposition 2.46 (Canonical Isomorphisms). Let M, N, P be A -modules. Then there exist unique isomorphisms

$$(i) \quad M \otimes N \rightarrow N \otimes M,$$

$$(ii) \quad (M \otimes N) \otimes P \rightarrow M \otimes (N \otimes P) \rightarrow M \otimes N \otimes P,$$

$$(iii) \quad (M \oplus N) \otimes P \rightarrow (M \otimes P) \oplus (N \otimes P),$$

$$(iv) \quad A \otimes M \rightarrow M,$$

such that, respectively,

$$(i) \quad x \otimes y \mapsto y \otimes x,$$

$$(ii) \quad (x \otimes y) \otimes z \mapsto x \otimes (y \otimes z) \mapsto x \otimes y \otimes z,$$

$$(iii) (x, y) \otimes z \longmapsto (x \otimes z, y \otimes z),$$

$$(iv) a \otimes x \longmapsto ax.$$

Proof. We partially show (ii), the rest of the assertions are left to the reader as an exercise. We construct A -linear maps $f: (M \otimes N) \otimes P \rightarrow M \otimes N \otimes P$ and $g: M \otimes N \otimes P \rightarrow (M \otimes N) \otimes P$ such that

$$f((x \otimes y) \otimes z) = x \otimes y \otimes z, \quad g(x \otimes y \otimes z) = (x \otimes y) \otimes z.$$

To construct f , we fix $z \in P$. The map

$$M \times N \longrightarrow M \otimes N \otimes P, \quad (x, y) \longmapsto x \otimes y \otimes z$$

is A -bilinear, so it induces an A -linear map $f_z: M \otimes N \rightarrow M \otimes N \otimes P$ such that $f_z(x \otimes y) = x \otimes y \otimes z$. Next, we consider the map

$$F: (M \otimes N) \times P \longrightarrow M \otimes N \otimes P, \quad (t, z) \longmapsto f_z(t).$$

This map is A -linear in t since f_z is so; we show that F is also A -linear in z . Namely, we have to show that for $a, b \in A$ and $z, z' \in P$ it holds

$$f_{az+bz'}(t) = af_z(t) + bf_{z'}(t) \quad \text{for all } t \in M \otimes N.$$

It is enough to show the previous equality for $t = x \otimes y$ since if two A -linear maps agree on a set of generators, they are equal. Now, we have

$$f_{az+bz'}(x \otimes y) = x \otimes y \otimes (az + bz') = a(x \otimes y \otimes z) + b(x \otimes y \otimes z') = af_z(x \otimes y) + bf_{z'}(x \otimes y),$$

and this shows that F is A -linear in z , hence A -bilinear. Thus, F induces an A -linear map

$$f: (M \otimes N) \otimes P \longrightarrow M \otimes N \otimes P, \quad (x \otimes y) \otimes z \longmapsto x \otimes y \otimes z.$$

To construct g , consider the map

$$M \times N \times P \longrightarrow (M \otimes N) \otimes P, \quad (x, y, z) \longmapsto (x \otimes y) \otimes z.$$

This map is A -linear in each variable, so it induces an A -linear map

$$g: M \otimes N \otimes P \longrightarrow (M \otimes N) \otimes P, \quad x \otimes y \otimes z \longmapsto (x \otimes y) \otimes z.$$

The A -linear maps $f \circ g$ and $\text{id}_{M \otimes N \otimes P}$ agree on a set of generators, hence $f \circ g = \text{id}_{M \otimes N \otimes P}$. Similarly, $g \circ f = \text{id}_{(M \otimes N) \otimes P}$, and thus f and g are isomorphisms. The uniqueness of f (respectively g) is clear since the condition $(x \otimes y) \otimes z \mapsto x \otimes y \otimes z$ (respectively $x \otimes y \otimes z \mapsto (x \otimes y) \otimes z$) uniquely determines f (respectively g). $\square$

Definition 2.47. Let $f: M \rightarrow M'$ and $g: N \rightarrow N'$ be two A -module homomorphisms. We define

$$h: M \times N \rightarrow M' \otimes N', \quad (x, y) \mapsto f(x) \otimes g(y).$$

One checks easily that h is A -bilinear, thus it factors through an A -module homomorphism

$$h': M \otimes N \longrightarrow M' \otimes N'.$$

We call this map *the tensor product of f and g* , and denote it by $f \otimes g$. Note that

$$(f \otimes g)(x \otimes y) = f(x) \otimes g(y).$$

2.7 Algebras, Restriction and Extension of Scalars

Definition 2.48. Let A, B be rings and let $f: A \rightarrow B$ be a ring homomorphism. Then B has a natural A -module structure as follows:

$$a \cdot b := f(a)b \quad \text{for all } a \in A, b \in B.$$

Hence, B has both a ring structure and an A -module structure which are compatible, i.e.,

$$a \cdot (bc) = (a \cdot b)c = b(a \cdot c).$$

The ring B together with the ring homomorphism f is called an *A -algebra*.

Definition 2.49. Let B and C be A -algebras with ring homomorphisms $f: A \rightarrow B$ and $g: A \rightarrow C$, respectively. A ring homomorphism $h: B \rightarrow C$ is called an *A -algebra homomorphism* if h is also A -linear. Equivalently, a ring homomorphism $h: B \rightarrow C$ is called an *A -algebra homomorphism* if $h \circ f = g$, i.e., if we have a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{h} & C \\ & \swarrow f \quad \searrow g & \\ & A & \end{array}$$

Definition 2.50. (a) A ring homomorphism $f: A \rightarrow B$ is called *finite* and B is called a *finite A -algebra* if B is finitely generated as an A -module.

(b) A ring homomorphism $f: A \rightarrow B$ is called *of finite type* and B is called a *finitely generated A -algebra* if there exist $b_1, \dots, b_n \in B$ such that every element in B is a polynomial expression in $b_1, \dots, b_n$ with coefficients in A . In other words, $B = A[b_1, \dots, b_n]$, where

$$A[b_1, \dots, b_n] := \left\{ g(b_1, \dots, b_n) \mid g \in A[X_1, \dots, X_n] \right\} \subseteq B$$

is in general a subring of B . Equivalently, B is called a finitely generated A -algebra if there exists a surjective ring homomorphism $A[X_1, \dots, X_n] \twoheadrightarrow B$.

Example 2.51. Let K be a field. The polynomial ring $K[X_1, \dots, X_n]$ is a finitely generated K -algebra, but it is not finite over K . Indeed, it admits the following direct sum decomposition as a K -vector space

$$K[X_1, \dots, X_n] = \bigoplus_{d \geq 0} K[X_1, \dots, X_n]_d,$$

where $K[X_1, \dots, X_n]_d$ denotes the K -vector space of homogeneous polynomials of degree d .

Definition 2.52 (Restriction of Scalars). Let B be an A -algebra with ring homomorphism $f: A \rightarrow B$. Let N be a B -module. Then there is an induced A -module structure on N : for all $a \in A$ and for all $x \in N$ we define

$$a \cdot x := f(a) \cdot x.$$

The A -module structure on N is called *the restriction of scalars of N to A* .

Let B be an A -algebra with ring homomorphism $f: A \rightarrow B$. If N is a finitely generated B -module, it is natural to ask if it is also finitely generated as an A -module. This holds true if B is a finite A -algebra.

Proposition 2.53. *Let B be an A -algebra with ring homomorphism $f: A \rightarrow B$ and let N be a finitely generated B -module. Assume that B is a finite A -algebra. Then N is finitely generated as an A -module.*

Proof. Let $\{x_1, \dots, x_n\} \subseteq N$ be a generating set of the B -module N and let $\{y_1, \dots, y_m\}$ be a generating set for the A -module B . Then the set

$$\{y_i x_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

is a generating set of the A -module N . □

Definition 2.54 (Extension of Scalars). Let B be an A -algebra with ring homomorphism $f: A \rightarrow B$. Let M be an A -module. The tensor product $M_B := M \otimes_A B$ has a natural structure as a B -module: For all $b \in B$, $\beta \in B$ and $m \in M$ we set

$$b \cdot (m \otimes \beta) := m \otimes (b\beta).$$

The B -module M_B is called *the extension of scalars of M to B* .

Definition 2.55. Let A, B be rings. An (A, B) -module N is an abelian group which is both an A -module and a B -module and the two structures are compatible, i.e., for all $a \in A$, $b \in B$ and $x \in N$ it holds

$$a \cdot (b \cdot x) = b \cdot (a \cdot x).$$

A map $f: N \rightarrow N'$ between (A, B) -modules is an (A, B) -module homomorphism if it is A -linear as well as B -linear.

Remark 2.56. Let A, B be rings. Let M be an A -module, let N be an (A, B) -module and let P be a B -module. Then $M \otimes_A N$ and $N \otimes_B P$ have a natural structure as (A, B) -modules.

Remark 2.57. Let B be an A -algebra.

- (a) Let N be a B -module. Then N together with its restriction of scalars to A is clearly an (A, B) -module. In particular, B is itself an (A, B) -module.
- (b) Let M be an A -module. Then the extension of scalars $M_B = M \otimes_A B$ is an (A, B) -module by Remark 2.56 since B is an (A, B) -module.

The following proposition gives an important and useful canonical isomorphism.

Proposition 2.58. *Let A, B be rings. Let M be an A -module, let N be an (A, B) -module and let P be a B -module. Then there exists a unique (A, B) -module isomorphism*

$$(M \otimes_A N) \otimes_B P \xrightarrow{\sim} M \otimes_A (N \otimes_B P)$$

such that $(x \otimes y) \otimes z \mapsto x \otimes (y \otimes z)$.

Proof. Exercise. □

Corollary 2.59. *Let B be an A -algebra. Let M be an A -module and let N, P be B -modules. Then there exists a unique B -linear isomorphism*

$$(M \otimes_A N) \otimes_B P \xrightarrow{\sim} M \otimes_A (N \otimes_B P)$$

such that $(x \otimes y) \otimes z \mapsto x \otimes (y \otimes z)$.

2.8 Tensor Product of Algebras

Let B and C be two A -algebras with ring homomorphisms $f: A \rightarrow B$ and $g: A \rightarrow C$, respectively. Consider the tensor product

$$D := B \otimes_A C.$$

We would like to endow D with an A -algebra structure. For this purpose, we need to define a product $\cdot : D \times D \rightarrow D$ which is A -bilinear. The natural definition would be

$$\cdot : D \times D \rightarrow D, \quad (b \otimes c, b' \otimes c') \mapsto bb' \otimes cc'.$$

Indeed, this is the correct definition, provided that it is well-defined! Hence, we need an argument to show it. Consider the map

$$B \times C \times B \times C \rightarrow D, \quad (b, b', c, c') \mapsto bb' \otimes cc'.$$

This is a 4-linear map over A , hence, by the universal property of the tensor product (Proposition 2.44), it induces a unique A -linear map

$$B \otimes_A C \otimes_A B \otimes_A C \rightarrow D,$$

Now, by the canonical isomorphism (ii) from Proposition 2.46, we have an A -linear map

$$D \otimes D \rightarrow D, \quad (b \otimes c) \otimes (b' \otimes c') \mapsto bb' \otimes cc'.$$

By the universal property of the tensor product (Proposition 2.39), such an A -linear map induces a unique A -bilinear map

$$\cdot : D \times D \rightarrow D, \quad (b \otimes c, b' \otimes c') \mapsto bb' \otimes cc'.$$

Hence, we have a well-defined product on D :

$$\left(\sum_i b_i \otimes c_i \right) \cdot \left(\sum_j b_j \otimes c_j \right) := \sum_{i,j} b_i b'_j \otimes c_i c'_j.$$

It is easy to see that $(D, +, \cdot)$ is a ring: the associativity (resp. the commutativity) of $\cdot$ is a consequence of the associativity (resp. the commutativity) of the products in B and C , the distributivity laws hold by A -bilinearity, and $1_D = 1_B \otimes 1_C$. Moreover, for all $a \in A$, $b \in B$ and $c \in C$ it holds

$$\begin{aligned} a \cdot (b \otimes c) &= (a \cdot b) \otimes c = f(a) \otimes c \\ a \cdot (b \otimes c) &= b \otimes (a \cdot c) = b \otimes g(a)c \end{aligned}$$

In particular, if $b = 1_B$ and $c = 1_C$, we have

$$f(a) \otimes 1_C = 1_B \otimes g(a).$$

Therefore, D is an A -algebra with ring homomorphism

$$A \rightarrow D, \quad a \mapsto f(a) \otimes 1_C = 1_B \otimes g(a),$$

and we have the following commutative diagram (of ring homomorphisms)

$$\begin{array}{ccc} & B & \\ f \nearrow & & \searrow u \\ A & & D \\ g \searrow & & \nearrow v \\ & C & \end{array}$$

where $u : B \rightarrow D$, $b \mapsto b \otimes 1_C$ and $v : C \rightarrow D$, $c \mapsto 1_B \otimes c$.

2.9 Exactness of Tensor Products

Let $M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3$ be an exact sequence of A -modules and let N be an A -module. Is the sequence $M_1 \otimes N \xrightarrow{f \otimes \text{id}_N} M_2 \otimes N \xrightarrow{g \otimes \text{id}_N} M_3 \otimes N$ exact? The answer is in general no, as shown by the following example.

Example 2.60. Let $A = \mathbb{Z}$ and let $N = \mathbb{Z}/2\mathbb{Z}$. Consider the exact sequence

$$\begin{aligned} 0 &\longrightarrow \mathbb{Z} \xrightarrow{f} \mathbb{Z} \\ x &\longmapsto 2x. \end{aligned}$$

Then by tensoring with N , we have the sequence

$$0 \longrightarrow \mathbb{Z} \otimes N \xrightarrow{f \otimes \text{id}_N} \mathbb{Z} \otimes N.$$

Note that $(f \otimes \text{id}_N)(x \otimes y) = f(x) \otimes y = (2x) \otimes y = x \otimes (2y) = x \otimes 0 = 0$, thus $f \otimes \text{id}_N$ is far from being injective and the sequence is not exact.

However, we can show that “tensoring is right exact”. To do so, we need the following lemma.

Lemma 2.61. *Let M, N, P be A -modules. Then we have a canonical isomorphism*

$$\text{Hom}_A(M \otimes_A N, P) \cong \text{Hom}_A(M, \text{Hom}_A(N, P)).$$

Proof. Let $f: M \otimes_A N \rightarrow P$ be an A -linear map. By the universal property of the tensor product (Proposition 2.39), there exists a unique A -bilinear map $f': M \times N \rightarrow P$ such that $f(m \otimes n) = f'(m, n)$. Now, we construct an A -linear map

$$\overline{f'}: M \longrightarrow \text{Hom}_A(N, P), \quad m \longmapsto (n \mapsto f'(m, n)).$$

On the other hand, if $g: M \rightarrow \text{Hom}_A(N, P)$ is an A -linear map, we define the A -bilinear map $\tilde{g}: M \times N \rightarrow P$, $(m, n) \mapsto [g(m)](n)$, which induces by the universal property of the tensor product a unique A -linear map

$$\hat{g}: M \otimes N \longrightarrow P, \quad m \otimes n \longmapsto [g(m)](n),$$

These two constructions are inverse to each other, yielding a one-to-one correspondence

$$\text{Hom}_A(M \otimes_A N, P) \rightarrow \text{Hom}_A(M, \text{Hom}_A(N, P)), \quad f \longmapsto \overline{f'},$$

which is indeed A -linear. □

Proposition 2.62 (“Tensoring is Right Exact”). *Let*

$$M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of A -modules. Furthermore, let N be an A -module. Then the sequence

$$M_1 \otimes N \longrightarrow M_2 \otimes N \longrightarrow M_3 \otimes N \longrightarrow 0$$

is exact.

Proof. Let P be an A -module. Then the sequence

$$0 \rightarrow \operatorname{Hom}(M_3, \operatorname{Hom}(N, P)) \rightarrow \operatorname{Hom}(M_2, \operatorname{Hom}(N, P)) \rightarrow \operatorname{Hom}(M_1, \operatorname{Hom}(N, P))$$

is exact by Proposition 2.32(i). Thus, by the previous lemma, we have an exact sequence

$$0 \longrightarrow \operatorname{Hom}(M_3 \otimes N, P) \longrightarrow \operatorname{Hom}(M_2 \otimes N, P) \longrightarrow \operatorname{Hom}(M_1 \otimes N, P).$$

Hence, since P is an arbitrary A -module, by Proposition 2.32(ii) the sequence

$$M_1 \otimes N \longrightarrow M_2 \otimes N \longrightarrow M_3 \otimes N \longrightarrow 0$$

is exact. □

Definition 2.63. An A -module N is called *flat* if tensoring with N is also left exact, i.e., for any short exact sequences $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ of A -modules, the sequence

$$0 \longrightarrow M_1 \otimes N \longrightarrow M_2 \otimes N \longrightarrow M_3 \otimes N \longrightarrow 0$$

is exact.

Proposition 2.64. *Let N be an A -module. Then the following are equivalent:*

- (i) N is flat.
- (ii) For any exact sequence $M_1 \rightarrow M_2 \rightarrow M_3$ of A -modules, the sequence $M_1 \otimes N \rightarrow M_2 \otimes N \rightarrow M_3 \otimes N$ is exact.
- (iii) If $f: M \rightarrow M'$ is an injective A -linear map, then $f \otimes \operatorname{id}_N: M \otimes N \rightarrow M' \otimes N$ is injective.
- (iv) If $f: M \rightarrow M'$ is an injective A -linear map where M and M' are finitely generated, then $f \otimes \operatorname{id}_N: M \otimes N \rightarrow M' \otimes N$ is injective.

Proof. (ii) $\Rightarrow$ (i). Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of A -modules. Split it into the three exact sequences

$$0 \rightarrow M_1 \rightarrow M_2, \quad M_1 \rightarrow M_2 \rightarrow M_3, \quad M_2 \rightarrow M_3 \rightarrow 0,$$

and then apply (ii) three times.

(i) $\Rightarrow$ (ii). Let $M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3$ be an exact sequence. Then the sequences

$$\begin{aligned} 0 &\rightarrow \ker(f) \rightarrow M_1 \rightarrow \operatorname{im}(f) \rightarrow 0, \\ 0 &\rightarrow \operatorname{im}(f) \rightarrow M_2 \rightarrow \operatorname{im}(g) \rightarrow 0 \\ 0 &\rightarrow \operatorname{im}(g) \rightarrow M_2 \rightarrow \operatorname{coker}(g) \rightarrow 0 \end{aligned}$$

Now apply (i) three times.

(i) $\Rightarrow$ (iii). Let $f: M \rightarrow M'$ be an injective A -linear map. Consider the short exact sequence $0 \rightarrow M \xrightarrow{f} M' \rightarrow M'/\text{im}(f) \rightarrow 0$. Since N is flat, the sequence

$$0 \rightarrow M \otimes N \xrightarrow{f \otimes \text{id}_N} M' \otimes N \rightarrow (M/M') \otimes N \rightarrow 0$$

is exact. In particular, $f \otimes \text{id}_N$ is injective.

(iii) $\Rightarrow$ (i). Let $0 \rightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \rightarrow 0$ be a short exact sequence. By Proposition 2.62 the sequence

$$M_1 \otimes N \xrightarrow{f \otimes \text{id}_N} M_2 \otimes N \xrightarrow{g \otimes \text{id}_N} M_3 \otimes N \rightarrow 0$$

is exact. By (iii) the map $f \otimes \text{id}_N$ is injective, hence the sequence

$$0 \rightarrow M_1 \otimes N \xrightarrow{f \otimes \text{id}_N} M_2 \otimes N \xrightarrow{g \otimes \text{id}_N} M_3 \otimes N \rightarrow 0$$

is exact.

(iii) $\Rightarrow$ (iv). This is obvious.

(iv) $\Rightarrow$ (iii). Let $f: M \rightarrow M'$ be an injective A -linear map. We have to show that $f \otimes \text{id}_N: M \otimes N \rightarrow M' \otimes N$ is also injective. Let $u = \sum_i x_i \otimes y_i \in \ker(f \otimes \text{id}_N)$, where $x_i \in M$ and $y_i \in N$. This means

$$(f \otimes \text{id}_N)(u) = \sum_i f(x_i) \otimes y_i = 0 \in M' \otimes N.$$

Let M_0 be the submodule of M generated by the elements x_i 's. Note that $f(M_0)$ is generated by the elements $f(x_i)$'s. By the same argument given in the proof of Proposition 2.42, there exists a finitely generated A -module $M'_0 \subseteq M'$ such that

$$\sum_i f(x_i) \otimes y_i = 0 \in M'_0 \otimes N.$$

Since M'_0 contains by construction the elements $f(x_i)$'s, we have

$$f(M_0) \subseteq M'_0.$$

Let u_0 denote $\sum_i x_i \otimes y_i$ as an element of $M_0 \otimes N$. In other words, if $j: M_0 \rightarrow M$ is the inclusion map, we have

$$(j \otimes \text{id}_N)(u_0) = u. \tag{2.7}$$

Since M_0 and M'_0 are finitely generated and the map $f|_{M_0}: M_0 \rightarrow M'_0$ is injective, by (iv) the map

$$f|_{M_0} \otimes \text{id}_N: M_0 \otimes N \rightarrow M'_0 \otimes N$$

is injective. Hence, $(f|_{M_0} \otimes \text{id}_N)(u_0) = \sum_i f(x_i) \otimes y_i = 0$ implies $u_0 = 0$, which in turn implies $u = 0$ by (2.7). Finally, we have just shown that $\ker(f \otimes \text{id}_N) = 0$, i.e., $f \otimes \text{id}_N$ is injective. $\square$

Flatness is preserved under extension of scalars.

Proposition 2.65. *Let $f: A \rightarrow B$ be a ring homomorphism and let M be a flat A -module. Then the extension of scalars $M_B = M \otimes_A B$ is a flat B -module.*

Proof. Exercise. □

Chapter 3

Localizations

Definition 3.1. Let A be a ring. A *multiplicative set* $S \subseteq R$ is a subset satisfying the following properties:

- (i) $1_A \in S$
- (ii) $xy \in S$ if $x, y \in S$.

Let A be a ring and let $S \subseteq A$ be a multiplicative set. We define an equivalence relation on $A \times S$ as follows:

$$(a, s) \sim (a', s') \iff \exists u \in S : u(as' - a's) = 0.$$

The relation $\sim$ is clearly reflexive and symmetric. We show the transitivity. Assume that $(a, s) \sim (a', s')$ and $(a', s') \sim (a'', s'')$. We have to show that $(a, s) \sim (a'', s'')$. By assumption, there exist $u, v \in S$ such that

$$u(as' - a's) = 0, \quad v(a's'' - a''s') = 0.$$

In other words,

$$uas' = ua's, \quad va's'' = va''s'.$$

Multiplying these equalities with vs'' and us respectively, we obtain

$$uvs'as'' = uvs'a''s \iff \underbrace{uvs'}_{\in S}(as'' - a''s) = 0,$$

whence $(a, s) \sim (a'', s'')$.

The set of equivalence classes $(A \times S)/\sim$ is denoted by $S^{-1}A$. For $a \in A$ and $s \in S$, we denote the equivalence class $[(a, s)]_\sim$ by a/s . Moreover, we define on $S^{-1}A$ the following operations:

$$\begin{aligned} \frac{a}{s} + \frac{a'}{s'} &:= \frac{as' + a's}{ss'}, \\ \frac{a}{s} \cdot \frac{a'}{s'} &:= \frac{aa'}{ss'}. \end{aligned}$$

It is easy to check that the operations above are well-defined and that $(S^{-1}A, +, \cdot)$ is a ring with $1_{S^{-1}A} = 1/1$.

Definition 3.2. The ring $S^{-1}A$ is called *the ring of fractions of A with respect to S* (or *the localization of A along S*).

Example 3.3. Let A be a ring and let $S \subseteq A$ be a multiplicative set.

- (a) If $0 \in S$, then $S^{-1}A = \{0\}$.
- (b) If A is an integral domain and $S = A \setminus \{0\}$, then $S^{-1}A = Q(A)$ is the field of fractions of A .
- (c) Let $\mathfrak{p}$ be a prime ideal in A and let $S = A \setminus \mathfrak{p}$. Then $S^{-1}A$ is called *the localization of A at $\mathfrak{p}$* and denoted by $A_{\mathfrak{p}}$.
- (d) Let $f \in A$ and let $S = \{f^n\}_{n \in \mathbb{N}} = \{1, f, f^2, \dots\}$. Then $S^{-1}A$ is called *the localization of A at f* and denoted by A_f .

Let A be a ring and let $S \subseteq A$ be a multiplicative set. There is a canonical ring homomorphism

$$f: A \rightarrow S^{-1}A, \quad a \mapsto \frac{a}{1}.$$

This homomorphism is in general not injective. The pair $(S^{-1}A, f)$ has a universal property.

Proposition 3.4 (Universal Property of $S^{-1}A$). *Let A be a ring, let $S \subseteq A$ be a multiplicative set and let $f: A \rightarrow S^{-1}A$, $a \mapsto a/1$. Let $g: A \rightarrow B$ be a ring homomorphism such that $g(s)$ is a unit in B for all $s \in S$. Then there exists a unique ring homomorphism $h: S^{-1}A \rightarrow B$ such that $h \circ f = g$, i.e., the following diagram commutes*

$$\begin{array}{ccc} & & S^{-1}A \\ & \nearrow f & \downarrow h \\ A & \xrightarrow{g} & B \end{array}$$

Proof. Uniqueness. Assume that $h: S^{-1}A \rightarrow B$ is a ring homomorphism fulfilling the hypothesis. Then

$$h(a/s) = h(a/1 \cdot 1/s) = h(a/1)h((s/1)^{-1}) = g(a)g(s)^{-1}.$$

Hence, h is uniquely determined.

Existence. If we define the map h as above, we can easily see that it is well-defined. Assume $a/s = a'/s'$, i.e., $u(as' - a's) = 0$ for some $u \in S$. Then $g(u)[g(a')g(s) - g(a)g(s')] = 0$, and since $g(u)$ is invertible we have

$$g(a')g(s) - g(a)g(s') = 0 \quad \Longleftrightarrow \quad g(a')g(s')^{-1} = g(a)g(s)^{-1}.$$

□

The ring homomorphism $f: A \rightarrow S^{-1}A$ has the following properties:

- (i) $f(s)$ is a unit in $S^{-1}A$ for all $s \in S$,
- (ii) $f(a) = 0$ implies $as = 0$ for some $s \in S$,
- (iii) Every element in $S^{-1}A$ is of the form $f(a)f(s)^{-1}$ for some $a \in A$, $s \in S$.

Indeed, the three properties above characterize $S^{-1}A$.

Proposition 3.5. *Let A be a ring, let $S \subseteq A$ be a multiplicative set and let $f: A \rightarrow S^{-1}A$, $a \mapsto a/1$. Let $g: A \rightarrow B$ be a ring homomorphism such that*

- (i) $g(s)$ is a unit in B for all $s \in S$,
- (ii) $g(a) = 0$ implies $as = 0$ for some $s \in S$,
- (iii) Every element in B is of the form $g(a)g(s)^{-1}$ for some $a \in A$, $s \in S$.

Then there exists a unique ring isomorphism $h: S^{-1}A \rightarrow B$ such that $h \circ f = g$, i.e., the following diagram commutes

$$\begin{array}{ccc} & & S^{-1}A \\ & \nearrow f & \downarrow h \\ A & \xrightarrow{g} & B \end{array}$$

Proof. Since (i) holds, we already have the existence and uniqueness of

$$h: S^{-1}A \rightarrow B, \quad a/s \mapsto g(a)g(s)^{-1}$$

by Proposition 3.4. Property (iii) implies that h is surjective, so it remains to show that h is injective. Let $a/s \in \ker h$. Then $h(a/s) = g(a)g(s)^{-1} = 0$, which implies $g(a) = 0$. Hence, by (ii) we have $at = 0$ for some $t \in S$. But this means $a/1 = 0$, whence $a/s = 0$. Finally, we have just shown that $\ker h = 0$, i.e., h is injective. $\square$

3.1 Localization of a Module

Let A be a ring, let $S \subseteq A$ be a multiplicative set and let M be an A -module. We define on $M \times A$ the following equivalence relation:

$$(m, s) \sim (m', s') \iff \exists u \in S : u(s'm - sm') = 0.$$

As before, it is easy to check that this is a well-defined equivalence relation. We denote the equivalence class of a pair (m, s) by m/s . We also denote the set of equivalence classes $(M \times S)/\sim$ by $S^{-1}M$ and define on it the following operations:

$$\begin{aligned} \frac{m}{s} + \frac{m'}{s'} &:= \frac{s'm + sm'}{ss'}, \\ \frac{a}{s} \cdot \frac{m}{t} &:= \frac{am}{st}. \end{aligned}$$

It is easy to check that the operations above are well-defined and make $S^{-1}M$ into an $(S^{-1}A)$ -module.

Definition 3.6. The $(S^{-1}A)$ -module $S^{-1}M$ is called *the module of fractions of M with respect to S (or the localization of M along S)*.

Example 3.7. Let M be an A -module and let $S \subseteq A$ be a multiplicative set.

- (a) If $0 \in S$, then $S^{-1}M = \{0\}$.
- (b) Let $\mathfrak{p}$ be a prime ideal in A and let $S = A \setminus \mathfrak{p}$. Then $S^{-1}M$ is called *the localization of M at $\mathfrak{p}$* and denoted by $M_{\mathfrak{p}}$.
- (c) Let $f \in A$ and let $S = \{f^n\}_{n \in \mathbb{N}} = \{1, f, f^2, \dots\}$. Then $S^{-1}M$ is called *the localization of M at f* and denoted by M_f .

Definition 3.8. Let $u: M \rightarrow N$ be an A -module homomorphism and let $S \subseteq A$ be a multiplicative set. Then we have the induced $(S^{-1}A)$ -linear map

$$S^{-1}u: S^{-1}M \longrightarrow S^{-1}N, \quad m/s \longmapsto u(m)/s.$$

Remark 3.9. Let $u: M \rightarrow N$ and $v: N \rightarrow P$ be A -module homomorphisms and let $S \subseteq A$ be a multiplicative set. Then

$$S^{-1}(v \circ u) = S^{-1}v \circ S^{-1}u.$$

Proposition 3.10. Let A be a ring and let $S \subseteq A$ be a multiplicative set. Furthermore, let $M_1 \xrightarrow{u} M_2 \xrightarrow{v} M_3$ be an exact sequence of A -modules. Then the sequence

$$S^{-1}M_1 \xrightarrow{S^{-1}u} S^{-1}M_2 \xrightarrow{S^{-1}v} S^{-1}M_3$$

is also exact.

Proof. We have to show that $\text{im } S^{-1}u = \ker S^{-1}v$.

$\subseteq$. By assumption we know that $\text{im } u = \ker v$, whence $v \circ u = 0$. Thus, by Remark 3.9 we have that $S^{-1}v \circ S^{-1}u = S^{-1}(v \circ u) = S^{-1}(0) = 0$, which implies $\text{im } S^{-1}u \subseteq \ker S^{-1}v$.

$\supseteq$. Let $m_2/s \in \ker S^{-1}v$. By definition, $v(m_2)/s = 0$, i.e., there exists $t \in S$ such that $tv(m_2) = v(tm_2) = 0$. Since $\ker v = \text{im } u$, there exists $m_1 \in M_1$ such that $u(m_1) = tm_2$. Then $m_2/s = tm_2/st = u(m_1)/st = (S^{-1}u)(m_1/st)$. In other words, $m_2/s \in \text{im } S^{-1}u$. $\square$

As a result, formation of fraction preserves short exact sequences.

Corollary 3.11. Let A be a ring and let $S \subseteq A$ be a multiplicative set. Let $0 \rightarrow M_1 \xrightarrow{u} M_2 \xrightarrow{v} M_3 \rightarrow 0$ be a short exact sequence of A -modules. Then the sequence

$$0 \longrightarrow S^{-1}M_1 \xrightarrow{S^{-1}u} S^{-1}M_2 \xrightarrow{S^{-1}v} S^{-1}M_3 \longrightarrow 0$$

is also exact.

Proof. Split the sequence $0 \rightarrow M_1 \xrightarrow{u} M_2 \xrightarrow{v} M_3 \rightarrow 0$ into

$$0 \rightarrow M_1 \xrightarrow{u} M_2, \quad M_1 \xrightarrow{u} M_2 \xrightarrow{v} M_3, \quad M_2 \xrightarrow{v} M_3 \rightarrow 0$$

and apply Proposition 3.10 three times. $\square$

3.1.1 Submodules and Canonical Isomorphisms

Let M be an A -module, let $N \subseteq M$ be a submodule and let $S \subseteq A$ be a multiplicative set. Proposition 3.10 implies in particular that $S^{-1}N$ can be regarded as a submodule of $S^{-1}M$. Hence, it makes sense to study the behavior of formation of fractions with respect to operations on submodules. Indeed, we show that formation of fractions commutes with sums, finite intersections and quotients.

Proposition 3.12. *Let A be a ring and let $S \subseteq A$ be a multiplicative set. Let M be an A -module, let $N, T \subseteq M$ be submodules and let $\{N_i\}_{i \in I}$ be a family of submodules. Then it holds*

$$(i) \quad S^{-1}(N \cap T) = S^{-1}N \cap S^{-1}T.$$

$$(ii) \quad S^{-1}(\sum_{i \in I} N_i) = \sum_{i \in I} S^{-1}N_i.$$

$$(iii) \quad S^{-1}(M/N) \cong S^{-1}M/S^{-1}N.$$

Proof. (i) The inclusion $\subseteq$ is obvious. Let us show the inclusion $\supseteq$. Let $x \in S^{-1}N \cap S^{-1}T$. Then $x = n/s_1 = t/s_2$ with $n \in N$, $t \in T$ and $s_1, s_2 \in S$. By definition, there is $q \in S$ such that $q(s_1t - s_2n) = 0$. Then $y := qs_1t = qs_2n \in N \cap T$ and $x = n/s_1 = y/(s_1s_2q) \in S^{-1}(N \cap T)$.

(ii) $\subseteq$. Let $z = (x_{i_1} + \dots + x_{i_k})/s \in S^{-1}(\sum_{i \in I} N_i)$. Then $z = x_{i_1}/s + \dots + x_{i_k}/s \in \sum_{i \in I} S^{-1}N_i$.

$\supseteq$. Let $y = x_{i_1}/s_{i_1} + \dots + x_{i_k}/s_{i_k} \in \sum_{i \in I} S^{-1}N_i$. Then for $1 \leq j \leq k$ set $\hat{s}_{i_j} := \prod_{l \neq j} s_{i_l}$ and $s := \prod_{l=1}^k s_{i_l}$. Then

$$y = \frac{\hat{s}_{i_1}x_{i_1} + \dots + \hat{s}_{i_k}x_{i_k}}{s} \in S^{-1}(\sum_{i \in I} N_i).$$

(iii) Applying Corollary 3.11 to the exact sequence $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$, we obtain the short exact sequence

$$0 \longrightarrow S^{-1}N \longrightarrow S^{-1}M \longrightarrow S^{-1}(M/N) \longrightarrow 0.$$

Hence, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$. $\square$

Remark 3.13. Note that formation of fraction does not commute with infinite intersections of submodules. Indeed, let $A := \mathbb{Z}$, let $S := \mathbb{Z} \setminus \{0\}$ and for $n \in \mathbb{N}_{>0}$ let $M_n := n\mathbb{Z}$. It is easy to see that $S^{-1}A = S^{-1}M_n = \mathbb{Q}$ for all $n \in \mathbb{N}_{>0}$, while $\bigcap_{n \in \mathbb{N}_{>0}} M_n = 0$. Thus, we have

$$S^{-1}\left(\bigcap_{n \in \mathbb{N}_{>0}} M_n\right) = S^{-1}0 = 0 \neq \mathbb{Q} = \bigcap_{n \in \mathbb{N}_{>0}} S^{-1}M_n.$$

Proposition 3.14. *Let M be an A -module and let $S \subseteq A$ be a multiplicative set. Then there exists a unique $(S^{-1}A)$ -module homomorphism*

$$S^{-1}A \otimes_A M \rightarrow S^{-1}M$$

such that $a/s \otimes m \mapsto am/s$.

Proof. Consider the A -bilinear map

$$S^{-1}A \times M \rightarrow S^{-1}M, \quad (a/s, m) \mapsto am/s.$$

By the universal property of the tensor product, this map induces an A -linear map

$$f: S^{-1}A \otimes_A M \rightarrow S^{-1}M, \quad a/s \otimes m \mapsto am/s$$

which is surjective since $m/s = f(1/s \otimes m)$. To show that f is also injective, we show first that every element in $S^{-1}A \otimes_A M$ is of the form $1/s \otimes m$. To see this, let $x = \sum_i a_i/s_i \otimes m_i \in S^{-1}A \otimes_A M$. Then, if $s := \prod_i s_i$ and $\hat{s}_i := \prod_{j \neq i} s_j$, we have

$$\begin{aligned} x &= \sum_i a_i/s_i \otimes m_i = \sum_i 1/s_i \otimes a_i m_i = \sum_i \hat{s}_i/s \otimes a_i m_i = \\ &= \sum_i 1/s \otimes \hat{s}_i a_i m_i = 1/s \otimes \sum_i \hat{s}_i a_i m_i = 1/s \otimes m, \end{aligned}$$

where $m := \sum_i \hat{s}_i a_i m_i$. Now let $1/s \otimes m \in \ker f$. Then we have

$$\frac{0}{1} = f\left(\frac{1}{s} \otimes m\right) = \frac{m}{s} \implies \exists t \in S : tm = 0 \implies \frac{1}{s} \otimes m = \frac{t}{st} \otimes m = \frac{1}{st} \otimes tm = 0.$$

Hence, f is injective. Finally, it is easy to see that f is also $S^{-1}A$ -linear. $\square$

Corollary 3.15. *Let A be a ring and let $\{M_i\}_{i \in I}$ be a family of modules. Then*

$$S^{-1}\left(\bigoplus_{i \in I} M_i\right) \cong \bigoplus_{i \in I} S^{-1}M_i.$$

Proof. We have

$$S^{-1}\left(\bigoplus_{i \in I} M_i\right) \cong S^{-1}A \otimes_A \left(\bigoplus_{i \in I} M_i\right) \cong \bigoplus_{i \in I} \left(S^{-1}A \otimes_A M_i\right) \cong \bigoplus_{i \in I} S^{-1}M_i,$$

where the first and the last isomorphisms are given by Proposition 3.14, while the second is the canonical isomorphism given by (iii) of Proposition 2.46. $\square$

Corollary 3.16. *Let A be a ring and let $S \subseteq A$ be a multiplicative set. Then $S^{-1}A$ is a flat A -module.*

Proof. Let $0 \rightarrow M_1 \xrightarrow{f} M_2$ be an exact sequence of A -modules. Then, by tensoring with $S^{-1}A$ we get the following commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & M_1 \otimes_A S^{-1}A & \xrightarrow{f \otimes \text{id}} & M_2 \otimes_A S^{-1}A \\ & & \downarrow \simeq & & \downarrow \simeq \\ 0 & \longrightarrow & S^{-1}M_1 & \xrightarrow{S^{-1}f} & S^{-1}M_2 \end{array}$$

where the second row is exact by Proposition 3.10 and the vertical maps are the isomorphisms given by Proposition 3.14. Hence, we deduce that the first row is also exact, i.e., $f \otimes \text{id}: M_1 \otimes_A S^{-1}A \rightarrow M_2 \otimes_A S^{-1}A$ is injective. This shows that $S^{-1}A$ is a flat A -module. $\square$

Proposition 3.17. *Let A be a ring and let $S \subseteq A$ be a multiplicative set. Let M and N be two A -modules. Then there exists a unique $(S^{-1}A)$ -module homomorphism*

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \rightarrow S^{-1}(M \otimes_A N)$$

such that $m/s \otimes n/t \mapsto (m \otimes n)/st$.

Corollary 3.18. *Let A be a ring, let $P \subseteq A$ be a prime ideal and let M and N be two A -modules. Then*

$$(M \otimes_A N)_P \cong M_P \otimes_{A_P} N_P.$$

3.2 Local Properties of Rings and Modules

Definition 3.19. A property Φ of a ring A (resp. of an A -module M) is called *local* if it holds

A (resp. M) fulfills $\Phi \iff A_P$ (resp. M_P) fulfills Φ for all $P \subseteq A$ prime ideals.

Proposition 3.20. *Let A be a ring and let M be an A -module. Then the following are equivalent:*

- (i) $M = \{0\}$,
- (ii) $M_P = \{0\}$ for all prime ideals $P \subseteq A$,
- (iii) $M_{\mathfrak{m}} = \{0\}$ for all maximal ideals $\mathfrak{m} \subseteq A$.

Proof. The implications “(i) $\Rightarrow$ (ii)” and “(ii) $\Rightarrow$ (iii)” are obvious.

“(iii) $\Rightarrow$ (i)”: Assume by contradiction that $M \neq \{0\}$. Let $x \in M$ be any element and consider the ideal

$$\text{Ann}(x) := \{a \in A \mid ax = 0\}.$$

Now let $x \in M$ be a nonzero element. Hence, $\text{Ann}(x)$ is a proper ideal of A , and thus there exists a maximal ideal $\mathfrak{m} \subseteq A$ such that $\text{Ann}(x) \subseteq \mathfrak{m}$. Since (iii) holds, $M_{\mathfrak{m}} = \{0\}$ and so $x/1 = 0$ in $M_{\mathfrak{m}}$. Thus, there is $t \in A \setminus \mathfrak{m}$ such that $tx = 0$, a contradiction since $\text{Ann}(x) \subseteq \mathfrak{m}$. $\square$

Proposition 3.21. *Let $\phi: M \rightarrow N$ be an A -module homomorphism. Then the following are equivalent:*

- (i) ϕ is injective (respectively surjective),
- (ii) $\phi_P: M_P \rightarrow N_P$ is injective for all prime ideals $P \subseteq A$ (respectively surjective),
- (iii) $\phi_{\mathfrak{m}}: M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ is injective for all maximal ideals $\mathfrak{m} \subseteq A$ (respectively surjective).

Proposition 3.22 (Flatness is a Local Property). *Let A be a ring and M be an A -module. The following are equivalent:*

- (i) M is a flat A -module,
- (ii) M_P is a flat A_P -module for all prime ideals $P \subseteq A$,
- (iii) $M_{\mathfrak{m}}$ is a flat $A_{\mathfrak{m}}$ -module for all maximal ideals $\mathfrak{m} \subseteq A$.

3.3 Ideals in Ring of Fractions

Definition 3.23. Let $f: A \rightarrow B$ be a ring homomorphism.

- (a) Let J be an ideal of B . Then $J^c := f^{-1}(J)$ is an ideal of A called *the contraction of J via f* .
- (b) Let I be an ideal of A . The set

$$f(I)B := \left\{ \sum_{i=1}^n b_i f(a_i) \mid n \in \mathbb{N}, a_i \in A, b_i \in B \right\}$$

is an ideal in B called *the extension of I via f* . This is the smallest ideal in B containing $f(I)$ and is denoted by I^e .

Remark 3.24. Let A be a ring, let $S \subseteq A$ be a multiplicative set and let $f: A \rightarrow S^{-1}A$ be the canonical map. Let I be an ideal in A .

- (a) The extended ideal I^e coincides with the $(S^{-1}A)$ -module $S^{-1}I$, i.e.,

$$I^e = S^{-1}I = \{a/s \mid a \in I, s \in S\}.$$

The inclusion $\supseteq$ is clear since $a/s = 1/s \cdot a/1 = 1/s \cdot f(a)$.

We show the inclusion $\subseteq$. Let $x \in I^e$. This means that there are $b_i \in A$, $s_i \in S$ and $a_i \in I$ such that

$$x = \frac{b_1}{s_1} \frac{a_1}{1} + \dots + \frac{b_n}{s_n} \frac{a_n}{1}.$$

For $1 \leq i \leq n$ set $\widehat{s}_i := \prod_{j \neq i} s_j$ and $s := \prod_{j=1}^n s_j$. Then we have

$$x = \frac{\overbrace{\widehat{s}_1 b_1 a_1}^{\in I} + \dots + \overbrace{\widehat{s}_n b_n a_n}^{\in I}}{s} \in S^{-1}I.$$

- (b) If $S = A \setminus \mathfrak{p}$ for some prime ideal $\mathfrak{p}$, then we denote the extended ideal I^e by $IA_{\mathfrak{p}}$.
- (c) If $S = \{f^n\}_{n \in \mathbb{N}}$ for some element $f \in A$, then we denote the extended ideal I^e by IA_f .

Proposition 3.25. *Let A be a ring, let $S \subseteq A$ be a multiplicative set and let $f: A \rightarrow S^{-1}A$ be the natural map.*

- (i) *Every ideal in $S^{-1}A$ has the form $S^{-1}I$ for some ideal I in A .*
- (ii) *There is a one-to-one inclusion-preserving correspondence between prime ideals in A which are disjoint from S and prime ideals in $S^{-1}A$.*
- (iii) *Let $I, J \subseteq A$ be ideals. Let $\{I_{\alpha}\}_{\alpha \in A}$ be a family of ideals of A . Then it holds*

$$(a) \quad S^{-1}\left(\sum_{\alpha \in A} I_{\alpha}\right) = \sum_{\alpha \in A} S^{-1}I_{\alpha},$$

$$(b) \quad S^{-1}(I \cap J) = S^{-1}I \cap S^{-1}J,$$

$$(c) \quad S^{-1}(IJ) = S^{-1}I \cdot S^{-1}J,$$

$$(d) \quad S^{-1}(\sqrt{I}) = \sqrt{S^{-1}I}.$$

For short, the localization operation S^{-1} commutes with formation of sums, finite intersections, finite products and radicals.

Corollary 3.26. *Let A be a ring and let $S \subseteq A$ be a multiplicative set. Then it holds*

$$S^{-1}N_A = N_{S^{-1}A}.$$

Proof. Apply (iii)(d) of Proposition 3.25 to the ideal $\{0\}$. $\square$

Corollary 3.27. *Let A be a ring and let $P \subseteq A$ be a prime ideal. Then there is a one-to-one inclusion-preserving correspondence between prime ideals of A_P and prime ideals in A contained in P .*

Proof. Apply (ii) of Proposition 3.25 with $S = A \setminus P$. $\square$

Proposition 3.28. *Let A be a ring and let $S \subseteq A$ be a multiplicative set. Let M be a finitely generated A -module. Then*

$$S^{-1}(\text{Ann } M) = \text{Ann}(S^{-1}M).$$

Remark 3.29. Let A be a ring and let $P \subseteq Q$ be prime ideals of A . By Proposition 1.16 and Corollary 3.27 there is a one-to-one inclusion-preserving correspondence between the set

$$\{P' \subseteq A \text{ prime ideal} \mid P \subseteq P' \subseteq Q\}$$

and the set of prime ideals of

$$(A/P)_{Q/P} \cong A_Q/PA_Q,$$

where the ring isomorphism is given by

$$\frac{a + Q/P}{s + Q/P} \mapsto \frac{a}{s} + PA_Q.$$

Chapter 4

Integral Dependence

Definition 4.1. Let A and B be rings such that A is a subring of B . Recall that this means $A \subseteq B$ and $1_B \in A$. In such a situation, $A \subseteq B$ is called a *ring extension*.

- (a) A ring extension $A \subseteq B$ is called *finite* if B is a finitely generated A -module.
- (b) A ring extension $A \subseteq B$ is called *of finite type* if B is a finitely generated A -algebra.
- (c) Let $A \subseteq B$ be a ring extension. An element $b \in B$ is called *integral over A* if there exists a monic polynomial

$$f := x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in A[x]$$

such that $f(b) = 0$.

Remark 4.2. Let $A \subseteq B$ be a ring extension and let $a \in A$. Then a is integral over A since $x - a \in A[x]$ vanishes at a .

Remark 4.3. Let $A \subseteq B$ be a ring extension and let $b_1, \dots, b_n \in B$. Then we have the following tower of ring extensions

$$A \subseteq A[b_1, \dots, b_n] \subseteq B,$$

where $A[b_1, \dots, b_n]$ is the ring of polynomial expressions in $b_1, \dots, b_n$ with coefficients in A , introduced in Definition 2.50(b).

Proposition 4.4. Let $A \subseteq B$ be a ring extension and let $b \in B$. Then the following are equivalent:

- (i) b is integral over A ,
- (ii) $A \subseteq A[b]$ is a finite ring extension,

- (iii) There exists a subring $C \subseteq B$ containing $A[b]$ and such that $A \subseteq C$ is a finite ring extension.
- (iv) There exists a faithful $A[b]$ -module M which is finitely generated as an A -module.

Corollary 4.5. Let $A \subseteq B$ be a ring extension and let $b_1, \dots, b_n$ be elements of B which are integral over A . Then the extension $A \subseteq A[b_1, \dots, b_n]$ is finite.

Corollary 4.6. Let $A \subseteq B$ be a ring extension and let C be the set of all elements of B which are integral over A . Then C is a subring of B which contains A .

Definition 4.7. Let $A \subseteq B$ be a ring extension. The ring C from Corollary 4.6 is called the *integral closure of A in B* .

- (a) If $A = C$, then A is called *integrally closed in B* .
- (b) If $B = C$, then B is called *integral over A* and the ring extension $A \subseteq B$ is called *integral*.
- (c) If A is an integral domain and $B = Q(A)$, then C is called the *integral closure of A* . If $A = C$, then A is called an *integrally closed integral domain*.

Corollary 4.8. Let $A \subseteq B$ be a ring extension. Then

$$A \subseteq B \text{ is finite} \iff A \subseteq B \text{ is integral and of finite type.}$$

Proposition 4.9 (Transitivity of Integrality). Let $A \subseteq B \subseteq C$ be a tower of ring extensions such that C is integral over B and B is integral over A . Then C is integral over A .

Corollary 4.10. Let $A \subseteq B \subseteq C$ be a tower of ring extensions. Then

$$A \subseteq C \text{ integral} \iff A \subseteq B \text{ and } B \subseteq C \text{ integral.}$$

Corollary 4.11. Let $A \subseteq B$ be a ring extension and let C be the integral closure of A in B . Then C is integrally closed in B .

4.1 Going-Up Theorem

Lemma 4.12. Let $A \subseteq B$ be an integral ring extension.

- (i) Let Q be an ideal in B and $P := A \cap Q$. Then B/Q is integral over A/P .
- (ii) If $S \subseteq A$ is a multiplicative set, then $S^{-1}B$ is integral over $S^{-1}A$.

Lemma 4.13. Let $A \subseteq B$ be an integral ring extension, where A and B are integral domains. Then A is a field if and only if B is a field.

Proposition 4.14. *Let $A \subseteq B$ be an integral ring extension. Let $Q \subseteq B$ be a prime ideal and set $P := Q \cap A$. Then P is maximal in A if and only if Q is maximal in B .*

Proposition 4.15 (Incomparability). *Let $A \subseteq B$ be an integral ring extension. Let Q and Q' be prime ideals in B such that $Q \subseteq Q'$ and $Q \cap A = Q' \cap A$. Then $Q = Q'$.*

Theorem 4.16 (Lying-Over Theorem). *Let $A \subseteq B$ be an integral ring extension. Let $P \subseteq A$ be a prime ideal. Then there exists a prime ideal $Q \subseteq B$ such that $Q \cap A = P$.*

Proof. Let $S := A \setminus P$ and denote $S^{-1}B$ by B_P . By Lemma 4.12(ii) B_P is integral over A_P , hence we have the following commutative diagram

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \lambda_A \downarrow & & \downarrow \lambda_B \\ A_P & \hookrightarrow & B_P \end{array}$$

where λ_A and λ_B are the natural maps. Let $\mathfrak{n}$ be a maximal ideal in B_P and let $\mathfrak{m} := A_P \cap \mathfrak{n}$. By Proposition 4.14 $\mathfrak{m}$ is maximal in A_P and since A_P is local, it holds $\mathfrak{m} = PA_P$, hence $\lambda_A^{-1}(\mathfrak{m}) = P$. Set $Q := \lambda_B^{-1}(\mathfrak{n})$. Then Q is prime in B (because $\mathfrak{n}$ is prime) and since the diagram above commutes we have

$$Q \cap A = \lambda_B^{-1}(\mathfrak{n}) \cap A = \lambda_A^{-1}(\mathfrak{n} \cap A_P) = \lambda_A^{-1}(\mathfrak{m}) = P.$$

□

Theorem 4.17 (Going-Up Theorem). *Let $A \subseteq B$ be an integral ring extension. Let $P_1 \subseteq P_2 \subseteq \cdots \subseteq P_n$ be a chain of prime ideals in A and let $Q_1 \subseteq Q_2 \subseteq \cdots \subseteq Q_m$ (where $m < n$) be a chain of prime ideals in B such that $Q_i \cap A = P_i$ for $1 \leq i \leq m$. Then we can extend this sequence to a sequence $Q_1 \subseteq Q_2 \subseteq \cdots \subseteq Q_m \subseteq Q_{m+1} \subseteq \cdots \subseteq Q_n$ of prime ideals in B such that $Q_i \cap A = P_i$ for $1 \leq i \leq n$.*

Proof. By induction it suffices to show the statement for $n = 2$, $m = 1$. We have the following commutative diagram

$$\begin{array}{ccc} A & \xhookrightarrow{i} & B \\ \pi_A \downarrow & & \downarrow \pi_B \\ A/P_1 & \xhookrightarrow{j} & B/Q_1 \end{array}$$

where i is the inclusion giving the ring extension $A \subseteq B$, $j: A/P_1 \hookrightarrow B/Q_1$, $[a]_{P_1} \mapsto [a]_{Q_1}$, and π_A and π_B are the canonical projections. By Lemma 4.12 B/Q_1 is integral over A/P_1 . By Proposition 1.16 $\tilde{P}_2 = \pi_A(P_2)$ is a prime ideal in A/P_1 .

Hence, by the lying-over theorem (Theorem 4.16) there is a prime ideal $\tilde{Q}_2$ in B/Q_1 such that $\tilde{Q}_2 \cap (A/P_1) = \tilde{P}_2$. Set $Q_2 := \pi_B^{-1}(\tilde{Q}_2)$. Thus, by Proposition 1.16 Q_2 is a prime ideal in B such that $Q_1 \subseteq Q_2$ and since the diagram above commutes we have

$$Q_2 \cap A = \pi_B^{-1}(\tilde{Q}_2) \cap A = \pi_A^{-1}(\tilde{Q}_2 \cap (A/P_1)) = \pi_A^{-1}(\tilde{P}_2) = P_2.$$

□

4.2 Integral Closure

Proposition 4.18. *Let $A \subseteq B$ be a ring extension and let C be the integral closure of A in B . Let $S \subseteq A$ be a multiplicative set. Then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.*

Proof. By (ii) of Lemma 4.12 we know that $S^{-1}C$ is contained in the integral closure of $S^{-1}A$ in $S^{-1}B$. For the other inclusion, take $x/s \in S^{-1}B$ which is integral over $S^{-1}A$. Then we have

$$(x/s)^n + (a_{n-1}/s_{n-1})(x/s)^{n-1} + \cdots + (a_0/s_0) = 0,$$

where $a_i \in A$ and $s_i \in S$. Set $s' := s_{n-1} \cdots s_0$. By multiplying the equality above with $(s \cdot s')^n$, we obtain in $S^{-1}B$

$$(x \cdot s')^n + \sum_{i=1}^n a_{n-i} s^i s_{n-1}^{i-1} \cdots s_{n-i}^{i-1} \cdots s_0^i (x \cdot s')^{n-i} = 0$$

By definition of equivalence classes in $S^{-1}B$ there is $t \in S$ such that

$$t \left((x \cdot s')^n + \sum_{i=1}^n a_{n-i} s^i s_{n-1}^{i-1} \cdots s_{n-i}^{i-1} \cdots s_0^i (x \cdot s')^{n-i} \right) = 0$$

in B , thus multiplying with t^{n-1} we get in B

$$(x \cdot t \cdot s')^n + \sum_{i=1}^n b_{n-i} s^i s_{n-1}^{i-1} \cdots s_{n-i}^{i-1} \cdots s_0^i (x \cdot t \cdot s')^{n-i} = 0,$$

where $b_{n-i} = a_{n-i} t^i \in A$. Hence, $x \cdot t \cdot s'$ is integral over A and thus $x \cdot t \cdot s' \in C$. Finally,

$$\frac{x}{s} = \frac{x \cdot t \cdot s'}{s \cdot t \cdot s'} \in S^{-1}C.$$

□

For an integral domain being integrally closed is a local property.

Theorem 4.19 (Integral Closure is a Local Property). *Let A be an integral domain. Then the following are equivalent:*

- (i) A is integrally closed,
- (ii) A_P is integrally closed for all $P \subseteq A$ prime ideals,
- (iii) $A_{\mathfrak{m}}$ is integrally closed for all $\mathfrak{m} \subseteq A$ maximal ideals.

Proof. Let C be the integral closure of A in $Q(A)$ and let $f: A \rightarrow C$ be the inclusion map. Observe that f is an A -module homomorphism. We have

$$A \text{ integrally closed} \iff f \text{ is surjective.}$$

Let P be a prime ideal in A and let $S := A \setminus P$. By Proposition 4.18 the integral closure of $S^{-1}A = A_P$ in $S^{-1}Q(A) \cong Q(A) \cong Q(A_P)$ is $S^{-1}C$. Hence,

$$A_P \text{ integrally closed} \iff f_P: A_P \rightarrow S^{-1}C \text{ is surjective.}$$

Now by Proposition 3.21 we have

$$\begin{aligned} f \text{ surjective} &\iff f_P \text{ surjective for all } P \subseteq A \text{ prime ideals} \\ &\iff f_{\mathfrak{m}} \text{ surjective for all } \mathfrak{m} \subseteq A \text{ maximal ideals.} \end{aligned}$$

□

Example 4.20. Let k be a field. Then $k[x_1, \dots, x_n]$ is an integrally closed integral domain.

Now we extend the notion of integrality by introducing integrality relative to an ideal.

Definition 4.21. Let $A \subseteq B$ a ring extension and let $I \subseteq A$ be an ideal. An element $b \in B$ is called *integral over I* if there exists a monic polynomial

$$f := x^n + a_{n-1}x^{n-1} + \dots + a_0 \in A[x]$$

with $a_i \in I$ for all $i \in \{0, \dots, n-1\}$ such that $f(b) = 0$. The *integral closure of I in B* is the set of all elements in B which are integral over I .

Remark 4.22. Let $A \subseteq B$ be a ring extension and let I be an ideal in A . Let C be the integral closure of A in B and let D be the integral closure of I in B . Any $a \in I$ is integral over I since $x - a \in A[x]$ vanishes at a and $a \in I$. Hence, $I \subseteq D$. Note that at this stage $D \subseteq C$ is just a subset of the ring C . Indeed, D is an ideal of C , as shown by the next lemma.

Lemma 4.23. *Let $A \subseteq B$ be a ring extension, let C be the integral closure of A in B and let $I \subseteq A$ be an ideal of A and let I^e denote the extension of I in C . Then the integral closure of I in B is $\sqrt{I^e}$. In particular, sums and products of elements of B which are integral over I are again integral over I .*

Proof. $\subseteq$. Let $b \in B$ be an element which is integral over I . In particular, b is integral over A , and thus $b \in C$. This means that there exist $n \in \mathbb{N}$ and $a_i \in I$ such that $b^n + a_{n-1}b^{n-1} + \dots + a_0 = 0$. Since

$$b^n = -a_{n-1}b^{n-1} - \dots - a_0 \in I \cdot C = I^e,$$

it holds $b \in \sqrt{I^e}$.

$\supseteq$. Let $b \in \sqrt{I^e}$. Then $b^n = \sum_{i=1}^k a_i c_i \in I^e = I \cdot C$ with $a_i \in I$ and $c_i \in C$. Now consider $M := A[c_1, \dots, c_k]$. Since the elements c_i 's are integral over A , the ring extension $A \subseteq M$ is finite by Corollary 4.5. Hence, $M = Am_1 + \dots + Am_r$ for some $m_i \in M$. Set $x := b^n$ and observe that $x \cdot m_1, \dots, x \cdot m_r \in IM$. Thus, similarly as in the proof of Proposition 4.4 (see (iv) $\Rightarrow$ (i)), we obtain

$$f := x^r + \alpha_{r-1}x^{r-1} + \dots + \alpha_0$$

with $\alpha_i \in I$ for $0 \leq i \leq r-1$ such that $f \cdot m = 0$ for all $m \in M$. Hence, in particular $f = f \cdot 1 = 0$ and we have

$$(b^n)^r + \alpha_{r-1}(b^n)^{r-1} + \dots + \alpha_0 = 0,$$

which implies that b is integral over I . $\square$

Remark 4.24. Let $A \subseteq B$ be a ring extension where A and B are integral domains. Then we have the following commutative diagram

$$\begin{array}{ccc} A & \xhookrightarrow{i} & B \\ \downarrow \alpha & & \downarrow \beta \\ Q(A) & \xhookrightarrow{j} & Q(B) \end{array}$$

where $i: A \rightarrow B$ is the inclusion giving the ring extension, $\alpha: A \rightarrow Q(A)$, $a \mapsto a/1$, $\beta: B \rightarrow Q(B)$, $b \mapsto b/1$ and $j: Q(A) \rightarrow Q(B)$, $a/s \mapsto a/s$.

Proposition 4.25. Let $A \subseteq B$ be a ring extension where A and B are integral domains and A is integrally closed. Let I be an ideal of A and let $b \in B$ be integral over I . Then b is algebraic over $Q(A)$ and if its minimal polynomial (over $Q(A)$) is $t^n + a_{n-1}t^{n-1} + \dots + a_0$, then $a_i \in \sqrt{I}$.

Proof. The element b (regarded as an element of the field $Q(B)$, see Remark 4.24) is obviously algebraic over $Q(A)$. Let $\mu = t^n + \sum_{i=0}^{n-1} a_i t^i$ be the minimal polynomial of b over $Q(A)$ and let $L \supseteq Q(A)$ be a splitting field of μ over $Q(A)$. Then $\mu = (t - b_1) \cdots (t - b_n) \in L[t]$ and there is $1 \leq i \leq n$ such that $b = b_i$. Without loss of generality, we may assume $b = b_1$. If λ is the monic polynomial with coefficients in I such that $\lambda(b) = 0$, then $\mu \mid \lambda$, hence $\lambda(b_i) = 0$ for $1 \leq i \leq n$. Thus, $b_1, \dots, b_n$ are elements of L which are integral over I . The coefficients a_i of μ are (up to a sign) elementary symmetric polynomials in $b_1, \dots, b_n$. Hence, by Lemma 4.23 applied to the ring extension $A \subseteq L$ the a_i 's are integral over I . Finally, since $a_i \in Q(A)$ for all i and A is integrally closed, by Lemma 4.23 applied to the ring extension $A \subseteq Q(A)$ we have $a_i \in \sqrt{I}$. $\square$

Corollary 4.26. *Let $A \subseteq B$ be a ring extension where A and B are integral domains and A is integrally closed. Let $b \in B$ be integral over A . Then b is algebraic over $Q(A)$ and if its minimal polynomial (over $Q(A)$) is $t^n + a_{n-1}t^{n-1} + \dots + a_0$, then $a_i \in A$.*

Proof. Take $I = A$ in Proposition 4.25. □

4.3 Going-Down Theorem

Proposition 4.27. *Let $f: A \rightarrow B$ be a ring homomorphism and $P \subseteq A$ be a prime ideal. Then there is a prime ideal $Q \subseteq B$ such that $P = f^{-1}(Q)$ if and only if $P = f^{-1}(f(P)B)$.*

Proof. The implication $\Rightarrow$ is easy. Let us show $\Leftarrow$. Let $S := f(A \setminus P) \subseteq B$. Then S is a multiplicative set in B and $S \cap f(P)B = \emptyset$ since by assumption $P = f^{-1}(f(P)B)$. Hence, $S^{-1}[f(P)B]$ is a proper ideal of $S^{-1}B$. Let $\mathfrak{m}$ be a maximal ideal of $S^{-1}B$ such that $\mathfrak{m} \supseteq S^{-1}[f(P)B]$ and set $Q := \pi^{-1}(\mathfrak{m})$, where $\pi: B \rightarrow S^{-1}B$ is the natural map. Hence, Q is a prime ideal of B such that $Q \supseteq f(P)B$ and $Q \cap S = \emptyset$. Since $Q \supseteq f(P)B$, we have $f^{-1}(Q) \supseteq P$. On the other hand, since $Q \cap S = \emptyset$ and $S = f(A \setminus P)$, it holds

$$\emptyset = f^{-1}(Q) \cap f^{-1}(f(A \setminus P)) \supseteq f^{-1}(Q) \cap (A \setminus P),$$

hence $f^{-1}(Q) \cap (A \setminus P) = \emptyset$, i.e., $f(Q) \subseteq P$. Finally, we have $f^{-1}(Q) = P$. □

Theorem 4.28 (Going-Down Theorem). *Let $A \subseteq B$ be an integral ring extension, where A and B are integral domains and A is integrally closed. Let $P_1 \supseteq P_2 \supseteq \dots \supseteq P_n$ be a chain of prime ideals in A and let $Q_1 \supseteq Q_2 \supseteq \dots \supseteq Q_m$ (where $m < n$) be a chain of prime ideals in B such that $Q_i \cap A = P_i$ for $1 \leq i \leq m$. Then we can extend this sequence to a sequence $Q_1 \supseteq Q_2 \supseteq \dots \supseteq Q_m \supseteq Q_{m+1} \supseteq \dots \supseteq Q_n$ of prime ideals in B such that $Q_i \cap A = P_i$ for $1 \leq i \leq n$.*

Proof. As in the proof of the going-up theorem, we may assume $n = 2$, $m = 1$. We have the following commutative diagram

$$\begin{array}{ccc} A & \xhookrightarrow{i} & B \\ \lambda_A \downarrow & \searrow f & \downarrow \lambda_B \\ A_{P_1} & \xhookrightarrow{j} & B_{Q_1} \end{array} \quad (4.1)$$

where i is the inclusion giving the ring extension, the natural maps λ_A and λ_B are injective because A and B are integral domains and $j: A_{P_1} \rightarrow B_{Q_1}$, $a/s \mapsto a/s$, is a well-defined map since $A \setminus P_1 \subseteq B \setminus Q_1$ and it is injective since B is an integral domain. We set $f := j \circ \lambda_A = \lambda_B \circ i$.

We claim that there is a prime ideal $\tilde{Q}_2 \subseteq B_{Q_1}$ such that $P_2 = f^{-1}(\tilde{Q}_2)$. If we show this, the assertion of the theorem follows, since $Q_2 := \lambda_B^{-1}(\tilde{Q}_2)$ fulfills $P_2 = Q_2 \cap A$ (because the diagram is commutative!) and $Q_2 \subseteq Q_1$ by Corollary 3.27.

By Proposition 4.27, it suffices to show that $P_2 = f^{-1}(f(P_2)B_{Q_1})$. Since f is injective, we have to show that $P_2 = A \cap P_2B_{Q_1}$. Clearly, $P_2 \subseteq A \cap P_2B_{Q_1}$. Let us show the inclusion $\supseteq$. Take $x \in A \cap P_2B_{Q_1}$. Since $x \in P_2B_{Q_1}$, then $x = y/s$ with $y \in P_2B$ and $s \in B \setminus Q_1$. The element y is integral over P_2 by Lemma 4.23 applied to the integral extension $A \subseteq B$. Hence, by Proposition 4.25 the minimal polynomial of y over $Q(A)$ is of the form

$$t^n + \alpha_{n-1}t^{n-1} + \dots + \alpha_0 \in Q(A)[t]$$

with $\alpha_i \in P_2$. Since in $Q(B)$ we have $s = yx^{-1}$ with $x^{-1} \in Q(A)$ (because $x \in A$), the minimal polynomial of s over $Q(A)$ is

$$t^n + \beta_{n-1}t^{n-1} + \dots + \beta_1t + \beta_0 \in Q(A)[t]$$

where $\beta_i = \alpha_i x^{-(n-i)}$ for $0 \leq i \leq n-1$. Since $s \in B$ is integral over A , then by Corollary 4.26 we have $\beta_{n-i} \in A$. Moreover, it holds

$$\beta_i x^{n-i} = \alpha_i \in P_2 \quad \text{for all } 0 \leq i \leq n-1.$$

If we assume by contradiction that $x \notin P_2$, then $\beta_i \in P_2$ for $0 \leq i \leq n-1$ since P_2 is prime. But then $s^n = -\beta_{n-1}s^{n-1} - \beta_{n-2}s^{n-2} - \dots - \beta_0 \in P_2B \subseteq P_1B \subseteq Q_1$, thus $s \in Q_1$ since Q_1 is prime. This contradicts $s \in B \setminus Q_1$, and hence it must hold $x \in P_2$. Finally, we showed $P_2 = A \cap P_2B_{Q_1}$. $\square$

Remark 4.29 (Fake Proof of the Going-Down Theorem). A natural (but wrong!) attempt of proving the going-down theorem might have been the following.

Consider diagram (4.1). Take the extension $P_2A_{P_1}$ of P_2 via λ_A . Since $P_2A_{P_1}$ is prime, by the lying-over theorem (Theorem 4.16) there exists a prime $\tilde{Q}_2$ in B_{Q_1} such that $\tilde{Q}_2 \cap A_{P_1} = P_2A_{P_1}$. Set $Q_2 := \lambda_B^{-1}(\tilde{Q}_2)$. By Corollary 3.27 Q_2 is a prime ideal such that $Q_2 \subseteq Q_1$. Moreover, since diagram (4.1) commutes we have

$$Q_2 \cap A = \lambda_B^{-1}(\tilde{Q}_2) \cap A = \lambda_A^{-1}(\tilde{Q}_2 \cap A_{P_1}) = \lambda_A^{-1}(P_2A_{P_1}) = P_2.$$

Everything seems to be correct, but there is a subtle mistake. The problem is that we are not allowed to apply the lying-over theorem since we do not know if the ring extension $A_{P_1} \subseteq B_{Q_1}$ is integral! In fact, we localize A and B with respect two different multiplicative sets, namely $A \setminus P_1$ and $B \setminus Q_1$, respectively. Hence, we cannot apply Lemma 4.12(ii) to the ring extension $A_{P_1} \subseteq B_{Q_1}$.

Chapter 5

Noetherian Modules and Rings

We start with two easy, but important lemmata.

Lemma 5.1. *Let $(\Sigma, \leq)$ be a partially ordered set. The following are equivalent:*

- (i) Σ satisfies the ascending chain condition (ACC), that is, if we have an ascending chain $x_1 \leq x_2 \leq \dots$, then there exists $n \in \mathbb{N}$ such that $x_m = x_n$ for all $m \geq n$,
- (ii) Σ satisfies the maximal condition, that is, every nonempty set in Σ has a maximal element.

Lemma 5.2. *Let $(\Sigma, \leq)$ be a partially ordered set. The following are equivalent:*

- (i) Σ satisfies the descending chain condition (DCC), that is, if we have a descending chain $x_1 \geq x_2 \geq \dots$, then there exists $n \in \mathbb{N}$ such that $x_m = x_n$ for all $m \geq n$,
- (ii) Σ satisfies the minimal condition, that is, every nonempty set in Σ has a minimal element.

Example 5.3. (a) Let $\Sigma := \{1/n \mid n \in \mathbb{N}_{>0}\}$ and let $\leq$ be the usual $\leq$ between real numbers. Then $(\Sigma, \leq)$ satisfies the ACC (or equivalently, the maximal condition).

- (b) Let $\Sigma := \{1 - 1/n \mid n \in \mathbb{N}_{>0}\}$ and let $\leq$ be the usual $\leq$ between real numbers. Then $(\Sigma, \leq)$ satisfies the DCC (or equivalently, the minimal condition).

Definition 5.4. Let A be a ring and let M be an A -module. Let Σ be the set of submodules of M ordered by inclusion.

- (a) M is called *Noetherian* (after Emmy Noether) if Σ satisfies the ACC (or equivalently, the maximal condition).
- (b) M is called *Artinian* (after Emil Artin) if Σ satisfies the DCC (or equivalently, the minimal condition).

Proposition 5.5. *Let M be an A -module. Then M is Noetherian if and only if every submodule of M is finitely generated.*

Proof. “ $\Rightarrow$ ”. Assume by contradiction there exists a submodule $N \subseteq M$ that is not finitely generated. Pick $n_1 \in N$, then $n_2 \in N \setminus An_1$ and so on. The sequence $An_1 \subsetneq An_1 + An_2 \subsetneq \dots$ is an ascending chain of submodules which does not stabilize, a contradiction since M is Noetherian.

“ $\Leftarrow$ ”. Let $N_1 \subseteq N_2 \subseteq \dots$ be an ascending chain of submodules of M . Since the submodule $\bigcup_{i=1}^{\infty} N_i$ is finitely generated, there are generators $n_1, \dots, n_k$. Thus, for $1 \leq i \leq k$ there is $j_i \in \mathbb{N}$ such that $n_i \in N_{j_i}$ and for $j := \max\{j_i \mid 1 \leq i \leq k\}$ it holds $n_1, \dots, n_k \in N_j$. Hence, $\bigcup_{i=1}^{\infty} N_i = N_j$ and $N_m = N_j$ for all $m \geq j$. $\square$

Proposition 5.6. *Let M_1, M_2, M_3 be A -modules and let*

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3 \longrightarrow 0$$

be a short exact sequence. Then

(i) *M_2 is Noetherian if and only if M_1 and M_3 are Noetherian.*

(ii) *M_2 is Artinian if and only if M_1 and M_3 are Artinian.*

Proof. Both statements are proven in precisely the same way. We will just prove (i).

“ $\Rightarrow$ ”. Assume $N_1 \subseteq N_2 \subseteq \dots$ is an ascending chain of submodules in M_3 . Then $\beta^{-1}(N_1) \subseteq \beta^{-1}(N_2) \subseteq \dots$ is an ascending chain of submodules in M_2 , which is stationary since M_2 is Noetherian. Since β is surjective, for a subset $X \subseteq M_3$ it holds $\beta(\beta^{-1}(X)) = X$. Thus, the sequence $N_1 \subseteq N_2 \subseteq \dots$ must be stationary.

If $P_1 \subseteq P_2 \subseteq \dots$ is an ascending chain of submodules in M_1 , then $\alpha(P_1) \subseteq \alpha(P_2) \subseteq \dots$ is a sequence of submodules in M_2 which stabilizes since M_2 is Noetherian. Since α is injective, for a subset $Y \subseteq M_1$ it holds $\alpha^{-1}(\alpha(Y)) = Y$. Thus, $P_1 \subseteq P_2 \subseteq \dots$ must stabilize.

“ $\Leftarrow$ ”. By Proposition 5.5 it is enough to show that any submodule of M_2 is finitely generated. Let $N_2 \subseteq M_2$ be a submodule. Define $N_1 := \alpha^{-1}(N_2)$ and $N_3 := \beta(N_2)$. Hence, the sequence

$$0 \longrightarrow N_1 \xrightarrow{\alpha|_{N_1}} N_2 \xrightarrow{\beta|_{N_2}} N_3 \longrightarrow 0$$

is exact by construction. The modules N_1 and N_3 are finitely generated by Proposition 5.5 since they are submodules of Noetherian modules. Hence, by Exercise 4(a) from Exercise Sheet 3 the module N_2 is finitely generated and so M_2 is Noetherian.

Alternatively, let $N_1 \subseteq N_2 \subseteq \dots$ be an ascending chain of submodules in M_2 . Then $\alpha^{-1}(N_1) \subseteq \alpha^{-1}(N_2) \subseteq \dots$ is an ascending chain of submodules in M_1 , which is stationary since M_1 is Noetherian. Hence, there exists $k_1 \in \mathbb{N}$ such

that $\alpha^{-1}(N_m) = \alpha^{-1}(N_{k_1})$ for all $m \geq k_1$. Moreover, $\beta(N_1) \subseteq \beta(N_2) \subseteq \dots$ is an ascending chain of submodules in M_3 , which is stationary since M_3 is Noetherian. Hence, there exists $k_2 \in \mathbb{N}$ such that $\beta(N_m) = \beta(N_{k_2})$ for all $m \geq k_2$. Set $k := \max\{k_1, k_2\}$. Then for all $m \geq k$ it holds

$$\alpha^{-1}(N_m) = \alpha^{-1}(N_k) \quad \text{and} \quad \beta(N_m) = \beta(N_k).$$

Now we show that $N_m = N_k$ for all $m \geq k$. Since $N_k \subseteq N_m$ already holds, we need to show only $N_m \subseteq N_k$. Let $x \in N_m$. Then $\beta(x) \in \beta(N_m) = \beta(N_k)$, thus there exists $y \in N_k$ such that $\beta(x) = \beta(y)$. This implies that $x - y \in \ker \beta = \operatorname{im} \alpha$, hence there exists $z \in M_1$ such that $x - y = \alpha(z)$. Since $x, y \in N_m$, then $z \in \alpha^{-1}(N_m) = \alpha^{-1}(N_k)$, and thus $\alpha(z) \in N_k$. Finally, we have $x = y + \alpha(z) \in N_k$. $\square$

Corollary 5.7. *Let A be a ring, let I be a finite index set and let $\{M_i\}_{i \in I}$ be family of Noetherian (respectively Artinian) A -modules. Then $\bigoplus_{i \in I} M_i$ is Noetherian (respectively Artinian) as well.*

Proof. By induction, it suffices to show the statement for $n = 2$. We have the short exact sequence $0 \rightarrow M_1 \rightarrow M_1 \oplus M_2 \rightarrow M_2 \rightarrow 0$. Now use Proposition 5.6. $\square$

Definition 5.8. A ring A is called *Noetherian* (respectively *Artinian*) if it is Noetherian (respectively Artinian) as a module over itself.

Proposition 5.9. *Let A be a Noetherian (respectively Artinian) ring and M be a finitely generated A -module. Then M is Noetherian (respectively Artinian) as well.*

Proof. By Corollary 5.7, $A^n = \bigoplus_{i=1}^n A$ is Noetherian (respectively Artinian) for every $n \in \mathbb{N}$. By Proposition 2.21 every finitely generated A -module is a quotient of A^n for some $n \in \mathbb{N}$. Finally, use Proposition 5.6. $\square$

Proposition 5.10. *Let A be a Noetherian (respectively Artinian) ring and let M and N be finitely generated A -modules. Then $\operatorname{Hom}_A(M, N)$ is a Noetherian (respectively Artinian) A -module.*

Proof. Both statements are proven in precisely the same way. We will just prove the Noetherian case. Since M is finitely generated, there is a surjective homomorphism $\varphi: A^n \twoheadrightarrow M$, i.e. $M \cong A^n / \ker \varphi$. Thus, we have an injective A -linear map (see Proposition 2.32(i)):

$$\bar{\varphi}: \operatorname{Hom}_A(M, N) \hookrightarrow \operatorname{Hom}_A(A^n, N), \quad \theta \longmapsto \theta \circ \varphi.$$

By Proposition 5.6, it suffices to show that $\operatorname{Hom}_A(A^n, N)$ is Noetherian. Observe that, if $e_1, \dots, e_n$ are the canonical generators of A^n , we have an A -module isomorphism

$$\operatorname{Hom}_A(A^n, N) \xrightarrow{\sim} N^n, \quad f \longmapsto (f(e_1), \dots, f(e_n)).$$

Now, since A is Noetherian and N is finitely generated, N is Noetherian by Proposition 5.9. Finally, by Corollary 5.7 N^n is Noetherian, and we are done. $\square$

Corollary 5.11. *Let A be a Noetherian (respectively Artinian) ring and let M be a finitely generated A -module. Then the dual module $M^* := \text{Hom}_A(M, A)$ is a Noetherian (respectively Artinian) A -module.*

Proof. Apply Proposition 5.10 with $N = A$. □

Let A be a Noetherian ring. By Hilbert's basis theorem (Theorem 1.34) the polynomial ring $A[X_1, \dots, X_n]$ is also Noetherian. This is a way to construct new Noetherian rings from a given one. We now discuss some further constructions that produce new Noetherian (respectively Artinian) rings from existing ones.

Proposition 5.12. *Let A be a Noetherian (respectively Artinian) ring and let $I \subseteq A$ be an ideal. Then A/I is a Noetherian ring (respectively Artinian).*

Proof. Consider the short exact sequence of A -modules $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$. By Proposition 5.6 A/I is a Noetherian (respectively Artinian) A -module. Since the submodules of A/I as an A -module coincide with the ideals of A/I , we have that A/I is a Noetherian (respectively Artinian) ring. □

Corollary 5.13. *Let $\phi: A \rightarrow B$ be a surjective ring homomorphism. If A is a Noetherian (respectively Artinian) ring, then B is a Noetherian (respectively Artinian) ring as well.*

Proof. Observe that the ring B is isomorphic to $A/\ker \phi$. Apply now Proposition 5.12. □

Proposition 5.14. *Let A be a Noetherian (respectively Artinian) ring and let $S \subseteq A$ be a multiplicative set. Then $S^{-1}A$ is a Noetherian (respectively Artinian) ring.*

Proof. Both statements are proven in precisely the same way. We will just prove the Noetherian case. Let $f: A \rightarrow S^{-1}A$, $a \mapsto a/1$, and let $I_1 \subseteq I_2 \subseteq \dots$ be an ascending chain of ideals in $S^{-1}A$. Then $f^{-1}(I_1) \subseteq f^{-1}(I_2) \subseteq \dots$ is an ascending sequence of ideals in A which stabilizes because A is Noetherian. Since $S^{-1}(f^{-1}(I_j)) = I_j$ for all $j \in \mathbb{N}$, we have that $I_1 \subseteq I_2 \subseteq \dots$ stabilizes as well. □

Corollary 5.15. *Let A be a Noetherian (respectively Artinian) ring.*

- (i) *Let $P \subseteq A$ be a prime ideal. Then A_P is a Noetherian (respectively Artinian) ring.*
- (ii) *Let $f \in A$. Then A_f is a Noetherian (respectively Artinian) ring.*

Proposition 5.16. *Let A be a Noetherian (respectively Artinian) ring and let B be a finite A -algebra. Then B is a Noetherian (respectively Artinian) ring as well.*

Proof. Both statements are proven in precisely the same way. We will just prove the Noetherian case. Since B is finitely generated as an A -module and A is Noetherian, by Proposition 5.9 B is Noetherian as an A -module. Let $I_1 \subseteq I_2 \subseteq \dots$ be an ascending chain of ideals in B . Since the I_j 's are also A -modules and B is a Noetherian A -module, the sequence $I_1 \subseteq I_2 \subseteq \dots$ stabilizes. Hence, B is a Noetherian ring. $\square$

Proposition 5.17. *Let A be a Noetherian ring and let B be a finitely generated A -algebra. Then B is a Noetherian ring as well. In particular, every finitely generated algebra over a field is a Noetherian ring.*

Proof. By definition, there is a surjective ring homomorphism

$$\phi: A[X_1, \dots, X_n] \twoheadrightarrow B,$$

where $A[X_1, \dots, X_n]$ is a polynomial ring. By Hilbert's basis theorem (Theorem 1.34) $A[X_1, \dots, X_n]$ is Noetherian since A is Noetherian. Now by Corollary 5.13 B is a Noetherian ring. Since any field is Noetherian, the second part of the statement follows. $\square$

The next lemma will be used to prove Hilbert's Nullstellensatz.

Lemma 5.18 (Artin-Tate Lemma). *Let $A \subseteq B \subseteq C$ be a tower of ring extensions. Assume that A is Noetherian and that C is finitely generated as an A -algebra and finitely generated as a B -module. Then B is finitely generated as an A -algebra.*

Proof. By assumption, there are $x_1, \dots, x_n, y_1, \dots, y_m \in C$ such that

$$C = A[x_1, \dots, x_n] \quad \text{and} \quad C = By_1 + \dots + By_m.$$

Then for each $1 \leq i \leq n$ we can write

$$x_i = \sum_{j=1}^m b_{ij} y_j, \quad \text{for some } b_{ij} \in B \tag{5.1}$$

and for all $1 \leq \alpha, \beta \leq m$ we have

$$y_\alpha \cdot y_\beta = \sum_{\gamma=1}^m d_{\alpha\beta\gamma} y_\gamma, \quad \text{for some } d_{\alpha\beta\gamma} \in B. \tag{5.2}$$

Let B_0 be the A -algebra generated by the set of b_{ij} 's and $d_{\alpha\beta\gamma}$'s, namely

$$B_0 := A[\{b_{ij}, d_{\alpha\beta\gamma} \mid 1 \leq i \leq n, 1 \leq j, \alpha, \beta, \gamma \leq m\}] \subseteq B.$$

Since A is Noetherian, by Proposition 5.17 B_0 is Noetherian. Now let $z \in C$. Then z is a polynomial expression in $x_1, \dots, x_n$ with coefficients in A . Then, by

substituting (5.1) and making repeated use of (5.2), we can write z as a linear combination of $y_1, \dots, y_m$ with coefficients in B_0 . In other words, it holds

$$C = B_0 y_1 + \dots + B_0 y_m.$$

Hence, since B_0 is a Noetherian ring and C is a finitely generated B_0 -module, by Proposition 5.9 C is a Noetherian B_0 -module. Moreover, we have the following tower of ring extensions

$$A \subseteq B_0 \subseteq B \subseteq C.$$

Thus, since B is a submodule of C (as a B_0 -module) and C is a Noetherian B_0 -module, by Proposition 5.5 B is finitely generated as a B_0 -module, i.e., there are $b'_1, \dots, b'_r \in B$ such that

$$B = B_0 b'_1 + \dots + B_0 b'_r.$$

Finally, since B_0 is by construction a finitely generated A -algebra, it follows that B is finitely generated as a A -algebra: namely, we have

$$B = A[\{b_{ij}, d_{\alpha\beta\gamma}, b'_t \mid 1 \leq i \leq n, 1 \leq j, \alpha, \beta, \gamma \leq m, 1 \leq t \leq r\}].$$

□

Proposition 5.19. *Let A be a Noetherian ring and let $I \subseteq A$ be an ideal. Then there exists $m \in \mathbb{N}$ such that $(\sqrt{I})^m \subseteq I$. In particular, the nilradical N_A is nilpotent.*

Proof. Since A is Noetherian, the radical $\sqrt{I}$ is a finitely generated ideal of A , i.e., we can write $\sqrt{I} = (x_1, \dots, x_k)$. For each $1 \leq i \leq k$, there is $n_i \in \mathbb{N}$ such that $x_i^{n_i} \in I$. Observe that for an integer $m \in \mathbb{N}$ we have that $(\sqrt{I})^m$ is generated by the products of the form $x_1^{r_1} \cdots x_k^{r_k}$ with $m = r_1 + \dots + r_k$. Now set $m := n_1 + \dots + n_k$. By the choice of m , for each product $x_1^{r_1} \cdots x_k^{r_k}$ there exists $j \in \{1, \dots, k\}$ such that $r_j \geq n_j$. Thus, $x_j^{r_j} \in I$ and $x_1^{r_1} \cdots x_k^{r_k} \in I$. Since all the generators of $(\sqrt{I})^m$ are in I , the statement follows. □

Definition 5.20. Let A be a ring.

- (i) Let $P \subseteq A$ be a prime ideal. The *height* P is the supremum of all integers n such that there exists a chain $P_0 \subsetneq \dots \subsetneq P_n = P$ of distinct prime ideals. We denote it by $\text{ht}(P)$.
- (ii) The *Krull dimension* of A is the supremum of the heights of all prime ideals in A . We denote it by $\dim A$.

Example 5.21. (a) A field k has $\dim k = 0$.

(b) A principal ideal domain A has $\dim A = 1$. In particular, $\dim \mathbb{Z} = 1$.

Remark 5.22. As the reader might have noticed, there is an apparent symmetry between Noetherian rings and Artinian rings. However, this is misleading since Artinian rings are indeed Noetherian rings of a special kind, as the following Proposition clarifies.

Proposition 5.23. *Let A be a ring. Then A is Artinian if and only if A is Noetherian and $\dim A = 0$.*

5.1 Modules of Finite Length

Definition 5.24. Let A be a ring and let M be an A -module.

- (i) A *chain of submodules* of M is a sequence

$$M = M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots \supsetneq M_n = \{0\},$$

where $n \in \mathbb{N}$ is the number of strict inclusions and is called the *length of the chain*.

- (ii) A *composition series* of M is a chain of submodules which is maximal in the sense that no extra submodules can be inserted. This is equivalent to saying that for each $0 \leq i \leq n-1$, the modules M_i/M_{i+1} are *simple*, i.e., the only submodules of M_i/M_{i+1} are $\{0\}$ and M_i/M_{i+1} .
- (iii) Assume that M has a composition series. We denote by $\ell(M)$ the minimum length of a composition series of M . The module M is called a *module of finite length* and $\ell(M)$ is called the *length of M* .

Lemma 5.25. *Let A be a ring and let M be an A -module with a composition series. Let $N \subseteq M$ be a submodule. Then N has a composition series as well. Furthermore, the following statements hold:*

- (i) *If $N \subsetneq M$, then $\ell(N) < \ell(M)$.*

- (ii) *Any chain of submodules of M has length $\leq \ell(M)$.*

Proof. Let $M = M_0 \supsetneq \cdots \supsetneq M_n = \{0\}$ be a composition series of M of length n . For $0 \leq i \leq n$ set $N_i := M_i \cap N$. Note that $N_0 = N$ and $N_n = \{0\}$. Since for each $0 \leq i \leq n-1$ we have an induced embedding $N_i/N_{i+1} \hookrightarrow M_i/M_{i+1}$ and the latter is a simple module, we have either $N_i/N_{i+1} \cong M_i/M_{i+1}$ or $N_i/N_{i+1} = \{0\}$, that is $N_i = N_{i+1}$. Hence, after possibly removing repeated terms, we get a composition series of N of length $m \leq n$.

- (i) Assume that $n = \ell(M)$. Then $\ell(N) \leq m \leq n = \ell(M)$, whence $\ell(N) \leq \ell(M)$. If $\ell(N) = \ell(M)$, then for each $0 \leq i \leq n-1$ we have $N_i/N_{i+1} \cong M_i/M_{i+1}$. In particular, $N_{n-1}/\{0\} \cong M_{n-1}/\{0\}$, which gives $N_{n-1} = M_{n-1}$. Since $N_{n-2}/N_{n-1} \cong M_{n-2}/M_{n-1}$ and $N_{n-1} = M_{n-1}$, we have $N_{n-2} = M_{n-2}$ and so on. Finally, we get $N = N_0 = M_0 = M$.

- (ii) Let $M = M_0 \supsetneq \cdots \supsetneq M_k = \{0\}$ be a chain of submodules. Then by (i) we have

$$\ell(M) > \ell(M_1) > \cdots > \ell(M_k) = 0,$$

hence for each $0 \leq i \leq k$ we get inductively $\ell(M_{k-i}) \geq i$, and in particular for $i = k$ we have $\ell(M) \geq k$.

□

Proposition 5.26. *Let A be a ring and let M be an A -module. Assume that M has a composition series. Then all composition series of M have length $\ell(M)$. Moreover, let $M = M_0 \supsetneq \cdots \supsetneq M_n = \{0\}$ be a chain of submodules of M of length n . Then this chain is a composition series of M if and only if $n = \ell(M)$.*

Proof. Let $M = M_0 \supsetneq \cdots \supsetneq M_k = \{0\}$ be any composition series of M of length k . By (ii) of Lemma 5.25 we have $k \leq \ell(M)$. Then by the definition of $\ell(M)$ we have indeed $k = \ell(M)$. Hence, all composition series have indeed length $\ell(M)$. Now consider the chain $M = M_0 \supsetneq \cdots \supsetneq M_n = \{0\}$. By (ii) of Lemma 5.25 we have $n \leq \ell(M)$; if $n = \ell(M)$ the chain is maximal, i.e., it is a composition series, while if $n < \ell(M)$, it cannot be a composition series of M since we proved that all composition series have length $\ell(M)$. □

Proposition 5.27. *Let A be a ring and let M be an A -module M . Then M has a composition series if and only if M is Noetherian and Artinian.*

Proof. $\Rightarrow$. If M has a composition series, any sequence of submodules of M is finite, and hence M trivially satisfies both ACC and DCC.

$\Leftarrow$. Let $M = M_0$. If $M = \{0\}$, there is nothing to show. Hence, assume $M \neq \{0\}$. Since M satisfies ACC, by Lemma 5.1 M satisfies the maximal condition. Hence, there is a maximal proper submodule M_1 of M . If $M_1 = \{0\}$, we are done, otherwise consider the set of all proper submodules of M strictly contained in M_1 , that is nonempty since $\{0\} \subsetneq M_1$. Hence, by the maximal condition this set has a maximal element $M_2 \subsetneq M_1$. Inductively, there are maximal proper submodules $M_{i+1} \subsetneq M_i$, thus we have a descending sequence $M_0 \supsetneq M_1 \supsetneq \cdots$ which must be finite, since M satisfies also DCC. Finally, we obtained a composition series

$$M = M_0 \supsetneq \cdots \supsetneq M_n = \{0\}.$$

□

Proposition 5.28. *Let A be a ring. The function $\ell(\cdot)$ is an additive function on the class of all A -modules of finite length, i.e., if*

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

is a short exact sequence of A -modules of finite length, then

$$\ell(M_2) = \ell(M_1) + \ell(M_3).$$

Example 5.29. If $A = k$ is a field, the modules of finite length over k coincide with the finite-dimensional k -vector spaces and the function $\ell(\cdot)$ coincides with $\dim_k(\cdot)$.

Chapter 6

Dimension Theory

Proposition 6.1. *Let $A \subseteq B$ be an integral ring extension. Then $\dim A = \dim B$.*

Proof. Let $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$ be a chain of prime ideals in A . By the lying-over theorem (Theorem 4.16) there exists a prime ideal Q_0 such that $Q_0 \cap A = P_0$. Then by the going-up theorem (Theorem 4.17) we get a sequence $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_n$ of prime ideals of B . This shows that $\dim A \leq \dim B$. Now let $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_n$ be a sequence of prime ideals of B . Set $P_i := Q_i \cap A$. By incomparability (Proposition 4.15) we obtain a sequence $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$ of prime ideals of A . This shows $\dim B \leq \dim A$. $\square$

Proposition 6.2. *Let $A \subseteq B$ be an integral extension where A and B are integral domains with A integrally closed. Let $\mathfrak{m}$ be a maximal ideal of B and let $\mathfrak{n} = \mathfrak{m} \cap A$. Then $\mathfrak{n}$ is a maximal ideal in A and $\dim A_{\mathfrak{n}} = \dim B_{\mathfrak{m}}$.*

Proof. By Proposition 4.14 the ideal $\mathfrak{n}$ is maximal. We can lift a chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m = \mathfrak{n}$ in A to a chain $Q_0 \subsetneq \cdots \subsetneq Q_m = \mathfrak{m}$ in B by the going-down theorem. By Proposition 3.27 we have then $\dim A_{\mathfrak{n}} \leq \dim B_{\mathfrak{m}}$. Similarly, if $Q_0 \subsetneq \cdots \subsetneq Q_m = \mathfrak{m}$ is a chain of prime ideals in B , setting $\mathfrak{p}_i := Q_i \cap A$ we get a chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m = \mathfrak{n}$ in A where all inclusions are strict by incomparability (Proposition 4.15). Again by Proposition 3.27 we have $\dim B_{\mathfrak{m}} \leq \dim A_{\mathfrak{n}}$. $\square$

Theorem 6.3. *Let A be a Noetherian local ring. Then $\dim A$ is finite.*

Theorem 6.4. *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Then $\dim A \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$.*

Definition 6.5. Let A be a Noetherian local ring. We say that A is *regular* if $\dim A = \dim_k \mathfrak{m}/\mathfrak{m}^2$.

Definition 6.6. Let A be a ring and let $I \subsetneq A$ be a proper ideal. A *minimal prime ideal* of I is a prime ideal $P \supseteq I$ which is minimal among those prime ideals which contain I .

Remark 6.7. Let A be a ring and let $I \subsetneq A$ be a proper ideal. Consider the set Σ_I of all prime ideals containing I ordered by reverse inclusion $\supseteq$. Since there is at least one maximal ideal $\mathfrak{m}$ such that $I \subseteq \mathfrak{m}$, Σ_I is nonempty. Hence, since the intersection of a chain of prime ideals is a prime ideal, by Zorn's Lemma the ideal I has at least one minimal prime ideal.

Proposition 6.8. *Let A be a Noetherian ring and let $I \subsetneq A$ be a proper ideal. Then the set of minimal prime ideals of I is finite.*

Theorem 6.9 (Krull's Height Theorem). *Let A be a Noetherian ring and let $x_1, \dots, x_r \in A$. Then every minimal prime ideal $\mathfrak{p}$ of $(x_1, \dots, x_r)$ has height $\text{ht}(\mathfrak{p}) \leq r$.*

Corollary 6.10 (Krull's Principal Ideal Theorem). *Let A be a Noetherian ring and let $x \in A$ be a non-unit and a non-zero-divisor. Then every minimal prime ideal $\mathfrak{p}$ of (x) has $\text{ht}(\mathfrak{p}) = 1$.*

Corollary 6.11. *Let A be a Noetherian local ring and let x be an element of $\mathfrak{m}$ which is not a zero-divisor. Then $\dim A/(x) = \dim A - 1$.*

6.1 Noether's Normalization Lemma

Definition 6.12. Let k be a field and let A be a k -algebra. A subset $S \subseteq A$ is called *algebraically independent over k* if for any finite subset $\{x_1, \dots, x_m\} \subseteq S$ there exists no nonzero polynomial $f \in k[T_1, \dots, T_m]$ such that $f(x_1, \dots, x_m) = 0$. In particular, $S = \emptyset$ is algebraically independent over k .

Theorem 6.13 (Noether's Normalization Lemma). *Let k be a field and let A be a finitely generated k -algebra. Then there exist elements $x_1, \dots, x_m$ and $y_1, \dots, y_n$ in A such that $x_1, \dots, x_m$ are algebraically independent over k , $y_1, \dots, y_n$ are integral over $k[x_1, \dots, x_m]$ and $A = k[x_1, \dots, x_m, y_1, \dots, y_n]$.*

Proof. Let $z_1, \dots, z_\ell \in A$ such that $A = k[z_1, \dots, z_\ell]$. The proof is done by induction on ℓ . If $\ell = 0$, then $A = k$ and the statement is trivially true with $n = m = 0$. If $\ell > 0$, we distinguish two cases.

- (1) If $z_1, \dots, z_\ell$ are algebraically independent over k , we are done by choosing $m = \ell$, $n = 0$ and $x_i = z_i$ for all $1 \leq i \leq m$.
- (2) If $z_1, \dots, z_\ell$ are not algebraically independent, there is some nonzero polynomial $f \in k[T_1, \dots, T_\ell]$ such that $f(z_1, \dots, z_\ell) = 0$. This f is of the form

$$f = \sum_{I=(\alpha_1, \dots, \alpha_\ell)} \alpha_I T_1^{\alpha_1} \cdots T_\ell^{\alpha_\ell}, \quad \alpha_I \in k.$$

Let r be a positive integer and for $2 \leq k \leq \ell$ consider $w_k = z_k - z_1^{r^k} \in A$. Then $z_k = w_k + z_1^{r^k}$. Since $f(z_1, \dots, z_\ell) = 0$, we have

$$\begin{aligned} 0 &= f(z_1, w_2 + z_1^{r^2}, \dots, w_\ell + z_1^{r^\ell}) = \sum_I \alpha_I z_1^{\alpha_1} (w_2 + z_1^{r^2})^{\alpha_2} \cdots (w_\ell + z_1^{r^\ell})^{\alpha_\ell} \\ &= g(z_1, w_2, \dots, w_\ell) = g_p(w_2, \dots, w_\ell) z_1^p + \cdots + g_0(w_2, \dots, w_\ell). \end{aligned}$$

For a fixed multi-index I , the summand with the largest degree in z_1 is the term

$$\alpha_I z_1^{\alpha_1 + r^2 \alpha_2 + \cdots + r^\ell \alpha_\ell}.$$

For two distinct multi-indices $I = (\alpha_1, \dots, \alpha_\ell)$ and $J = (\beta_1, \dots, \beta_\ell)$ appearing in the expression of f , denote by S_{IJ} the set of positive integral roots of the polynomial

$$(\alpha_1 - \beta_1) + Y^2(\alpha_2 - \beta_2) + \cdots + Y^\ell(\alpha_\ell - \beta_\ell) \in \mathbb{Z}[Y],$$

and let $S := \bigcup_{I,J} S_{IJ}$. Since $S \subseteq \mathbb{N}$ is a finite set, if we choose $r > \max S$, then for any two distinct multi-indices $I = (\alpha_1, \dots, \alpha_\ell)$ and $J = (\beta_1, \dots, \beta_\ell)$ appearing in the expression of f we have by construction

$$\alpha_1 + r^2 \alpha_2 + \cdots + r^\ell \alpha_\ell \neq \beta_1 + r^2 \beta_2 + \cdots + r^\ell \beta_\ell.$$

This implies that $g_p(w_2, \dots, w_\ell) = \alpha_I \in k$ for some multi-index I . Thus, g_p is a unit and z_1 is integral over $k[w_2, \dots, w_\ell]$, hence for each $k \geq 2$ we have that $z_1^{r^k}$ is integral over $k[w_2, \dots, w_\ell]$. Since each w_k is integral over $k[w_2, \dots, w_\ell]$, then each $z_k = w_k + z_1^{r^k}$ is integral over $k[w_2, \dots, w_\ell]$. Thus, $A = k[z_1, \dots, z_\ell]$ is integral over $A' := k[w_2, \dots, w_\ell]$. By inductive hypothesis, there exist elements $x_1, \dots, x_m \in A'$ algebraically independent over k and $y'_1, \dots, y'_s \in A'$ integral over $k[x_1, \dots, x_m]$ such that $A' = k[x_1, \dots, x_m, y'_1, \dots, y'_s]$. Finally, we have

$$k[x_1, \dots, x_m] \subseteq A' \subseteq A,$$

where A is integral over A' and A' is integral over $k[x_1, \dots, x_m]$. By transitivity of integrality (Proposition 4.9) we have that A is integral over $k[x_1, \dots, x_m]$. Finally, since A is obviously finitely generated over $k[x_1, \dots, x_m]$, there are $y_1, \dots, y_n \in A$ integral over $k[x_1, \dots, x_m]$ such that

$$A = k[x_1, \dots, x_m, y_1, \dots, y_n].$$

□

Definition 6.14. Let $L \subseteq K$ be a field extension.

- (i) A maximal algebraically independent subset of K over L is called a *transcendence basis of $L \subseteq K$* .

- (ii) The *transcendence degree* of $L \subseteq K$, denoted by $\text{tr deg}_L K$, is defined as the cardinality of a transcendence basis of $L \subseteq K$.

Remark 6.15. Let $L \subseteq K$ be a field extension.

- (a) The previous definition is well-posed. Indeed, one shows by using Zorn's Lemma that there exists a transcendence basis of $L \subseteq K$. Moreover, any two transcendence bases have the same cardinality (see for example Chapter VIII of the book "*Algebra*" by Serge Lang). Note that the notions of a transcendence basis and transcendence degree of a field extension are analogous to those of a basis and dimension of a vector space, respectively.
- (b) An algebraically independent subset $S \subseteq K$ is a transcendence basis of $L \subseteq K$ if and only if the field extension $L(S) \subseteq K$ is algebraic. In particular, the field extension $L \subseteq K$ is algebraic if and only if $\text{tr deg}_L K = 0$.
- (c) Let k be a field and let $k(X_1, \dots, X_n)$ be the field of rational functions in the indeterminates $X_1, \dots, X_n$ over k . Then $\text{tr deg}_k k(X_1, \dots, X_n) = n$.

Remark 6.16. Let k be a field and let A be a finitely generated k -algebra.

- (a) The integer $m \geq 0$ in Noether's Normalization Lemma is independent of the choice of x_i and equals $\dim A$. Indeed, applying the theorem we have an integral extension

$$k[x_1, \dots, x_m] \subseteq A,$$

where $k[x_1, \dots, x_m]$ is a polynomial ring over k . If we assume to know that $\dim k[x_1, \dots, x_m] = m$, then $\dim A = m$ by Proposition 6.1.

- (b) If A is an integral domain, then $m = \dim A = \text{tr deg}_k Q(A)$. Indeed, we have a tower of ring extensions

$$k \subseteq k[x_1, \dots, x_m] \subseteq A,$$

where $x_1, \dots, x_m$ are algebraically independent over k and $k[x_1, \dots, x_m] \subseteq A$ is integral. Hence, passing to the fields of fractions we have

$$k \subseteq k(x_1, \dots, x_m) \subseteq Q(A),$$

where $k \subseteq k(x_1, \dots, x_m)$ is purely transcendental (i.e., generated by an algebraically independent set) and $k(x_1, \dots, x_m) \subseteq Q(A)$ is algebraic. Hence, by Remark 6.15(b) we have

$$\text{tr deg}_k Q(A) = \text{tr deg}_k k(x_1, \dots, x_m) = m.$$

The previous remark proves the following theorem.

Theorem 6.17. *Let k be a field and let A be a finitely generated k -algebra. Then A has finite Krull dimension which equals the integer m of Noether's Normalization Lemma (Theorem 6.13). Moreover, if A is also an integral domain, then*

$$\dim A = \text{tr deg}_k Q(A).$$

Chapter 7

Valuation Rings

Definition 7.1. Let R be an integral domain with field of fractions $K = Q(R)$. We say that R is a *valuation ring* of K if for every $x \in K \setminus \{0\}$ we have either $x \in R$ or $x^{-1} \in R$ (or both).

Example 7.2. Let K be a field. Then K is a valuation ring for K .

Proposition 7.3. Let R be a valuation ring for $K = Q(R)$. Then the following hold:

- (i) R is a local ring,
- (ii) For all rings R' such that $R \subseteq R' \subseteq K$, R' is a valuation ring of K ,
- (iii) R is integrally closed in K .

Proof. (i) Let $\mathfrak{m} := R \setminus R^*$ be the set of non-units. We want to show that $\mathfrak{m}$ is an ideal. Thus, let $x \in \mathfrak{m}$, $r \in R$ and suppose by contradiction that $rx \in R^*$. Then there is $a \in R$ such that $a(rx) = (ar)x = 1$, hence $x \in R^*$, a contradiction. Therefore, $rx \in \mathfrak{m}$. Let now $x, y \in \mathfrak{m}$. Since R is a valuation ring of K and $x/y \in K$, we have either $x/y \in R$ or $y/x \in R$. Assume $x/y \in R$. Then $1 + (x/y) = (y + x)/y \in R$, thus by absorption property which we have just shown above we have $x + y = y \cdot (x + y)/y \in \mathfrak{m}$. This shows that $\mathfrak{m}$ is an ideal. Finally, by Proposition 1.28(i) R is a local ring with unique maximal ideal $\mathfrak{m}$.

(ii) Trivial.

(iii) Assume by contradiction that there exists $x \in K \setminus R$ integral over R . Then $1/x \in R$ and x satisfies an equation $x^n + \alpha_{n-1}x^{n-1} + \cdots + \alpha_0 = 0$ with $\alpha_i \in R$. Thus, by multiplying with $(1/x)^{n-1}$ we have

$$x = -\alpha_{n-1} - \alpha_{n-2}\frac{1}{x} - \cdots - \frac{1}{x^{n-1}}\alpha_0 \in R,$$

which is a contradiction. □

Our next goal is to show that there are many valuation rings for any given field.

Let K be a field. For an algebraically closed field L let Σ_L be the set consisting of all pairs (R, f) where $R \subseteq K$ is a subring of K and $f: R \rightarrow L$ is a ring homomorphism. We say

$$(R, f) \leq (R', f') \iff \begin{array}{l} \text{def} \\ R \subseteq R' \text{ is a subring and } f'|_R = f. \end{array}$$

$(\Sigma_L, \leq)$ is a partially ordered set. Assume $\Sigma_L \neq \emptyset$. Then by Zorn's Lemma Σ_L has a maximal element (R, f) .

Lemma 7.4. *R is a local ring with unique maximal ideal $\mathfrak{m} = \ker f$.*

Proof. Let $\mathfrak{m} = \ker f$. Note that $f(R)$ is an integral domain since L is a field, hence the ideal $\mathfrak{m}$ is prime by the first isomorphism theorem. As $f(R \setminus \mathfrak{m}) \subseteq L^*$, the universal property of the localization (Proposition 3.4) gives a ring homomorphism

$$g: R_{\mathfrak{m}} \longrightarrow L, \quad \frac{r}{s} \longmapsto \frac{f(r)}{f(s)}.$$

Since R is an integral domain ($R \subseteq K$), the localization map $R \rightarrow R_{\mathfrak{m}}$ is injective and moreover $Q(R_{\mathfrak{m}}) \cong Q(R) = K$. Thus, by abuse of notation we can write $R \subseteq R_{\mathfrak{m}} \subseteq K$ and $(R_{\mathfrak{m}}, g) \in \Sigma_L$. Because $(R, f) \leq (R_{\mathfrak{m}}, g)$ and (R, f) is maximal in Σ_L , we have $R = R_{\mathfrak{m}}$, and therefore $\mathfrak{m}$ is the unique maximal ideal of R . $\square$

Let $x \in K \setminus \{0\}$. We denote by $\mathfrak{m}[x]$ the extended ideal $\mathfrak{m}R[x]$ in $R[x]$.

Lemma 7.5. *Let $x \in K \setminus \{0\}$. Then at least one of the following holds:*

$$\mathfrak{m}[x] \neq R[x], \quad \mathfrak{m}[1/x] \neq R[1/x].$$

Proof. For any $x \in K \setminus \{0\}$ the elements of $\mathfrak{m}[x]$ are of the form $\sum_i \beta_i x^i$, where $\beta_i \in \mathfrak{m}$. If $\mathfrak{m}[x] = R[x]$, then there are $\gamma_0, \gamma_1, \dots, \gamma_k \in \mathfrak{m}$ such that

$$1 = \gamma_0 + \gamma_1 x + \dots + \gamma_k x^k. \tag{7.1}$$

We assume k to be the minimal integer such that there exists such polynomial expression. On the other hand, if $\mathfrak{m}[1/x] = R[1/x]$, then there are $\delta_0, \delta_1, \dots, \delta_n \in \mathfrak{m}$ such that

$$1 = \delta_0 + \delta_1 \frac{1}{x} + \dots + \delta_n \frac{1}{x^n}. \tag{7.2}$$

As above, we assume n to be minimal. Without loss of generality we have $n \leq k$. Then

$$x^n = \delta_0 x^n + \delta_1 x^{n-1} + \dots + \delta_n \iff (1 - \delta_0)x^n = \delta_1 x^{n-1} + \dots + \delta_{n-1}x + \delta_n.$$

Since $\delta_0 \in \mathfrak{m}$ and $\mathfrak{m}$ is a proper ideal, we deduce $1 - \delta_0 \notin \mathfrak{m}$. By Lemma 7.4, we know that $R \setminus \mathfrak{m}$ is the set of units in R , hence $1 - \delta_0 \in R^*$, and thus we may write

$$x^n = \frac{\delta_1}{1 - \delta_0} x^{n-1} + \cdots + \frac{\delta_{n-1}}{1 - \delta_0} x + \frac{\delta_n}{1 - \delta_0}. \quad (7.3)$$

If we now replace $\gamma_k x^k$ by $\gamma_k x^{k-n} x^n$ in (7.1), then by (7.3) we see that

$$\gamma_k x^k = \gamma_k x^{k-n} \sum_{i=1}^n \delta_i / (1 - \delta_0) x^{n-i},$$

which contradicts the minimality of k . $\square$

Theorem 7.6. *R is a valuation ring for K .*

Proof. Let $x \in K \setminus 0$. We have to show $x \in R$ or $1/x \in R$. Without loss of generality, by Lemma 7.5 we can assume $\mathfrak{m}[x] \neq R[x]$. Then $\mathfrak{m}[x]$ is a proper ideal in $R' := R[x]$ and there is a maximal ideal $\mathfrak{m}' \subseteq R'$ containing $\mathfrak{m}[x]$. Since $\mathfrak{m}'$ is a proper ideal, the intersection $\mathfrak{m}' \cap R$ is an ideal in R containing $\mathfrak{m}$ but not 1, hence $\mathfrak{m}' \cap R = \mathfrak{m}$ since $\mathfrak{m}$ is maximal. Let now $k := R/\mathfrak{m}$ and $k' := R'/\mathfrak{m}'$. It is easy to see that we have the well-defined inclusion

$$k \hookrightarrow k', \quad r + \mathfrak{m} \mapsto r + \mathfrak{m}'.$$

Let $r' = \sum_{i=0}^{\ell} a_i x^i \in R' = R[x]$. Then

$$r' + \mathfrak{m}' = \sum_{i=0}^{\ell} (a_i + \mathfrak{m}')(x + \mathfrak{m}')^i = \sum_{i=0}^{\ell} (a_i + \mathfrak{m}') \bar{x}^i,$$

where $\bar{x} := x + \mathfrak{m}'$. Thus, $k' = k[\bar{x}] = k(\bar{x})$, i.e., $\bar{x}$ is algebraic over k . By 7.4 we have the commutative diagram

$$\begin{array}{ccc} R & \longrightarrow & L \\ & \searrow f & \uparrow g \\ & & k = R/\mathfrak{m} \end{array}$$

and thus $g: k \hookrightarrow L$. Since k' is a finite algebraic extension of k and L is algebraically closed, there exists $g': k' \hookrightarrow L$ which extends g ; hence, we have the diagram

$$R' \twoheadrightarrow R'/\mathfrak{m}' = k' \xrightarrow{g'} L$$

which gives a map $f': R' \rightarrow L$ with $f'|_R = f$. Since (R, f) is a maximal element in Σ_L , we have $R' = R$. In particular, we have $x \in R$. If we assume at the beginning of the proof $\mathfrak{m}[1/x] \neq R[1/x]$, we show in the same way $1/x \in R$. $\square$

Corollary 7.7. *Let R be an integral domain. Then the integral closure of R is the intersection of all valuation rings of $Q(R)$ which contain R .*

Proof. Let $\overline{R}$ be the integral closure of R .

$\subseteq$. Let R' be a valuation ring of $Q(R)$ such that $R \subseteq R'$. Since by Proposition 7.3(iii) R' is integrally closed, we have $\overline{R} \subseteq R'$ and this shows “ $\subseteq$ ”.

$\supseteq$. By contradiction assume there is an element x in the intersection of all valuation rings of $Q(R)$ containing R and $x \notin \overline{R}$. Consider the ring $R' := R[1/x] \subseteq Q(R)$ and observe that $x \notin R'$, otherwise x is a polynomial in $1/x$, say $x = r_n(1/x)^n + \cdots + r_1(1/x) + r_0$, and by multiplying with x^n we have a monic equation for x , a contradiction since $x \in \overline{R}$. Since $x \notin R'$, the element $1/x$ is not a unit in R' , and thus there is some maximal ideal $\mathfrak{m}' \subseteq R'$ such that $1/x \in \mathfrak{m}'$. Denote by L the algebraic closure of the field $K' = R'/\mathfrak{m}'$. Then we have

$$\begin{array}{ccccc} R' & \longrightarrow & R'/\mathfrak{m}' & \hookrightarrow & L \\ & \searrow & & \nearrow & \\ & & f' & & \end{array}$$

Thus, the poset Σ_L is nonempty and there exists a maximal element $(S, g) \in \Sigma_L$ which dominates (R', f') . By Theorem 7.6 the ring S is a valuation ring of $Q(R)$. Because $R' \subseteq S$, we have $1/x \in S$. Assume by contradiction $x \in S$. Then we have $g(1/x) = g|_{R'}(1/x) = f'(1/x) = 0$ by construction, and thus $g(1) = g(x \cdot 1/x) = g(x)g(1/x) = 0$, which is a contradiction. Hence, $x \notin S \supseteq R$, against the assumption that x is in all valuation rings containing R . $\square$

7.1 Discrete Valuation Rings

Definition 7.8. Let K be a field. A function $\nu: K \rightarrow \mathbb{Z} \cup \{+\infty\}$ is called a *discrete valuation* if the following properties hold:

- (i) $\nu(0) = +\infty$,
- (ii) $\nu(K \setminus \{0\}) = \mathbb{Z}$,
- (iii) $\nu(ab) = \nu(a) + \nu(b)$,
- (iv) $\nu(a + b) \geq \min\{\nu(a), \nu(b)\}$.

Let K be a field and let $\nu: K \rightarrow \mathbb{Z} \cup \{+\infty\}$ be a discrete valuation. Then

$$R := \{a \in K \mid \nu(a) \geq 0\}$$

is a subring of K (indeed an integral domain since K is a field), and is called the *valuation ring of ν* . By (iii) of the previous definition, $\nu(1) = 0$ and thus for $x \in R$ we have

$$0 = \nu(1) = \nu\left(x \cdot \frac{1}{x}\right) = \nu(x) + \nu\left(\frac{1}{x}\right),$$

which is equivalent to $\nu(x) = -\nu(1/x)$. Hence, $K = Q(R)$ by Proposition 3.5 and R is a valuation ring of K . By Lemma 7.3(i) R is a local ring. An element $x \in R$ is a unit if and only if $\nu(x) = 0$, hence

$$\mathfrak{m} = \{a \in R \mid \nu(a) \geq 1\}$$

is the maximal ideal of R .

Definition 7.9. A ring R is a *discrete valuation ring (DVR)* if it is an integral domain and there exists a discrete valuation $\nu: Q(R) \rightarrow \mathbb{Z} \cup \{+\infty\}$ such that

$$R = \{a \in Q(R) \mid \nu(a) \geq 0\}.$$

Let R be a discrete valuation ring and let $I \subseteq R$ be an ideal. Let $x \in I$ be an element such that $\nu(x)$ is minimal. Then

$$I = \{y \in R \mid \nu(y) \geq \nu(x)\} = (x).$$

The inclusions $(x) \subseteq I \subseteq \{y \in R \mid \nu(y) \geq \nu(x)\}$ are obvious. As for the inclusion $\{y \in R \mid \nu(y) \geq \nu(x)\} \subseteq (x)$, let $y \in R$ such that $\nu(y) \geq \nu(x)$. Then $\nu(y/x) = \nu(y) - \nu(x) \geq 0$, i.e., $y/x \in R$ and thus $(y/x) \cdot x = y \in (x)$.

Therefore, we have just shown the following proposition.

Proposition 7.10. *Any DVR (discrete valuation ring) is a PID (principle ideal domain).*

Indeed, given a discrete valuation ring R , we know all the ideals of R .

Proposition 7.11. *Let R be a discrete valuation ring and for $k \in \mathbb{N}$ let*

$$I_k := \{y \in R \mid \nu(y) \geq k\}.$$

Then $\{I_k\}_{k \in \mathbb{N}}$ is the family of all nonzero ideals in R which yields a unique maximal chain

$$R = I_0 \supsetneq I_1 \supsetneq I_2 \supsetneq \dots \supsetneq I_k \supsetneq \dots \supsetneq \{0\}. \quad (7.4)$$

Moreover, the unique nonzero prime ideal is the maximal ideal $\mathfrak{m} = I_1$, and for every $k \geq 1$ one has $I_k = \mathfrak{m}^k$. In particular, R is Noetherian and $\dim R = 1$.

Proof. By the discussion above, for every ideal I in R there exists $k \in \mathbb{N}$ such that $I = I_k$. Because $\nu: K \rightarrow \mathbb{Z} \cup \{+\infty\}$ is surjective, we deduce that the inclusions in the chain (7.4) are strict. Moreover, for $k \geq 2$, the ideal I_k is not prime and $I_k = \mathfrak{m}^k$. Indeed, take $a, b \in R$ such that $\nu(a) = k - 1$ and $\nu(b) = 1$. Then $ab \in I_k$, but $a \in I_{k-1} \setminus I_k$ and $b \in \mathfrak{m} \setminus I_k$, and this shows that I_k is not prime. Furthermore, it is easy to see by properties (iii) and (iv) of a discrete valuation that $I_{k+1} \subsetneq \mathfrak{m}^k \subseteq I_k$, and thus, since the chain (7.4) is maximal, we have $I_k = \mathfrak{m}^k$. $\square$

The following proposition gives some useful characterizations of a discrete valuation ring.

Proposition 7.12. *Let R be a Noetherian local integral domain, let $\mathfrak{m} \subseteq R$ be its maximal ideal and let $k = R/\mathfrak{m}$ be its residue field. Assume that $\dim R = 1$. Then the following are equivalent:*

- (i) R is a discrete valuation ring,
- (ii) R is integrally closed,
- (iii) $\mathfrak{m}$ is a principal ideal,
- (iv) $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 1$,
- (v) Every nonzero ideal is a power of $\mathfrak{m}$,
- (vi) There exists $x \in R$ such that every nonzero ideal of R is of the form (x^k) , where k is a natural number.

Remark 7.13. A discrete valuation ring R is by definition an integral domain corresponding to a discrete valuation of its field of fraction $Q(R)$. We could have given the definition of a valuation ring in a similar fashion, as we quickly explain below.

Let $(\Gamma, +)$ be a totally ordered abelian group and let K be a field. A function $\nu: K \rightarrow \Gamma \cup \{+\infty\}$ is called a *valuation of K with values in Γ* if it fulfills properties (i)-(iv) from Definition 7.8. One sees that $R := \{x \in K \mid \nu(x) \geq 0\}$ is a valuation ring of K .

Conversely, if R is a valuation ring of $K = Q(R)$, then the group of units R^* is a subgroup of the multiplicative abelian group K^* and $\Gamma := K^*/R^*$ is a totally ordered abelian group called *the value group of R* . Write it additively. Then the map $\nu: K \rightarrow \Gamma \cup \{+\infty\}$ induced by the canonical homomorphism $K^* \rightarrow \Gamma = K^*/R^*$ is a valuation of K with values in Γ and it holds $R = \{x \in K \mid \nu(x) \geq 0\}$.

Part II

Introduction to Algebraic Geometry

Chapter 8

Affine Varieties and Hilbert's Nullstellensatz

Let k be an algebraically closed field.

Definition 8.1. The *affine n -space over k* is the set of all n -tuples of elements of k , namely k^n . We denote it by $\mathbb{A}_k^n$ (or simply by $\mathbb{A}^n$ when the ground field is clear from the context). An element $p = (a_1, \dots, a_n) \in \mathbb{A}^n$ is called a *point* and a_i is called the i -th coordinate of p .

Let $A := k[X_1, \dots, X_n]$ be the polynomial ring in the indeterminates $X_1, \dots, X_n$ over k . Since k is algebraically closed, it is in particular infinite, hence we can regard each $f \in A$ as a function $f: \mathbb{A}_k^n \rightarrow k$ which sends a point $p = (a_1, \dots, a_n)$ to the evaluation $f(a_1, \dots, a_n)$.

Definition 8.2. Let $f \in A$. We define the *zero-set of f* as

$$Z(f) = \{p \in \mathbb{A}^n \mid f(p) = 0\}.$$

More generally, let $T \subseteq A$ be a subset. The *zero-set of T* is the set

$$Z(T) = \{p \in \mathbb{A}^n \mid f(p) = 0 \text{ for all } f \in T\}.$$

If $T = \{f_1, \dots, f_r\}$, we simply write $Z(f_1, \dots, f_r)$ instead of $Z(\{f_1, \dots, f_m\})$.

Remark 8.3. Let $T \subseteq A$ be a subset. It is easy to see that

$$Z(T) = Z(\langle T \rangle),$$

where

$$\langle T \rangle := \left\{ \sum_{i=1}^r g_i f_i \mid f_i \in T, g_i \in A, r \in \mathbb{N} \right\}$$

is the ideal generated by T in A . Therefore, from now on we will only consider zero-sets of ideals.

Definition 8.4. A subset $X \subseteq \mathbb{A}_k^n$ is called an *algebraic set* (of $\mathbb{A}_k^n$) if there exists an ideal $I \subseteq A$ such that $X = Z(I)$.

Remark 8.5. Let $I \subseteq A$ be an ideal. By Hilbert's basis theorem (Theorem 1.34) $A = k[X_1, \dots, X_n]$ is Noetherian, hence there exist $f_1, \dots, f_r \in A$ such that $I = (f_1, \dots, f_r)$, and thus

$$Z(I) = Z(f_1, \dots, f_r) = \bigcap_{i=1}^r Z(f_i).$$

Proposition 8.6. (i) *The union of two algebraic sets is algebraic.*

(ii) *The intersection of a family of algebraic sets is algebraic.*

(iii) *The empty set $\emptyset$ and $\mathbb{A}_k^n$ are algebraic sets.*

Proof. (i) Let $X_1 = Z(I_1)$ and $X_2 = Z(I_2)$ be algebraic sets, where $I_1, I_2 \subseteq k[X_1, \dots, X_n]$ are ideals. We want to show that

$$X_1 \cup X_2 = Z(I_1 I_2).$$

$\subseteq$. Let $p \in X_1 \cup X_2$, w.l.o.g. $p \in X_1$. Let $f = \sum_{i=1}^m a_i b_i \in I_1 I_2$ with $a_i \in I_1$ and $b_i \in I_2$ for each $1 \leq i \leq m$. Hence, we have

$$f(p) = \sum_{i=1}^m a_i(p) b_i(p) = 0$$

since $a_i(p) = 0$ for each $1 \leq i \leq m$, and this shows that $p \in Z(I_1 I_2)$.

$\supseteq$. Let $p \in Z(I_1 I_2)$ and assume by contradiction $p \notin X_1 \cup X_2$. Then $p \notin X_1$ and $p \notin X_2$, thus there are $f \in I_1$ with $f(p) \neq 0$ and $g \in I_2$ with $g(p) \neq 0$. Then $fg \in I_1 I_2$ and $fg(p) \neq 0$, hence, $p \notin Z(I_1 I_2)$, which is a contradiction.

(ii) If $X_i = Z(J_i)$, $i \in I$, then $\bigcap_{i \in I} X_i = Z(\sum_{i \in I} J_i)$ is an algebraic set.

(iii) $\emptyset = Z(1) = Z(k[X_1, \dots, X_n])$ and $\mathbb{A}_k^n = Z(0)$. □

The previous proposition shows that the family of algebraic sets satisfies the defining properties of the closed sets of a topology. Therefore, there exists a unique topology on $\mathbb{A}_k^n$ whose closed sets are precisely the algebraic sets. This justifies the following definition.

Definition 8.7. The *Zariski topology* on $\mathbb{A}_k^n$ is the topology whose open sets are complements of algebraic sets.

From now on, we consider $\mathbb{A}_k^n$ as a topological space with respect to the Zariski topology.

Example 8.8. The Zariski topology on $\mathbb{A}^1 = k$ is the cofinite topology, i.e., the topology whose open sets are the empty set and complements of finite sets. Indeed, an algebraic set $X \subseteq \mathbb{A}^1$ is of the form $X = Z(f)$ (since $k[X]$ is a PID), hence X is finite since the zeroes of a polynomial $f \in k[X]$ are finitely many.

Definition 8.9. Let Y be a nonempty topological space. We say that Y is *irreducible* if it cannot be written as the union of two proper closed subsets. More precisely, whenever

$$Y = Y_1 \cup Y_2,$$

with $Y_1, Y_2 \subseteq Y$ closed in Y , it follows that either $Y = Y_1$ or $Y = Y_2$ (or both).

A nonempty subset $X \subseteq Y$ is called *irreducible* if it is irreducible as a topological space with respect to the subspace topology.

By convention, the empty set is not irreducible.

Example 8.10. The affine space $\mathbb{A}_k^1$ is irreducible since the nonempty closed sets are the finite sets and $\mathbb{A}_k^1$ itself.

Proposition 8.11. *Let Y be a topological space and let $X \subseteq Y$ be a nonempty subset endowed with the subspace topology.*

(a) *If X is irreducible, then its closure in Y*

$$\overline{X}^Y := \bigcap \{ C \subseteq Y \mid C \text{ is closed in } Y \text{ and } X \subseteq C \}$$

is also irreducible.

(b) *The following statements are equivalent:*

- (i) *Y is irreducible.*
- (ii) *Every nonempty open subset $W \subseteq Y$ is dense in Y , i.e., $\overline{W}^Y = Y$.*
- (iii) *For every pair of nonempty open subsets $U, V \subseteq Y$ it holds $U \cap V \neq \emptyset$.*

(c) *If Y is irreducible, then every nonempty open subset of Y is irreducible.*

Proof. Exercise. □

Definition 8.12. An *affine variety* is an irreducible algebraic set of $\mathbb{A}_k^n$. A *quasi-affine variety* is an open subset of an affine variety.

Before proceeding further, we need to investigate more carefully the relationship between algebraic sets of $\mathbb{A}_k^n$ and ideals of $A = k[X_1, \dots, X_n]$. This will naturally lead us to Hilbert's Nullstellensatz.

8.1 Hilbert's Nullstellensatz

Definition 8.13. Let $S \subseteq \mathbb{A}_k^n$ be a subset. We define

$$I(S) := \{f \in A \mid f(p) = 0 \text{ for all } p \in S\}.$$

It is easy to see that $I(S)$ is a radical ideal in $A = k[X_1, \dots, X_n]$. Therefore, we have two functions:

$$\begin{array}{ccc} & I & \\ \swarrow & & \searrow \\ \left\{ \text{ideals in } A = k[X_1, \dots, X_n] \right\} & & \left\{ \text{algebraic sets of } \mathbb{A}_k^n \right\} \\ \searrow & & \swarrow \\ & Z & \end{array}$$

We might hope that $I(\cdot)$ and $Z(\cdot)$ are inverse to each other, but in general this is not the case, as the following remark explains.

Remark 8.14. Let $J \subseteq A$ be an ideal. By definition of $Z(J)$, it holds $I(Z(J)) \supseteq J$, but in general this inclusion is strict. For example, if $J = (X^2) \subseteq k[X]$, then $Z(J) = \{0\}$, but

$$I(Z(J)) = I(\{0\}) = (X) \supsetneq (X^2) = J.$$

On the other hand, since $I(Z(J))$ is a radical ideal, we have $I(Z(J)) \supseteq \sqrt{J}$.

The previous remark leads to the following natural question: given an ideal $J \subseteq A$, what is $I(Z(J))$? This is the content of Hilbert's Nullstellensatz.

Theorem 8.15 (Hilbert's Nullstellensatz). *Let k be an algebraically closed field. Let $J \subseteq k[X_1, \dots, X_n]$ be an ideal. A polynomial $g \in k[X_1, \dots, X_n]$ vanishes along $Z(J)$ if and only if $g^m \in J$ for some $m \in \mathbb{N}$. In other words, it holds $I(Z(J)) = \sqrt{J}$.*

Hilbert's Nullstellensatz follows from the following theorem.

Theorem 8.16 (Weak Nullstellensatz). *Let k be an algebraically closed field.*

(i) *An ideal $\mathfrak{m} \subseteq k[X_1, \dots, X_n]$ is maximal if and only if*

$$\mathfrak{m} = (X_1 - a_1, \dots, X_m - a_m)$$

for some $a_i \in k$, $1 \leq i \leq m$. Furthermore, we have $k[X_1, \dots, X_n]/\mathfrak{m} \cong k$.

(ii) *Let $I \subseteq k[X_1, \dots, X_n]$ be an ideal. If $Z(I) = \emptyset$, then $I = k[X_1, \dots, X_n]$.*

Remark 8.17. The hypothesis that k is algebraically closed is fundamental for both (i) and (ii) of the weak Nullstellensatz. Indeed, if $k = \mathbb{R}$ the ideal $(X^2 + 1) \subseteq \mathbb{R}[X]$ is maximal and $\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$. Moreover, $Z(X^2 + 1) = \emptyset$ even though $(X^2 + 1)$ is a proper ideal in $\mathbb{R}[X]$.

Lemma 8.18. *The weak Nullstellensatz implies Hilbert's Nullstellensatz.*

Proof. Assume the weak Nullstellensatz holds true. Let $J \subseteq k[X_1, \dots, X_n]$ be an ideal. By Hilbert's basis theorem, there are $f_1, \dots, f_m \in k[X_1, \dots, X_n]$ such that $J = (f_1, \dots, f_m)$. Let $g \in k[X_1, \dots, X_n]$ be a polynomial vanishing along $Z(J)$. We have to show that there exists $k \in \mathbb{N}$ such that $g^k \in J$. To do so, we use the so-called *Rabinowitsch trick*, that is, we introduce a new variable X_{n+1} . Consider the polynomials $f_1, \dots, f_m, gX_{n+1} - 1$ as elements of $k[X_1, \dots, X_n, X_{n+1}]$. Then $Y := Z(f_1, \dots, f_m, gX_{n+1} - 1) = \emptyset$. Assume by contradiction there is $p = (a_1, \dots, a_n, a_{n+1}) \in Y \subseteq \mathbb{A}_k^{n+1}$. This means $f_i(a_1, \dots, a_n) = 0$ for $1 \leq i \leq m$, thus $(a_1, \dots, a_n) \in Z(J)$. Hence, since g vanishes along $Z(J)$, we have $g(a_1, \dots, a_n) = 0$. But on the other hand, we have

$$0 = g(a_1, \dots, a_n)a_{n+1} - 1 = 0 \cdot a_{n+1} - 1,$$

a contradiction. This shows $Z(f_1, \dots, f_m, gX_{n+1} - 1) = \emptyset$. By (ii) of the weak Nullstellensatz, $(f_1, \dots, f_m, gX_{n+1} - 1) = k[X_1, \dots, X_{n+1}]$, thus there are polynomials $h_1, \dots, h_m, h \in k[X_1, \dots, X_{n+1}]$ such that

$$f_1 h_1 + \dots + f_m h_m + (gX_{n+1} - 1)h = 1. \quad (8.1)$$

If we replace X_{n+1} in (8.1) by $\frac{1}{g}$, then we get an equality in $k(X_1, \dots, X_n)$, namely

$$f_1(X_1, \dots, X_n)h_1\left(X_1, \dots, X_n, \frac{1}{g}\right) + \dots + f_m(X_1, \dots, X_n)h_m\left(X_1, \dots, X_n, \frac{1}{g}\right) = 1. \quad (8.2)$$

Now for each $1 \leq i \leq m$ there exist $p_i \in k[X_1, \dots, X_n]$ and $k_i \in \mathbb{N}$ such that

$$h_i\left(X_1, \dots, X_n, \frac{1}{g}\right) = \frac{p_i(X_1, \dots, X_n)}{g^{k_i}}.$$

Let $k \geq \max(k_1, \dots, k_m)$ be a positive integer and for each $1 \leq i \leq m$ let $\tilde{h}_i = g^{k-k_i}p_i$. By multiplying (8.2) with g^k , we finally obtain

$$g^k = f_1(X_1, \dots, X_n)\tilde{h}_1(X_1, \dots, X_n) + \dots + f_m(X_1, \dots, X_n)\tilde{h}_m(X_1, \dots, X_n) \in J.$$

□

In order to prove the weak Nullstellensatz, we need first the following proposition, which is a direct consequence of Artin-Tate Lemma (Lemma 5.18).

Proposition 8.19. *Let k be a field and let E be a finitely generated k -algebra which is a field. Then the field extension $k \subseteq E$ is algebraic.*

Proof. Let $E = k[x_1, \dots, x_n]$ and assume by contradiction that E is not algebraic over k . Then, up to relabeling the elements $x_1, \dots, x_n$, we see inductively that $x_1, \dots, x_r$ are algebraically independent over k for some $1 \leq r \leq n$ and $x_{r+1}, \dots, x_n$

are algebraic over $F := k(x_1, \dots, x_r)$, equivalently, $F \subseteq E$ is a finite field extension. Then we have a tower of extensions

$$k \subseteq F \subseteq E$$

where E is finitely generated as a k -algebra and finitely generated as a module over F . Hence, by Artin-Tate Lemma (Lemma 5.18) there are $y_1, \dots, y_s \in F$ such that

$$F = k[y_1, \dots, y_s].$$

Note that $k[x_1, \dots, x_r]$ is a polynomial ring since $x_1, \dots, x_r$ are algebraically independent over k , hence F is the field of rational functions in the r variables $x_1, \dots, x_r$ over k . Therefore, for each $1 \leq i \leq s$ we can write

$$y_i = f_i(x_1, \dots, x_r) / g_i(x_1, \dots, x_r)$$

with $f_i, g_i \in k[x_1, \dots, x_r]$ such that $\gcd(f_i, g_i) = 1$ for each $1 \leq i \leq s$ and at least one of the g_i 's is a non-constant polynomial (otherwise $F = k[y_1, \dots, y_s] \subseteq k[x_1, \dots, x_r]$). Let $h := g_1 \cdots g_s + 1 \in k[x_1, \dots, x_r]$. By construction, h is a non-constant polynomial and $\gcd(h, g_i) = 1$ for every $i \in \{1, \dots, s\}$.

Claim 1. $1/h \notin k[y_1, \dots, y_s]$.

Proof (Claim). Assume by contradiction that there exists a polynomial

$$H(T_1, \dots, T_s) = \sum_{I=(i_1, \dots, i_s)} \alpha_I T_1^{i_1} \cdots T_s^{i_s} \in k[T_1, \dots, T_s]$$

such that $1/h = H(y_1, \dots, y_s)$. Then

$$\frac{1}{h} = \sum_{I=(i_1, \dots, i_s)} \alpha_I \left(\frac{f_1}{g_1}\right)^{i_1} \cdots \left(\frac{f_s}{g_s}\right)^{i_s} = \frac{G}{g_1^{l_1} \cdots g_s^{l_s}} \quad (8.3)$$

for some $G \in k[x_1, \dots, x_r]$ and some integers $l_1, \dots, l_s$. Note that $g_1^{l_1} \cdots g_s^{l_s} \nmid G$, otherwise we would have $1 = hG'$ for some $G' \in k[x_1, \dots, x_r]$, a contradiction since h is a non-constant polynomial. Let $D := \gcd(G, g_1^{l_1} \cdots g_s^{l_s})$ and write

$$G = D\tilde{G} \quad \text{and} \quad g_1^{l_1} \cdots g_s^{l_s} = D \cdot p_1^{m_1} \cdots p_t^{m_t}, \quad (8.4)$$

where $p_1, \dots, p_t$ are (pairwise distinct) irreducible polynomials such that $p_j \nmid \tilde{G}$ for each $1 \leq j \leq t$. Thus, by substituting (8.4) in (8.3) we have

$$\frac{1}{h} = \frac{\tilde{G}}{p_1^{m_1} \cdots p_t^{m_t}} \iff p_1^{m_1} \cdots p_t^{m_t} = h\tilde{G}.$$

Finally, each p_j divides the first member of the last equality, hence $p_j \mid h$ since $p_j \nmid \tilde{G}$ and p_j is prime. As $p_j \mid g_{k_j}$ for some $1 \leq k_j \leq s$, we have $p_j \mid \gcd(h, g_{k_j})$, a contradiction since $\gcd(h, g_i) = 1$ for every $1 \leq i \leq s$. $\square$

Since $1/h \in F = k(x_1, \dots, x_r)$ and $F = k[y_1, \dots, y_s]$, the claim yields right away a contradiction, and therefore E is algebraic over k . $\square$

Corollary 8.20. *Let k be an algebraically closed field and let E be a finitely generated k -algebra which is a field. Then $E = k$.*

Proof. By applying Proposition 8.19 we have that $k \subseteq E$ is an algebraic field extension. Since an algebraically closed field has no proper algebraic extensions, we have $E = k$. $\square$

Now we are ready to prove the weak Nullstellensatz.

Proof of Theorem 8.16. (i) Let $\mathfrak{m} := (X_1 - a_1, \dots, X_n - a_n) \subseteq k[X_1, \dots, X_n]$ be an ideal for some $a_i \in k$, $1 \leq i \leq n$. We show that $\mathfrak{m}$ is maximal. Indeed, let $I \supsetneq \mathfrak{m}$ be an ideal and let $f \in I \setminus \mathfrak{m}$. By the Euclidean algorithm applied to f and $X_1 - a_1$ in $k(X_2, \dots, X_n)(X_1)$, since the leading coefficient of $X_1 - a_1$ is an element of k , we can write

$$f = (X_1 - a_1)q_1(X_1, \dots, X_n) + r_1(X_1, \dots, X_n),$$

where $q_1, r_1 \in k[X_1, \dots, X_n]$ and $\deg_{X_1} r_1 < \deg_{X_1}(X_1 - a_1) = 1$. Thus, r_1 is a polynomial in $k[X_2, \dots, X_n]$ and $r_1 = f - (X_1 - a_1)q_1 \in I \setminus \mathfrak{m}$. By applying the Euclidean algorithm to r_1 and $X_2 - a_2$ in $k(X_3, \dots, X_n)(X_2)$, again since the leading coefficient of $X_2 - a_2$ is an element of k , we can write

$$r_1 = (X_2 - a_2)q_2(X_2, \dots, X_n) + r_2(X_2, \dots, X_n),$$

where $q_2, r_2 \in k[X_2, \dots, X_n]$ and $\deg_{X_2} r_2 < \deg_{X_2}(X_2 - a_2) = 1$. Thus, r_2 is a polynomial in $k[X_3, \dots, X_n]$ and $r_2 = r_1 - (X_2 - a_2)q_2 \in I \setminus \mathfrak{m}$. After n steps, we find a constant polynomial $r_n \in k$ such that $r_n \in I \setminus \mathfrak{m}$, hence r_n is a unit and $I = k[X_1, \dots, X_n]$.

Let now $\mathfrak{m}$ be a maximal ideal of $k[X_1, \dots, X_n]$ and consider the field $K := k[X_1, \dots, X_n]/\mathfrak{m}$. Let $\pi: k[X_1, \dots, X_n] \twoheadrightarrow K$ be the canonical projection and for each $1 \leq i \leq n$ let $\bar{X}_i := \pi(X_i) = X_i + \mathfrak{m}$. Then $K = k[\bar{X}_1, \dots, \bar{X}_n]$ is a finitely generated k -algebra which is a field, hence by Corollary 8.20 $K = k$ and for each $1 \leq i \leq n$ we have $\bar{X}_i = a_i \in k$. Hence, $X_i - a_i \in \ker \pi = \mathfrak{m}$, and thus

$$(X_1 - a_1, \dots, X_n - a_n) \subseteq \mathfrak{m}.$$

Since we showed above that $(X_1 - a_1, \dots, X_n - a_n)$ is maximal, we have $\mathfrak{m} = (X_1 - a_1, \dots, X_n - a_n)$ and we are done.

(ii) Let $I \subseteq k[X_1, \dots, X_n]$ be an ideal such that $Z(I) = \emptyset$ and assume that I is proper. Then there exists a maximal ideal $\mathfrak{m}$ containing I . Hence, by (i) $\mathfrak{m} = (X_1 - a_1, \dots, X_n - a_n)$ and

$$(a_1, \dots, a_n) \in Z(\mathfrak{m}) \subseteq Z(I),$$

which is a contradiction. $\square$

Proposition 8.21. *Let k be algebraically closed. The following statements hold:*

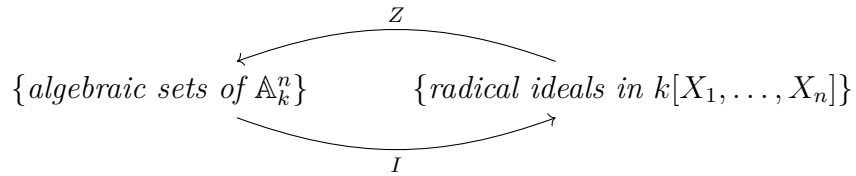
- (i) *Let X_1, X_2 be subsets of $\mathbb{A}_k^n$. Then $X_1 \subseteq X_2$ implies $I(X_1) \supseteq I(X_2)$.*
- (ii) *Let J_1, J_2 be ideals of $k[X_1, \dots, X_n]$. Then $J_1 \subseteq J_2$ implies $Z(J_1) \supseteq Z(J_2)$.*
- (iii) *For any two subsets $S_1, S_2 \subseteq \mathbb{A}_k^n$ we have $I(S_1 \cup S_2) = I(S_1) \cap I(S_2)$.*
- (iv) *For any ideal $J \subseteq k[X_1, \dots, X_n]$ we have $I(Z(J)) = \sqrt{J}$.*
- (v) *Let $J \subseteq k[X_1, \dots, X_n]$ be an ideal. Then $Z(J) = Z(\sqrt{J})$.*
- (vi) *For any subset $X \subseteq \mathbb{A}_k^n$, we have $Z(I(X)) = \overline{X}^{\mathbb{A}_k^n}$.*

Proof. Statements (i), (ii), (iii) are direct consequences of the definitions, statement (iv) follows from Hilbert's Nullstellensatz.

We show (v). Let $J \subseteq k[X_1, \dots, X_n]$ be an ideal. Since $J \subseteq \sqrt{J}$, we have by (ii) $Z(J) \supseteq Z(\sqrt{J})$. On the other hand, if $p \in Z(J)$, then $f(p) = 0$ for all $f \in J$. If $g \in \sqrt{J}$, then there exists $k \in \mathbb{N}$ such that $g^k \in J$. Thus, $g^k(p) = g(p)^k = 0$, whence $g(p) = 0$. Since $g \in \sqrt{J}$ is arbitrary, we have $p \in Z(\sqrt{J})$, and hence it holds $Z(J) \subseteq Z(\sqrt{J})$ as well.

Now we show (vi). Let $X \subseteq \mathbb{A}_k^n$ be a subset and let $\overline{X}^{\mathbb{A}_k^n}$ denote the closure of X in $\mathbb{A}_k^n$. Observe that $Z(I(X))$ is a closed set of $\mathbb{A}_k^n$ such that $X \subseteq Z(I(X))$, hence $\overline{X}^{\mathbb{A}_k^n} \subseteq Z(I(X))$. Conversely, let W be a closed subset of $\mathbb{A}_k^n$ such that $X \subseteq W$. Since W is closed, there is an ideal J such that $W = Z(J)$ and then $X \subseteq W = Z(J)$. Thus, by (i) we get $I(X) \supseteq I(Z(J))$. On the other hand, $J \subseteq I(Z(J))$, so we have $J \subseteq I(X)$ and by (ii) it holds $W = Z(J) \supseteq Z(I(X))$. Since this holds for any closed subset $W \subseteq \mathbb{A}_k^n$ is an arbitrary closed set containing X , we have $Z(I(X)) \subseteq \overline{X}^{\mathbb{A}_k^n}$. $\square$

Corollary 8.22. (i) *There are one-to-one inclusion-reversing correspondences*



- (ii) *Let $X \subseteq \mathbb{A}_k^n$ be a nonempty subset. Then X is irreducible if and only if $I(X)$ is prime.*

Proof. (i) Let $X = Z(J) \subseteq \mathbb{A}_k^n$ be an algebraic set. Then

$$Z(I(X)) = Z(I(Z(J))) = Z(\sqrt{J}) = Z(J) = X,$$

where the second last equality is given by Hilbert's Nullstellensatz and the last one follows by (v) from Proposition 8.21.

Let now $J = \sqrt{J}$ be a radical ideal in $k[X_1, \dots, X_n]$. Then

$$I(Z(J)) = \sqrt{J} = J$$

by Hilbert's Nullstellensatz. Hence, $I(\cdot)$ and $Z(\cdot)$ are one-to-one correspondences which are inverse to each other. By (i) and (ii) from Proposition 8.21 it follows that they are inclusion-reversing.

(ii) $\Rightarrow$. Assume X is irreducible and let $f, g \in k[X_1, \dots, X_n]$ such that $fg \in I(X)$. Then $(fg) \subseteq I(X)$, and thus by (ii) from Proposition 8.21 we have $Z(f) \cup Z(g) = Z(fg) \supseteq Z(I(X)) \supseteq X$. Hence, $X \subseteq Z(f) \cup Z(g)$, i.e.,

$$X = (X \cap Z(f)) \cup (X \cap Z(g)),$$

where both sets on the right-hand side are closed in X . Thus, because X is irreducible, it holds

$$X = X \cap Z(f) \quad \text{or} \quad X = X \cap Z(g),$$

i.e., $X \subseteq Z(f)$ or $X \subseteq Z(g)$. Therefore, $f \in I(X)$ or $g \in I(X)$, which shows that $I(X)$ is prime.

$\Leftarrow$. Assume $I(X)$ is prime and let $Z(J_1)$ and $Z(J_2)$ be algebraic sets of $\mathbb{A}_k^n$ such that

$$X \subseteq Z(J_1) \cup Z(J_2).$$

If we show that either $X \subseteq Z(J_1)$ or $X \subseteq Z(J_2)$ (or both), then X is irreducible and we are done. We can assume by (v) from Proposition 8.21 that J_1 and J_2 are radical ideals of $k[X_1, \dots, X_n]$. By (i), (iii) and (iv) from Proposition 8.21 we have

$$I(X) \supseteq I(Z(J_1) \cup Z(J_2)) = I(Z(J_1)) \cap I(Z(J_2)) = J_1 \cap J_2.$$

Since $I(X)$ is prime, by (ii) from Proposition 1.19 we have

$$I(X) \supseteq J_1 \quad \text{or} \quad I(X) \supseteq J_2.$$

Hence, by (ii) from Proposition 1.19 we have

$$X \subseteq Z(I(X)) \subseteq Z(J_1) \quad \text{or} \quad X \subseteq Z(I(X)) \subseteq Z(J_2),$$

which shows that X is irreducible. □

Example 8.23. (a) $\mathbb{A}_k^n$ is irreducible. Indeed, $I(\mathbb{A}_k^n) = \{0\}$ (see Exercise 4 of Exercise Sheet 1 of the course “Algebra (WS 2025/26)”), and thus it is prime.

(b) Let $f \in k[X_1, \dots, X_n]$ be an irreducible polynomial. Then (f) is prime. Indeed, if $gh \in (f)$, then $f \mid gh$. Since $k[X_1, \dots, X_n]$ is a unique factorization domain, every irreducible element is prime, hence $f \mid g$ or $f \mid h$, i.e., $g \in (f)$ or $h \in (f)$. Hence, by Corollary 8.22 $Z(f)$ is an affine variety. We say that $Z(f)$ is a *curve* if $n = 2$, a *surface* if $n = 3$ and a *hypersurface* if $n \geq 4$. The *degree* of $Z(f)$ is defined to be the degree of $f \in k[X_1, \dots, X_n]$.

Definition 8.24. Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set. We define the *affine coordinate ring* of X as

$$A(X) := k[X_1, \dots, X_n]/I(X).$$

Proposition 8.25. (i) If $X \subseteq \mathbb{A}_k^n$ is an affine variety, then its coordinate ring $A(X)$ is an integral domain which is finitely generated as a k -algebra.

(ii) Let R be an integral domain which is finitely generated as a k -algebra. Then there exists an affine variety X such that $R \cong A(X)$.

Proof. (i) If X is an affine variety, $I(X)$ is prime and thus $A(X)$ is an integral domain. Furthermore, if $\overline{X}_i$ is the class of X_i modulo $I(X)$, then $A(X) = k[X_1, \dots, X_n]/I(X) = k[\overline{X}_1, \dots, \overline{X}_n]$, and hence it is a finitely generated k -algebra.

(ii) Assume that R is generated as a k -algebra by $x_1, \dots, x_r \in R$, namely $R = k[x_1, \dots, x_r]$. Then there is a surjection $\varphi: k[X_1, \dots, X_r] \rightarrow R$, and thus $k[X_1, \dots, X_r]/J \cong R$ where $J := \ker(\varphi)$ is prime since R is an integral domain. Let $X := Z(J) \subseteq \mathbb{A}_k^r$. Since J is prime, hence radical, by Hilbert’s Nullstellensatz $I(X) = J$, and therefore by Corollary 8.22 X is an affine variety such that $A(X) \cong R$.

□

8.2 Irreducible Components

Definition 8.26. A topological space X is called *Noetherian* if the set of its closed subsets, ordered via inclusion, satisfies the DCC, or equivalently, if every nonempty family of closed sets has a minimal element (see Lemma 5.2).

Example 8.27. The affine space $\mathbb{A}_k^n$ is Noetherian. Indeed, if $X_1 \supseteq X_2 \supseteq X_3 \supseteq \dots$ is a descending chain of closed subsets of $\mathbb{A}_k^n$, then $I(X_1) \subseteq I(X_2) \subseteq I(X_3) \subseteq \dots$ is an ascending chain of ideals in $k[X_1, \dots, X_n]$, which stabilizes since $k[X_1, \dots, X_n]$ is a Noetherian ring. Hence, since $Z(I(X_n)) = X_n$ for every $n \in \mathbb{N}$, the descending chain of closed sets stabilizes as well.

Proposition 8.28. *Let X be a Noetherian topological space. Then every closed subset $Y \subseteq X$ can be written as a finite union of irreducible closed subsets, say $Y = \bigcup_{i=1}^n Y_i$. Furthermore, if we assume that $Y_i \not\subseteq Y_j$ for $i \neq j$, then this decomposition is unique (up to permutation of the factors). In this case, the irreducible closed subsets $Y_1, \dots, Y_n$ are called the irreducible components of Y .*

Proof. Let Σ be the set of all closed subsets of X which cannot be written as a finite union of irreducible closed subsets of X . We want to show that $\Sigma = \emptyset$.

Assume by contradiction $\Sigma \neq \emptyset$. Then, since X is Noetherian, Σ has a minimal element Z . Observe that Z is not irreducible since $Z \in \Sigma$, so by definition there are proper closed subsets $Z_1, Z_2 \subsetneq Z$ such that $Z = Z_1 \cup Z_2$. Since $Z_1, Z_2 \subsetneq Z$, we have $Z_1, Z_2 \notin \Sigma$. Thus, there are irreducible closed subsets $Z_{1,1}, \dots, Z_{1,p}$ and $Z_{2,1}, \dots, Z_{2,q}$ such that

$$Z_1 = \bigcup_{i=1}^p Z_{1,i} \quad \text{and} \quad Z_2 = \bigcup_{j=1}^q Z_{2,j}.$$

But we have then

$$Z = Z_1 \cup Z_2 = \bigcup_{i=1}^p Z_{1,i} \cup \bigcup_{j=1}^q Z_{2,j},$$

a contradiction since $Z \in \Sigma$. Hence, $\Sigma = \emptyset$.

Now assume we have two decompositions into irreducible components, i.e.,

$$Y = Y_1 \cup \dots \cup Y_p \quad \text{and} \quad Y = Y'_1 \cup \dots \cup Y'_q,$$

where for $i \neq j$ we have $Y_i \not\subseteq Y_j$ and $Y'_i \not\subseteq Y'_j$. Consider Y_1 and observe that $Y_1 \subseteq Y = Y'_1 \cup \dots \cup Y'_q$, in other words

$$Y_1 = (Y_1 \cap Y'_1) \cup \dots \cup (Y_1 \cap Y'_q),$$

and since Y_1 is irreducible there exists $1 \leq i \leq q$ such that $Y_1 = Y_1 \cap Y'_i$, i.e., $Y_1 \subseteq Y'_i$. Up to permuting $Y'_1, \dots, Y'_q$, we may assume $i = 1$. Now consider Y'_1 . Similarly as for Y_1 , there exists $1 \leq j \leq p$ such that $Y'_1 \subseteq Y_j$, and thus we have $Y_1 \subseteq Y'_1 \subseteq Y_j$. But $Y_1 \subseteq Y_j$ implies $j = 1$, and hence $Y_1 = Y'_1$. Then consider $W = Y_2 \cup \dots \cup Y_p$ and $W = Y'_2 \cup \dots \cup Y'_p$. Applying the same argument as above, we obtain $Y_2 = Y'_2$. Inductively, we find out that $p = q$ and (up to permuting $Y'_1, \dots, Y'_q$) $Y_i = Y'_i$ for all $1 \leq i \leq p$. $\square$

Corollary 8.29. *Each algebraic set X of $\mathbb{A}_k^n$ can be written as a finite union of affine varieties, no one containing another. Namely, $X = \bigcup_{i=1}^m X_i$ with $X_i \not\subseteq X_j$ for $i \neq j$. The affine varieties $X_1, \dots, X_m$ are called the irreducible components of the algebraic set X .*

Example 8.30. Let $X = \mathbb{A}_k^n$ and let f, g, h be three irreducible polynomials in $k[X_1, \dots, X_n]$. Then

$$Z(fgh) = Z(f) \cup Z(g) \cup Z(h)$$

and these are the irreducible components of $Z(fgh)$.

Definition 8.31. (i) Let X be a topological space. The *dimension of X* , denoted by $\dim X$, is the supremum of all $m \in \mathbb{N}$ such that there exists a strict chain $Z_0 \subsetneq Z_1 \subsetneq \dots \subsetneq Z_m$ of irreducible closed subsets of X .

(ii) The dimension of an algebraic set $Y \subseteq \mathbb{A}_k^n$ is its dimension as a topological space (with the subspace topology induced by the Zariski topology on $\mathbb{A}_k^n$).

Example 8.32. In $\mathbb{A}_k^1$ the irreducible closed subsets are $\mathbb{A}_k^1$ itself and the points. Hence, $\dim \mathbb{A}_k^1 = 1$.

Proposition 8.33. Let $Y \subseteq \mathbb{A}_k^n$ be an algebraic set. Then $\dim(Y) = \dim(A(Y))$.

Proof. Let $Z_0 \subsetneq Z_1 \subsetneq \dots \subsetneq Z_r$ be a chain of distinct irreducible closed subsets of Y . Since Y is algebraic, each Z_i is algebraic as well and $I(Z_0) \supsetneq I(Z_1) \supsetneq \dots \supsetneq I(Z_r)$ is a chain of distinct prime ideals of $A = k[X_1, \dots, X_n]$ where $I(Y) \subseteq I(Z_r)$. Hence, by the one-to-one inclusion-preserving correspondence between prime ideals of A containing $I(Y)$ and prime ideals of $A(Y)$, we obtain right away a chain of distinct prime ideals of length r in $A(Y)$. This shows $\dim(Y) \leq \dim A(Y)$.

Let now $P_0 \subsetneq P_1 \subsetneq \dots \subsetneq P_s$ be a chain of distinct prime ideals of A where $I(Y) \subseteq P_0$. Then $Z(P_0) \supsetneq Z(P_1) \supsetneq \dots \supsetneq Z(P_s)$ is a chain of distinct irreducible closed subsets of Y . This shows $\dim(A(Y)) \leq \dim Y$. $\square$

Corollary 8.34. The dimension of the affine n -space $\mathbb{A}_k^n$ over k is n .

Proof. By Proposition 8.33 we have $\dim \mathbb{A}_k^n = \dim k[X_1, \dots, X_n] = n$. $\square$

Theorem 8.35. Let k be a field and let A be an integral domain which is finitely generated as a k -algebra. Then, for any prime ideal $P \subseteq A$, it holds

$$\text{ht}(P) + \dim A/P = \dim A.$$

Remark 8.36. Note that in the previous theorem $\text{ht}(P)$, $\dim A/P$ and $\dim A$ are integers, and hence it makes sense to wonder if such an equality is true. Indeed, $\text{ht}(P) = \dim A_P$ which is finite by Theorem 6.3. Moreover, since A and A/P are finitely generated k -algebra, we have that $\dim A$ and $\dim A/P$ are finite by Theorem 6.17.

Corollary 8.37. Let X be an affine variety. Then $\dim X < \infty$.

Proof. By Proposition 8.33 we get $\dim X = \dim A(X)$. Since $A(X) = k[X_1, \dots, X_n]/P$, where P is a prime ideal of $k[X_1, \dots, X_n]$, we obtain by Theorem 8.35

$$\dim X = \dim A(X) = \dim k[X_1, \dots, X_n] - \text{ht } P \leq n.$$

$\square$

Lemma 8.38. Let X be a topological space and let $Z \subseteq Y \subseteq X$ be subsets.

$$(i) \quad \overline{Z}^X \cap Y = \overline{Z}^Y.$$

$$(ii) \quad \text{Assume that } Y \text{ is closed in } X. \text{ Then } \overline{Z}^Y = \overline{Z}^X.$$

Proof. Exercise. □

Lemma 8.39. *Let X be a topological space and let $U \subseteq X$ be a nonempty open subset. Let Z be an irreducible closed subset of X and suppose that $Z \cap U \neq \emptyset$. Then it holds $\overline{Z} \cap \overline{U}^X = Z$.*

Proof. Exercise. □

Proposition 8.40. *Let $Y \subseteq \mathbb{A}_k^n$ be a quasi-affine variety and let $\overline{Y}$ denote its closure in $\mathbb{A}_k^n$. Then $\dim Y = \dim \overline{Y}$.*

Proof. By definition, there exists an affine variety $X \subseteq \mathbb{A}_k^n$ and a nonempty open subset $U \subseteq \mathbb{A}_k^n$ such that $Y = U \cap X$. Hence, since X is closed in $\mathbb{A}_k^n$, we have by Lemma 8.38(ii)

$$\overline{Y} = \overline{Y}^{\mathbb{A}_k^n} = \overline{U \cap X}^{\mathbb{A}_k^n} = \overline{U \cap X}^X = X,$$

where the last equality holds by Proposition 8.11(b) since X is irreducible and $Y = U \cap X$ is a nonempty open subset of X . This shows that the affine variety of which Y is an open subset is exactly $\overline{Y}$.

Let $Z_0 \subsetneq \cdots \subsetneq Z_r$ be a chain of distinct irreducible closed subsets of Y and consider the sequence $\overline{Z}_0 \subsetneq \cdots \subsetneq \overline{Z}_r$ in $\overline{Y}$. For each $0 \leq i \leq r$ the set $\overline{Z}_i$ is closed in $\overline{Y}$ by construction and irreducible since it is the closure of an irreducible set (Proposition 8.11(a)). Moreover, each inclusion is still strict since for each $0 \leq i \leq r$ we have $\overline{Z}_i \cap Y = Z_i$ by Lemma 8.38(i), and hence $\overline{Z}_i = \overline{Z}_{i+1}$ would imply $Z_i = \overline{Z}_i \cap Y = \overline{Z}_{i+1} \cap Y = Z_{i+1}$. Therefore, by definition of dimension of a topological space, it holds $\dim Y \leq \dim \overline{Y}$; in particular, we have $\dim Y < +\infty$ since $\overline{Y}$ is an affine variety. Hence, we can consider a chain $Z_0 \subsetneq \cdots \subsetneq Z_r$ of distinct irreducible closed subsets of Y where $r = \dim Y$. Such a chain is obviously maximal, and in particular Z_0 must be a point in Y and $Z_r = Y$. Observe now that the point $Z_0 = \overline{Z}_0$ corresponds to a maximal ideal $\mathfrak{m}$ of $k[X_1, \dots, X_n]$ and the chain $\overline{Z}_0 \subsetneq \cdots \subsetneq \overline{Z}_r = \overline{Y}$ corresponds to the chain of prime ideals of $k[X_1, \dots, X_n]$

$$I(\overline{Y}) \subsetneq I(\overline{Z}_{r-1}) \subsetneq \cdots \subsetneq I(\overline{Z}_0) = \mathfrak{m},$$

which in turn yields a chain of prime ideals of length r in $A(\overline{Y})$, namely

$$0 \subsetneq I(\overline{Z}_{r-1})/I(\overline{Y}) \subsetneq \cdots \subsetneq \mathfrak{m}/I(\overline{Y}) =: \mathfrak{m}'.$$

This implies that $\text{ht } \mathfrak{m}' \geq r$. Assume by contradiction that $s := \text{ht } \mathfrak{m}' > r$. Then in $A(\overline{Y})$ there exists a chain of prime ideals

$$0 = P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_s = \mathfrak{m}'.$$

Hence, by the correspondence between prime ideals of $A(\overline{Y})$ and prime ideals of $k[X_1, \dots, X_n]$ containing $I(\overline{Y})$ and by the one-to-one inclusion-reversing correspondence between prime ideals of $k[X_1, \dots, X_n]$ and irreducible closed subsets of $\mathbb{A}_k^n$, we obtain a chain $Z_0 = W_0 \subsetneq W_1 \subsetneq \dots \subsetneq W_s = \overline{Y}$ of distinct irreducible closed subsets of $\overline{Y}$, which in turn gives a chain in Y , namely

$$Z_0 \subseteq W_1 \cap Y \subseteq \dots \subseteq W_s \cap Y = Y. \quad (8.5)$$

Note that for each $0 \leq i \leq s$ the set $W'_i := W_i \cap Y$ is a nonempty open subset of W_i (since Y is an open subset of $\overline{Y}$ and $\emptyset \neq Z_0 \subseteq W'_i$), hence W'_i is irreducible by Proposition 8.11(c). Moreover, W'_i is closed in Y and by Lemma 8.39 we have $\overline{W'_i} = W_i$, which implies that each inclusion in the chain (8.5) is strict. Indeed, if there is $0 \leq i \leq s-1$ such that $W'_i = W'_{i+1}$, we have

$$W_i = \overline{W'_i \cap Y} = \overline{W'_{i+1} \cap Y} = W_{i+1},$$

a contradiction. Thus, (8.5) is a chain of distinct irreducible closed subsets of Y , but this is a contradiction since $s > r = \dim Y$. Therefore, it must hold $\text{ht } \mathfrak{m}' = r$.

Finally, since $A(\overline{Y})$ is an integral domain which is a finitely generated k -algebra and $\mathfrak{m}'$ is a maximal ideal, hence prime, in $A(\overline{Y})$, we can apply the formula given by Theorem 8.35:

$$\text{ht } \mathfrak{m}' + \dim A(\overline{Y})/\mathfrak{m}' = \dim A(\overline{Y}).$$

Since $A(\overline{Y})/\mathfrak{m}' = k$, we have $\dim A(\overline{Y})/\mathfrak{m}' = 0$, and thus by Proposition 8.33

$$\dim \overline{Y} = \dim A(\overline{Y}) = \text{ht } \mathfrak{m}' = r = \dim Y.$$

□

Next we recall two important results in commutative algebra.

Theorem 8.41 (Krull's Principal Ideal Theorem (Corollary 6.10)). *Let A be a Noetherian ring and let $f \in A$ be an element which is neither a zero-divisor nor a unit. Then every minimal prime ideal P of (f) has $\text{ht}(P) = 1$.*

Theorem 8.42. *A Noetherian integral domain A is a unique factorization domain (UFD) if and only if every prime ideal of height 1 is principal.*

Now we are ready to prove the following result.

Proposition 8.43. *An affine variety $Y \subseteq \mathbb{A}^n$ has dimension $n-1$ if and only if $Y = Z(f)$, where $f \in k[X_1, \dots, X_n]$ is an irreducible polynomial.*

Proof. $\Leftarrow$. If f is an irreducible polynomial in $k[X_1, \dots, X_n]$, we have that $Y = Z(f)$ is an affine variety whose corresponding ideal is the prime ideal $P = (f)$, see Example 8.23(b). Then by applying Krull's Principal Ideal Theorem we have $\text{ht}(P) = 1$, and thus $\dim Z(f) = n-1$ by Theorem 8.35.

$\Rightarrow$. The affine variety Y of dimension $n-1$ corresponds to a prime ideal $P := I(Y) \subseteq k[X_1, \dots, X_n]$ of height 1 by Theorem 8.35. Hence, since $k[X_1, \dots, X_n]$ is a Noetherian UFD, by Theorem 8.42 P is a principal ideal, and so it is generated by an irreducible polynomial $f \in k[X_1, \dots, X_n]$. Finally, $Y = Z(f)$. □

Chapter 9

Projective Varieties

Let k be an algebraically closed field.

Definition 9.1. (i) Let $n \in \mathbb{N}$. Consider the set

$$\mathbb{A}^{n+1} \setminus \{0\} := \{(a_0, \dots, a_n) \in k^{n+1} \mid (a_0, \dots, a_n) \neq (0, \dots, 0)\}$$

and define on it the equivalence relation given by

$$(a_0, \dots, a_n) \sim (b_0, \dots, b_n) \stackrel{\text{def}}{\iff} \exists \lambda \in k^* : b_i = \lambda a_i \text{ for all } 0 \leq i \leq n.$$

The *projective n -space over k* , denoted by $\mathbb{P}_k^n$ (or briefly by $\mathbb{P}^n$), is the quotient set

$$\mathbb{P}_k^n := (\mathbb{A}_k^{n+1} \setminus \{0\}) / \sim.$$

In other words, $\sim$ is the equivalence relation identifying the points of $\mathbb{A}_k^n \setminus \{0\}$ lying on the same line through the origin.

- (ii) An element $p \in \mathbb{P}_k^n$ is called a *point* and any $(n+1)$ -tuple $(a_0, \dots, a_n)$ in the equivalence class of p is called a set of *homogeneous coordinates for p* . Moreover, we write

$$p = (a_0 : \dots : a_n) := [(a_0, \dots, a_n)]_{\sim}.$$

Definition 9.2. (i) A *graded ring* is a ring S together with a decomposition

$$S = \bigoplus_{d \geq 0} S_d$$

of S into a direct sum of abelian groups S_d such that for any two indices $d, e \geq 0$ it holds $S_d S_e \subseteq S_{d+e}$. An element of S_d is called a *homogeneous element of degree d* .

- (ii) An ideal $\mathfrak{a} \subseteq S$ is called *homogeneous* if $\mathfrak{a} = \bigoplus_{d \geq 0} (\mathfrak{a} \cap S_d)$.

Example 9.3. Let S be the polynomial ring $k[X_0, \dots, X_n]$. Let $d \in \mathbb{N}$ and define

$$S_d := \left\{ f \in S \mid f = \sum_{i_0 + \dots + i_n = d} \alpha_{i_0, \dots, i_n} X_0^{i_0} \cdots X_n^{i_n} \right\}.$$

Note that $S_0 = k$ and S_d is a k -vector space for each $d \geq 0$. A polynomial $f \in S_d$ is called *homogeneous of degree d* . It is easy to see that every polynomial $f \in S$ decomposes (in a unique way) as

$$f = f_0 + \dots + f_d,$$

where $d := \deg f$ and $f_j \in S_j$ for each $0 \leq j \leq d$. The f_j 's are called the *homogeneous parts of f* . Hence, we have a decomposition of S

$$S = \bigoplus_{d \geq 0} S_d$$

into k -vector spaces which gives S the structure of a graded ring.

Proposition 9.4. *Let $S = \bigoplus_{d \geq 0} S_d$ be a graded ring.*

- (i) *An ideal of S is homogeneous if and only if it is generated by homogeneous elements.*
- (ii) *The sum, intersection, product and radical of homogeneous ideals are homogeneous ideals.*
- (iii) *Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal. Suppose that for any pair of homogeneous elements $f, g \in S$ the following holds:*

$$fg \in \mathfrak{a} \implies f \in \mathfrak{a} \text{ or } g \in \mathfrak{a}.$$

Then $\mathfrak{a}$ is prime. In other words, the primality of a homogeneous ideal can be tested using homogeneous elements only.

Proof. Exercise. □

If $f \in S = k[X_0, \dots, X_n]$ is a polynomial, we cannot regard it as a k -valued function on $\mathbb{P}_k^n$ as we did in the affine case because of the non-uniqueness of homogeneous coordinates. But if f is a homogeneous polynomial of degree d , then for all $\lambda \in k^*$ it holds

$$f(\lambda a_0, \dots, \lambda a_n) = \lambda^d f(a_0, \dots, a_n).$$

Hence, the property of f being zero or not depends only on the equivalence class of $(a_0, \dots, a_n)$, and we have an induced function (which we still call f by abuse of notation)

$$f: \mathbb{P}_k^n \rightarrow \{0, 1\}, \quad p = (a_0 : \dots : a_n) \mapsto \begin{cases} 0, & \text{if } f(a_0, \dots, a_n) = 0, \\ 1, & \text{if } f(a_0, \dots, a_n) \neq 0, \end{cases}$$

Thus, we can talk about the *zeros of a homogeneous polynomial*.

Definition 9.5. Let $f \in S = k[X_0, \dots, X_n]$ be a homogeneous polynomial. We define the *zero-set* of f

$$Z(f) := \{p \in \mathbb{P}_k^n \mid f(p) = 0\}.$$

More generally, if $T \subseteq S$ is a set of homogeneous polynomials, we define the *zero-set* of T as

$$Z(T) := \{p \in \mathbb{P}_k^n \mid f(p) = 0 \text{ for all } f \in T\}.$$

If $\mathfrak{a} \subseteq S$ is a homogeneous ideal, we define $Z(\mathfrak{a}) = Z(T)$, where T is the set of all homogeneous elements in $\mathfrak{a}$.

Remark 9.6. (a) Let $T \subseteq S = k[X_0, \dots, X_n]$ be a set of homogeneous polynomials. Then the ideal

$$\langle T \rangle := \left\{ \sum_{i=1}^r g_i f_i \mid f_i \in T, g_i \in S, r \in \mathbb{N} \right\}$$

is homogeneous by Proposition 9.4(i) and it is easy to see that

$$Z(T) = Z(\langle T \rangle).$$

(b) Since $S = k[X_0, \dots, X_n]$ is Noetherian, by using (a) one can show that any set T of homogeneous polynomials has a finite subset $\{f_1, \dots, f_r\}$ such that $Z(T) = Z(f_1, \dots, f_r)$.

(c) If $\mathfrak{a} \subseteq S = k[X_0, \dots, X_n]$ is a homogeneous ideal, then there are homogeneous polynomials $f_1, \dots, f_r \in \mathfrak{a}$ such that $\mathfrak{a} = (f_1, \dots, f_r)$. Moreover, it holds

$$Z(\mathfrak{a}) = Z(f_1, \dots, f_r).$$

Definition 9.7. A subset X of $\mathbb{P}_k^n$ is an *algebraic set* if there exists a set T of homogeneous elements of $S = k[X_0, \dots, X_n]$ such that $X = Z(T)$.

Proposition 9.8. (i) *The union of two algebraic sets is an algebraic set.*

(ii) *The intersection of any family of algebraic sets is an algebraic set.*

(iii) *The empty set and $\mathbb{P}_k^n$ are algebraic sets.*

Proof. Exercise. □

Thanks to the previous proposition we can give the following definition.

Definition 9.9. The *Zariski topology* on $\mathbb{P}_k^n$ is the topology whose open sets are the complements of the algebraic sets.

Now that we have a topology on the projective n -space $\mathbb{P}_k^n$, we can talk about irreducibility and dimension of a subset of $\mathbb{P}_k^n$ (with respect to the subspace topology).

Definition 9.10. (i) A *projective algebraic variety* (or simply *projective variety*) is an irreducible algebraic set of $\mathbb{P}_k^n$.

(ii) A *quasi-projective variety* is an open subset of a projective variety.

(iii) The *dimension* of a (quasi-)projective variety is its dimension as a topological space.

Similarly as in the affine case, we can attach to any subset Y of $\mathbb{P}_k^n$ a homogeneous ideal $I(Y)$.

Definition 9.11. (i) Let Y be a subset of $\mathbb{P}_k^n$. We define the *homogeneous ideal of Y in $S = k[X_0, \dots, X_n]$* , denoted by $I(Y)$, to be the ideal generated by the set

$$\{f \in S \mid f \text{ is homogeneous and } f(p) = 0 \text{ for all } p \in Y\}.$$

(i) If Y is an algebraic set of $\mathbb{P}_k^n$, we define the *homogeneous coordinate ring of Y* to be

$$S(Y) := S/I(Y).$$

Proposition 9.12 (Homogeneous Nullstellensatz). *Let $\mathfrak{a} \subseteq S = k[X_0, \dots, X_n]$ be a homogeneous ideal and let $f \in S$ be a homogeneous polynomial of positive degree such that $f(p) = 0$ for all $p \in Z(\mathfrak{a}) \subseteq \mathbb{P}_k^n$. Then $f^q \in \mathfrak{a}$ for some $q > 0$.*

Proof. Exercise. □

Proposition 9.13. *Let $\mathfrak{a} \subseteq S = k[X_0, \dots, X_n]$ be a homogeneous ideal. Then the following are equivalent:*

(a) $Z(\mathfrak{a}) = \emptyset$,

(b) the radical of $\mathfrak{a}$ equals either S or the ideal $S_+ = \bigoplus_{d>0} S_d$,

(c) $\mathfrak{a} \supseteq S_d$ for some $d > 0$.

Proof. Exercise. □

Proposition 9.14. *Let $S = k[X_0, \dots, X_n]$ and let S^h be the set of homogeneous elements of S . Then the following statements hold:*

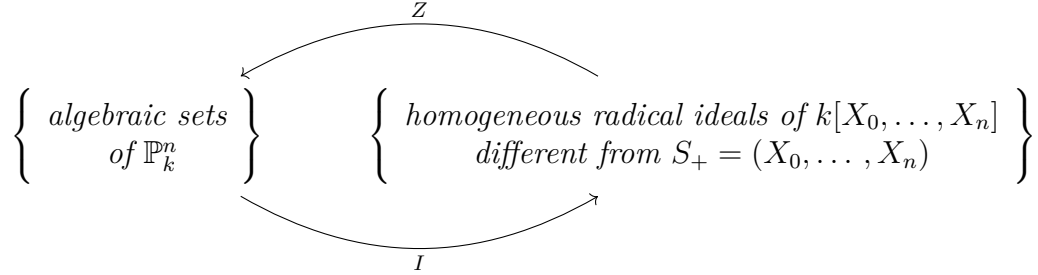
(a) Let $T_1 \subseteq T_2$ be subsets of S^h . Then $Z(T_1) \supseteq Z(T_2)$.

(b) Let $Y_1 \subseteq Y_2$ be subsets of $\mathbb{P}_k^n$. Then $I(Y_1) \supseteq I(Y_2)$.

- (c) Let Y_1, Y_2 be two subsets of $\mathbb{P}_k^n$. Then $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.
- (d) Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal with $Z(\mathfrak{a}) \neq \emptyset$. Then $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.
- (e) Let Y be a subset of $\mathbb{P}_k^n$. Then $Z(I(Y)) = \overline{Y}^{\mathbb{P}_k^n}$.

Proof. Exercise. □

Proposition 9.15. *There are one-to-one inclusion-reversing correspondences*



Proof. Exercise. □

Remark 9.16. Since the homogeneous ideal $S_+ = (X_0, \dots, X_n)$ is geometrically meaningless, it is called the *irrelevant maximal ideal* of $S = k[X_0, \dots, X_n]$.

Next, we show that the projective n -space $\mathbb{P}_k^n$ has an open covering by $n + 1$ affine n -spaces. As a result, every projective (respectively quasi-projective) variety has an open covering by affine (respectively quasi-affine) varieties.

Definition 9.17. Let $f \in S = k[X_0, \dots, X_n]$ be a linear homogeneous polynomial, namely $f = a_0x_0 + \dots + a_nx_n$. We call the zero-set of f a *hyperplane*.

We denote $H_i := Z(X_i)$ for $0 \leq i \leq n$. Let $U_i := \mathbb{P}_k^n \setminus H_i$. Note that this set is open in $\mathbb{P}_k^n$. Then $\mathbb{P}_k^n$ is covered by the open sets $U_0, \dots, U_n$: if $p = (a_0 : \dots : a_n)$ is a point in $\mathbb{P}_k^n$, then $a_i \neq 0$ for some i , and thus $p \in U_i$ for some i .

Now, for each $0 \leq i \leq n$, define a map

$$\varphi_i: U_i \longrightarrow \mathbb{A}_k^n, \quad (a_0 : \dots : a_n) \longmapsto \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i} \right),$$

where $\widehat{\cdot}$ means that $\cdot$ is omitted. The map φ_i is well-defined. Indeed, if we have $(a_0 : \dots : a_n) = (b_0 : \dots : b_n)$, then there is $\lambda \in k^*$ such that $a_j = \lambda b_j$ for $0 \leq j \leq n$ and then $a_j/a_i = \lambda b_j/\lambda b_i = b_j/b_i$.

Consider U_i and $\mathbb{A}_k^n$ equipped with the subspace topology and the (affine) Zariski topology, respectively.

Proposition 9.18. *The maps $\varphi_i: U_i \rightarrow \mathbb{A}_k^n$ are homeomorphisms.*

Proof. We have to show that φ_i is bijective, continuous and φ_i^{-1} is continuous as well.

The bijectivity of φ_i follows easily by the definition of φ_i . As for the continuity, we show that φ_i and φ_i^{-1} are closed maps. Without loss of generality, we may assume $i = 0$. For the rest of the proof we write $\varphi := \varphi_0$ and $U := U_0$.

Let $Y \subseteq U$ be a closed subset of U and consider its closure $\bar{Y}$ in $\mathbb{P}^n$. By definition of the Zariski topology on $\mathbb{P}_k^n$, $\bar{Y}$ is an algebraic set, i.e., $\bar{Y} = Z(T)$ for a set $T \subseteq S^h$, where S^h denotes the set of homogeneous elements of $S = k[X_0, \dots, X_n]$. We define

$$\alpha: S^h \longrightarrow k[Y_1, \dots, Y_n], \quad f \longmapsto \alpha(f) := f(1, Y_1, \dots, Y_n).$$

The polynomial $\alpha(f)$ is called the *dehomogenization of f (with respect to X_0)*. Let $T' := \alpha(T)$. We show that $\varphi(Y) = Z(T')$.

$\subseteq$. Let $f \in T$. Then $f(x_0, \dots, x_n) = 0$ for $(x_0 : \dots : x_n) \in \bar{Y} = Z(T)$. Let $p = (x_0 : \dots : x_n) \in Y = U \cap \bar{Y}$; then $p \in U$, i.e., $x_0 \neq 0$, and $f(x_0, \dots, x_n) = 0$. Note that $p = (1 : \frac{x_1}{x_0} : \dots : \frac{x_n}{x_0}) \in \mathbb{P}^n$. Moreover, if $k = \deg(f)$, then

$$f(X_0, \dots, X_n) = X_0^k \alpha(f) \left(1, \frac{X_1}{X_0}, \dots, \frac{X_n}{X_0} \right).$$

Hence, $\alpha(f)(\varphi(p)) = 0$, and thus $\varphi(Y) \subseteq Z(T')$.

$\supseteq$. Let $p = (y_1, \dots, y_n) \in Z(T')$. Let $f \in T$ and let $g := \alpha(f)$. Then $p = \varphi(q)$ for $q = (1 : y_1 : \dots : y_n)$ and

$$0 = g(y_1, \dots, y_n) = f(1, y_1, \dots, y_n).$$

Thus, $q \in \bar{Y} = Z(T)$ and by construction $q \in U$ as well. Finally, $q \in U \cap \bar{Y} = Y$, hence $p = \varphi(q) \in \varphi(Y)$ and $Z(T') \subseteq \varphi(Y)$.

We have just shown that φ^{-1} is continuous. Now we show that φ is continuous.

Let $W \subseteq \mathbb{A}^n$ be closed. Then $W = Z(T')$ for some $T' \subseteq A := k[Y_1, \dots, Y_n]$. Define

$$\beta: A \longrightarrow S^h$$

$$\beta(g)(X_0, \dots, X_n) := X_0^d g \left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0} \right),$$

where d is the degree of the polynomial $g(Y_1, \dots, Y_n)$. Note that $\beta(g)$ is a homogeneous polynomial of degree d called the *homogenization of g (with respect to X_0)*. By using similar arguments as above one can show that $\varphi^{-1}(W) = Z(\beta(T')) \cap U$. Thus, $\varphi^{-1}(W)$ is closed in U and φ is continuous. Finally, φ is a homeomorphism. $\square$

Corollary 9.19. *Let Y be a projective (respectively quasi-projective) variety. Then Y is covered by the open sets $Y \cap U_i$, $0 \leq i \leq n$, which are homeomorphic to affine (respectively quasi-affine) varieties via the maps φ_i defined above.*

Chapter 10

Morphisms

Definition 10.1. Let Y be a quasi-affine variety in $\mathbb{A}^n$. A function $f: Y \rightarrow k$ is called *regular at a point* $p \in Y$, if there exists an open neighbourhood $p \in U \subseteq Y$ together with two polynomials $g, h \in A = k[X_1, \dots, X_n]$ such that h is nowhere zero on U and $f = g/h$. f is *regular on* Y , if f is regular at every point.

Lemma 10.2. *A regular function is continuous (where we identify k with $\mathbb{A}_k^1$ equipped with the Zariski topology).*

Proof. Let $f: Y \rightarrow k$ be regular. We are going to show that preimages of closed sets under f are closed. Recall that closed sets of $\mathbb{A}^1$ are exactly sets with finitely many points. Thus it is satisfactory to show $f^{-1}(\{a\}) = \{p \in Y \mid f(p) = a\}$ is closed in Y for any $a \in k$.

We check this locally using the following fact: $Z \subseteq Y$ is closed in the topological space Y if and only if it can be covered by open $U \subseteq Y$ such that $Z \cap U$ is closed in U for every U .

Let U be an open set on which f can be represented as quotient g/h with $g, h \in A = k[X_1, \dots, X_n]$ and h is nowhere zero on U . Then we have:

$$f^{-1}(a) \cap U = \left\{ p \in U : \frac{g(p)}{h(p)} = a \right\}$$

but this means $g(p) = ah(p)$ or equivalently $(g - ah)(p) = 0$. This means $f^{-1}(a) \cap U = Z(g - ah) \cap U$ which is closed in U . Now we can conclude that $f^{-1}(a)$ is closed in Y and f is continuous. $\square$

Definition 10.3. Let now $Y \subseteq \mathbb{A}^n$ be a quasi-projective variety. A function $f: Y \rightarrow k$ is *regular at a point* p if there are an open neighbourhood $p \in U \subseteq Y$ and homogeneous polynomials $g, h \in S = k[X_0, \dots, X_n]$ of the same degree such that h is nowhere zero on U and $f = g/h$ on U . We say f is *regular on* Y if it is regular at every point in Y .

Note that g, h are not functions on $\mathbb{P}^n$, but their quotient is a well-defined function on $\mathbb{P}^n$, when $h \neq 0$.

Remark 10.4. 1. As above we can show that regular functions on quasi-projective varieties are continuous (the proof works analogously to the proof of 10.2).

2. If f and g are regular on a (quasi-)affine variety or (quasi-)projective variety X and $f = g$ holds on an open subset $U \subseteq X$, then $f = g$ on X .

Proof. Let $Z = \{p \in X \mid (f - g)(p) = 0\}$. Z is closed and it contains an open U . Because X is irreducible, we know from Exercise 1 (a) from Sheet 3, that $Z = X$ must hold and thus $f = g$ on the whole of X . $\square$

Definition 10.5. • A *variety over k* is any affine-, quasi-affine-, projective- or quasi-projective variety as we have defined it in the previous lectures.

- Let X, Y be two varieties. A *morphism* $\varphi: X \rightarrow Y$ is a continuous map such that for every open subset $V \subseteq Y$ and for every regular function $f: V \rightarrow K$, the function $f \circ \varphi: \varphi^{-1}(V) \rightarrow k$ is regular.

Remark 10.6. 1. The composition of two morphisms is also a morphism. This allows us to define the category $\mathcal{V}\text{ar}$ with varieties as objects and morphisms as defined in 10.5.

2. In particular, we have isomorphisms between varieties, namely the morphisms $\varphi: X \rightarrow Y$ that admit inverse morphisms $\psi: Y \rightarrow X$ such that $\psi \circ \varphi = \text{id}_X$ and $\varphi \circ \psi = \text{id}_Y$.

Definition 10.7.

- Let Y be a variety. We denote by $\Gamma(Y, \mathcal{O}_Y)$ the ring of all regular functions on Y (in literature, also $\mathcal{O}_Y$, $\mathcal{O}(Y)$ or $\mathcal{O}_Y(Y)$ are in use).
- If p is a point on Y , we define the *local ring of p on Y* , denoted by $\mathcal{O}_{Y,p}$ (in literature, $\mathcal{O}_p$ and $\mathcal{O}_{p,Y}$ are in use) to be the ring of *germs* of regular functions on Y near p . In other words, an element of $\mathcal{O}_{Y,p}$ is a pair $\langle U, f \rangle$, where U is an open subset of Y containing p and f is a regular function on U and $\langle U, f \rangle = \langle V, g \rangle$ if and only if $f = g$ on $U \cap V$ (note that this is an equivalence relation, i.e., an element $\langle U, f \rangle$ of $\mathcal{O}_{Y,p}$ is an equivalence class of functions).

Remark 10.8. • $\mathcal{O}_{Y,p}$ is a local ring with maximal ideal

$$\mathfrak{m} = \{\text{germs of regular functions vanishing at } p\}.$$

To see this, check Commutative Algebra Proposition I.5 (or Exercise sheet 1). Note that if $f(p) \neq 0$, then $1/f$ is regular at p .

- The residue field $\mathcal{O}_p/\mathfrak{m}$ is isomorphic to k .

To see this, consider the map $\mathcal{O}_{Y,p}/\mathfrak{m} \rightarrow k$, which sends the class of $\langle U, f \rangle$ to the evaluation $f(p) \in k$ (note that this is independent of the choices we have made). This map is surjective and injective: For surjectivity, if we take $c \in k$, then the function that is constantly c gets mapped to c under our map; for injectivity: Let $\langle U, f \rangle$ be a representative of an element of $\mathcal{O}_{Y,p}$ such that $f(p) = 0 \in k$. By definition of $\mathfrak{m}$, $\langle U, f \rangle \in \mathfrak{m}$ and thus the class of $\langle U, f \rangle$ is zero in $\mathcal{O}_{Y,p}/\mathfrak{m}$.

Definition 10.9. Let Y be a variety. We define the *function field* $K(Y)$ of Y as follows: An element of $K(Y)$ is an equivalence class of pairs $\langle U, f \rangle$, where U is a non-empty open subset of Y and f is a regular function on U . Two elements $\langle U, f \rangle, \langle V, g \rangle$ are the same if and only if $f = g$ on $U \cap V$. The elements of $K(Y)$ will be called *rational functions*.

Remark 10.10. $K(Y)$ is indeed a field. First of all, $K(Y)$ is a ring: If U_1, U_2 are two non-empty open sets, then $U_1 \cap U_2 \neq \emptyset$ because Y is irreducible. The operations of rational functions are defined on the intersections – check the ring axioms. Moreover, it is a field: If $\langle U, f \rangle \in K(Y)$ with $f \neq 0$, then f restricted to $V = U - (Z(f) \cap U)$ (which is open) never vanishes, thus $1/f$ is regular on V and $\langle V, 1/f \rangle = \langle U, f \rangle^{-1}$.

Remark 10.11. For all variety Y , we have defined the three rings $\Gamma(Y, \mathcal{O}_Y)$, $\mathcal{O}_{Y,p}$ and $K(Y)$. By restricting functions we get maps

$$\begin{aligned} \Gamma(Y, \mathcal{O}_Y) &\longrightarrow \mathcal{O}_{Y,p} \longrightarrow K(Y) \\ f &\longmapsto \langle U, f \rangle \\ \langle U, f \rangle &\longmapsto \langle U, f \rangle \end{aligned}$$

Those maps are injective. For the first one: If $\langle U, f \rangle = \langle V, g \rangle$ then $f = g$ on $U \cap V$ and thus by (Remark I.57) $f = g$. For the second one: If $\langle U, f \rangle = 0$, then $f(p) = 0$. Hence we can view $\Gamma(Y, \mathcal{O}_Y)$ and $\mathcal{O}_{Y,p}$ as subrings of $K(Y)$.

Remark 10.12. If Y is replaced by an isomorphic variety, then the corresponding rings are isomorphic as well. This follows immediately from the definition of morphisms. We say the three rings are *invariants of Y up to isomorphism*.

We are going to relate the rings $\Gamma(Y, \mathcal{O}_Y)$, $\mathcal{O}_{Y,p}$ and $K(Y)$ to $A(Y)$ (the affine coordinate ring of an affine variety) and $S(Y)$ (the homogeneous coordinate ring of a projective variety).

Theorem 10.13. *Let $Y \subseteq \mathbb{A}^n$ be an affine variety with affine coordinate ring $A(Y)$. Then the following statements are true:*

1. $A(Y) \cong \Gamma(Y, \mathcal{O}_Y)$,

2. For every point $p \in Y$, let $\mathfrak{m}_p \subseteq A(Y)$ be the ideal of functions vanishing at p . Then $p \mapsto \mathfrak{m}_p$ gives a one-to-one correspondence

$$\{\text{points of } Y\} \longleftrightarrow \{\text{maximal ideals of } A(Y)\}, \quad p \longmapsto \mathfrak{m}_p.$$

3. For every point $p \in Y$ we have $\mathcal{O}_{Y,p} \cong A(Y)_{\mathfrak{m}_p}$ and $\dim \mathcal{O}_{Y,p} = \dim A(Y)$.
4. The function field $K(Y)$ is isomorphic to the quotient field of $A(Y)$. Consequently $K(Y)$ is a finitely generated field extension of k and is of transcendence degree equal to $\dim Y$.

Proof. (1) Define a map $\alpha: A(Y) \rightarrow \Gamma(Y, \mathcal{O}_Y)$: Every $f \in A = k[X_1, \dots, X_n]$ defines a regular function f on Y , we thus have a ring homomorphism $A \rightarrow \Gamma(Y, \mathcal{O}_Y)$ with kernel $I(Y)$ (by the definition of $I(Y)$), hence we have a map

$$\alpha: A(Y) = A/I(Y) \longrightarrow \Gamma(Y, \mathcal{O}_Y).$$

- (2) We prove (ii).

By the Nullstellensatz, there is a one-to-one correspondence

$$\begin{aligned} \{\text{points of } Y\} &\longleftrightarrow \{\text{maximal ideals of } A \text{ containing } I(Y)\} \\ &\longleftrightarrow \{\text{maximal ideals of } A(Y)\}. \end{aligned}$$

Note that a maximal ideal of $A(Y)$ corresponds to $p = (X_1 - a_1, \dots, X_n - a_n)$ and the maximal ideal is $\mathfrak{m} = (X_1 - a_1, \dots, X_n - a_n)$. But this is equal to $\mathfrak{m}_p$: If $f \in \mathfrak{m}$, then $f(a_1, \dots, a_n) = 0$, hence $f \in \mathfrak{m}_p$ and if $f(p) = 0$, we take a representative $F \in K[X_1, \dots, X_n]$ for which $F(p) = 0$, i.e., $F \in \mathfrak{m}$ and we have the required correspondence.

- (3) We prove (iii): We have a natural map

$$\begin{aligned} \left\{ \frac{\text{functions of } A(Y)}{\text{functions of } A(Y) \text{ that don't vanish at } p} \right\} &= A(Y)_{\mathfrak{m}_p} \\ &\longrightarrow \mathcal{O}_{Y,p} = \Gamma(Y, \mathcal{O}_Y)_{\mathfrak{m}_p} \end{aligned}$$

coming from α from step (1). By a fact from commutative algebra: If α is injective, then the corresponding “localisation map” is also injective. By the definition of elements of $\mathcal{O}_{Y,p}$, this map is also surjective and hence

$$A(Y)_{\mathfrak{m}_p} \cong \mathcal{O}_{Y,p}.$$

Moreover, $\dim \mathcal{O}_{Y,p} = \dim A(Y)_{\mathfrak{m}_p} = \dim A(Y) = \dim Y$.

(4) *We prove (iv):* It holds $Q(A(Y)) \cong Q(\mathcal{O}_{Y,p})$ for all $p \in Y$. Furthermore $Q(\mathcal{O}_{Y,p}) = K(Y)$ for all $p \in Y$.

For the first isomorphism: Let R be an integral domain and let $\mathfrak{m}$ be a maximal ideal in R . Then $Q(R) \cong Q(R_{\mathfrak{m}})$: $R \subseteq R_{\mathfrak{m}}$ is an injection ($r \mapsto r/1$ is injective since R is an integral domain), hence $Q(R) \subseteq Q(R_{\mathfrak{m}})$. Conversely, let $a/b \in Q(R_{\mathfrak{m}})$ where $a, b \in R_{\mathfrak{m}}$, $b \neq 0$ and $a = a_1/a_2$, $b = b_1/b_2$ where $a_i, b_i \in R$ and $a_2, b_2 \notin \mathfrak{m}$. In particular,

$$\frac{a}{b} = \frac{a_1 b_2}{a_2 b_1} \in Q(R).$$

From (iii) we know that $\mathcal{O}_{Y,p} \cong A(Y)_{\mathfrak{m}_p}$ and by the considerations above we get $Q(A(Y)) = Q(A(Y)_{\mathfrak{m}_p})$, i.e., $Q(\mathcal{O}_{Y,p}) \cong Q(A(Y))$ for all $p \in Y$.

For the second isomorphism: By 10.11 we can view $\mathcal{O}_{Y,p}$ as a subset of $K(Y)$, hence $Q(\mathcal{O}_{Y,p}) \subseteq K(Y)$. Moreover, we have

$$A(Y) \xrightarrow{\alpha} \Gamma(Y, \mathcal{O}_Y) \longrightarrow \mathcal{O}_{Y,p} \longrightarrow K(Y),$$

thus we have $Q(A(Y)) \rightarrow Q(\mathcal{O}_{Y,p}) \rightarrow K(Y)$ which is independent of p . By the first isomorphism $Q(\mathcal{O}_{Y,p}) \cong Q(A(Y))$, hence $Q(\mathcal{O}_{Y,p})$ is the same for all $p \in Y$.

If f/g is a representative of an element of $K(Y)$ and g is as in the definition of f in (f) and g does not vanish at some p , for this p , $f/g \in Q(\mathcal{O}_{Y,p})$. Thus $K(Y) \cong Q(A(Y))$. Because $A(Y)$ is finitely generated as k -algebra, $K(Y)$ is finitely generated as field extension of k (as algebra).

Finally, by (Theorem I.36) the transcendence degree of $K(Y)/k$ is $\dim A(Y)$ and $\dim A(Y) = \dim Y$, which shows (iv).

(5) *We show (i):* Note that $\Gamma(Y, \mathcal{O}_Y) \subseteq \bigcap_{p \in Y} \mathcal{O}_{Y,p}$. By definition if $f \in \Gamma(Y, \mathcal{O}_Y)$, f is regular on Y and thus $f = g/h$ on an open neighbourhood U of p , where $h(p) \neq 0$ for all $p \in Y$. Eventually

$$\begin{aligned} A(Y) \xrightarrow{\alpha} \Gamma(Y, \mathcal{O}_Y) &\subseteq \bigcap_{p \in Y} \mathcal{O}_{Y,p} = \bigcap_{\mathfrak{m}_p} A(Y)_{\mathfrak{m}_p} \\ &= \bigcap \{A(Y)_{\mathfrak{m}} \mid \mathfrak{m} \text{ is maximal ideal of } A(Y)\}. \end{aligned} \quad (10.1)$$

If B is an integral domain, then $B = \bigcap \{B_{\mathfrak{m}} \mid \mathfrak{m} \text{ is maximal ideal of } B\}$. Suppose $z \notin B$ but $z \in B'_{\mathfrak{m}}$ for all maximal ideals $\mathfrak{m}'$ in I . Then $I = \{x \in B \mid xz \in B\}$ is an ideal in B and $1 \notin I$, hence there is a maximal ideal $\mathfrak{m}$ such that $I \subseteq \mathfrak{m}$. But since $z \in B_{\mathfrak{m}}$, $z = b/s \in B$, where $s \notin \mathfrak{m}$, i.e. $sz = b \in B$, i.e., $s \in I$ and $s \notin \mathfrak{m}$, which is a contradiction.

This shows that the equalities hold in 10.1, which proves (i). □

We deal with the projective case now.

Proposition 10.14. *Let U_i in $\mathbb{P}^n$ be the open sets $U_i = \mathbb{P}^n - H_i = \mathbb{P}^n - Z(X_i)$. Then the maps*

$$\varphi_i: U_i \longrightarrow \mathbb{A}^n, \quad p = (a_0, \dots, a_n) \longmapsto \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i} \right)$$

from (Proposition I.52) are isomorphisms of varieties.

Proof. We already know that φ_i is a homeomorphism. If f is regular on $V = \mathbb{A}^n$, f is locally a quotient of polynomials in $k[Y_1, \dots, Y_n]$. Using the maps α, β from (Proposition I.52), we can identify such f with regular functions on U_i which are locally quotients of homogeneous polynomials in $k[X_0, \dots, X_n]$ of same degree. $\square$

Notation

Let S be a graded ring and let P be a homogeneous prime ideal in S . Consider the set T of homogeneous elements of $S - P$. It is a multiplicative set; consider $T^{-1}S$, it is a ring with grading $\deg(f/g) = \deg(f) - \deg(g)$.

Let $S_{(P)}$ be the subring in $T^{-1}S$ of elements of degree 0, i.e.,

$$S_{(P)} = \{r/s = rs^{-1} \mid \deg(r) = \deg(s)\}.$$

Note: $S_{(P)}$ is a local ring with maximal ideal $(PT^{-1}S) \cap S_{(P)}$. In the special case that S is an integral domain and $P = (0)$, the maximal ideal in $S_{(P)}$ is (0) , so $S_{(0)}$ is a field. If $f \in S$ is homogeneous, denote by $S_{(f)}$ the subring of elements of degree 0 in the localised ring S_f .

Theorem 10.15. *Let $Y \subseteq \mathbb{P}^n$ be a projective variety with homogeneous coordinate ring $S(Y)$. Then the following statements hold:*

1. $\Gamma(Y, \mathcal{O}_Y) = k$,
2. For any point $p \in Y$, let $\mathfrak{m}_p \subseteq S(Y)$ be the ideal generated by the set of the homogeneous elements of $S(Y)$ that vanish at p . Then $\mathcal{O}_{Y,p} = S(Y)_{\mathfrak{m}_p}$,
3. The function field $K(Y)$ is isomorphic to $S(Y)_{(0)}$

Proof. (1) Let $Y_i = Y \cap U_i$ where $U_i = \mathbb{P}^n - Z(X_i) \subseteq \mathbb{P}^n$. Since $U_i \cap \mathbb{A}^n$, we can consider Y_i as an affine variety. Thus we have an isomorphism

$$k[Y_1, \dots, Y_n] \xrightarrow{\cong} k[X_0, \dots, X_n]_{(X_i)} = S_{(X_i)},$$

$$f(y_1, \dots, y_n) \longmapsto f\left(\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i}\right).$$

Now it holds

$$A(Y_i) = k[Y_1, \dots, Y_n]/I(Y_i) \cong S_{(X_i)}/I(Y)S_{(X_i)} = S(Y)_{(X_i)},$$

and thus $\varphi_i^*: A(Y_i) \cong S(Y)_{(X_i)}$.

(2) *We prove (ii):* Let $p \in Y$ and choose i such that $P \in Y_i = Y \cap U_i$. By (Theorem I.67), we have $\mathcal{O}_{Y,p} \cong A(Y_i)_{\mathfrak{m}'_p}$, where $\mathfrak{m}'_p$ is the maximal ideal of $A(Y_i)$ corresponding to P . By the definition of φ_i^* , it holds $\varphi_i^*(\mathfrak{m}'_p) = \mathfrak{m}_p S(Y)_{(X_i)}$. Thus we have

$$\mathcal{O}_{Y_i,p} \cong A(Y_i)_{\mathfrak{m}'_p} = (\varphi_i^* A(Y_i) \mathfrak{m}'_p)_{\varphi_i^*(\mathfrak{m}'_p)} = (S(Y)_{(X_i)})_{\mathfrak{m}_p S(Y)_{(X_i)}} = S(Y)_{\mathfrak{m}_p}.$$

which finishes (ii), since the local ring is “local” on open sets.

(3) It holds

$$\begin{aligned} (S(Y)_{(X_i)})_{\mathfrak{m}_p S(Y)_{(X_i)}} &= \left\{ \frac{\frac{r}{X_i^k}}{g} : r \in S(Y), \deg(r) = k, k \in \mathbb{N}, g \notin \mathfrak{m}_p S(Y)_{(X_i)} \right\} \\ &= \left\{ \frac{r}{X_i^k g} : r \in S(Y), x_i \notin \mathfrak{m}_p, x_i^k g \notin \mathfrak{m}_p S(Y)_{(X_i)} \right\}, \end{aligned}$$

on the other hand

$$S(Y)_{(\mathfrak{m}_p)} = \left\{ \frac{r'}{f} : r' \in S(Y), \deg(r') = \deg(f), f \notin \mathfrak{m}_p \right\}$$

those two rings are identified (send an element of the first ring to itself on the second ring).

(4) *We prove (iii):* It holds $K(Y_i) = K(Y)$ (again, because the definition of function fields is on open sets) and

$$K(Y_i) = a(A(Y_i)) = Q(S(Y)_{(X_i)}) = S(Y)_{(0)}.$$

Indeed,

$$\begin{aligned} Q(S(Y)_{(X_i)}) &= \left\{ \frac{a}{b} : a \in S(Y)_{(X_i)}, b \in S(Y)_{(X_i) - \{0\}}, a = \frac{a'}{X_i^M}, \right. \\ &\quad \left. \deg(a') = M \text{ and } b = \frac{b'}{X_i^k}, \deg(b') = k \right\} \\ &= \left\{ \frac{a' \cdot X_i^{M-k}}{b'} : \text{nominator and denominator have the same degree} \right\} \end{aligned}$$

and

$$S(Y)_{(0)} = \left\{ \frac{f}{g} : g \neq 0, \deg(f) = \deg(g) \right\}.$$

(5) *We prove (i):* We prove “ $\subseteq$ ”: Take $f \in \Gamma(Y, \mathcal{O}_Y)$. f is regular on Y_i , $1 \leq i \leq n$, i.e. by the previous theorem

$$f \in \Gamma(Y_i, \mathcal{O}_{Y_i}) = A(Y_i, \mathcal{O}_{Y_i}) = S(Y)_{(X_i)},$$

we thus can write $f = g_i/X_i^N$, where g_i is homogeneous and is of degree N .

Note that $\Gamma(Y, \mathcal{O}_Y) \subseteq K(Y)$ is a subring and $K(Y)$ locally looks like rational functions f/g where $g \neq 0$. We can thus view $\Gamma(Y, \mathcal{O}_Y)$, $K(Y)$ and $S(Y)$ as subrings of $Q(S(Y))$. Thus we can write $g_i = X_i^N f \in S(Y)_{N_i}$ ($\deg(f) = 0$). If we choose $N \geq \sum_i N_i$, we have the following: $S(Y)_N f \subseteq S(Y)_N$, i.e., $gf \in S(Y)_N$ for all $g \in S(Y)_N$. To see this, it suffices to check the claim where g is a monomial. Take a monomial of degree N , say $g = X_0^{i_0} \cdots X_n^{i_n}$, where $\sum_j i_j = N \geq \sum_i N_i$. $S(Y)_N$ (as k -vector space) is spanned by monomials of degree N in $X_0, \dots, X_n$ and in any such monomial we can find j such that $i_j \geq N_j$. Without loss of generality we may assume $i_0 \geq N_0$. If we now consider the product

$$X_0^{i_0} \cdots X_n^{i_n} f = (X_0^{i_0} f) X_1^{i_1} \cdots X_n^{i_n},$$

because of $i_0 \geq N_0$ and $f = \dots$, we get $X_0^{i_0} \cdots X_n^{i_n} f \in S(Y)_N$.

Furthermore, we have $S(Y)_N f^q \subseteq S(Y)_N$ for all $q \in \mathbb{N} - \{0\}$. In particular, $X_0^N f^q \subseteq S(Y)_N$ and $X_0^{-N} S(Y) \supseteq S(Y)[f]$, where $X_0^{-N} S(Y)$ is a finitely generated $S(Y)$ -module (with generator X_0^{-N}) and $S(Y)[f]$ is a polynomial ring.

$X_0^{-N} S(Y)$ is Noetherian (as finitely generated module); in particular $S(Y)[f]$ is Noetherian. By a commutative algebra fact, $S(Y)[f]$ is finitely generated and hence f is an *integral over* $S(Y)$, i.e., there are $a_1, \dots, a_m \in S(Y)$ such that $f^m + a_1 f^{m-1} + \cdots + a_m = 0$.

We now can replace the $a_1, \dots, a_m$ with their homogeneous components of degree 0. Indeed, we can write

$$a_i = a_i^{(0_i)} + \cdots a_i^{(l_i)},$$

where $a_i^{(j_i)}$ is the homogeneous component of a_i of degree j_i . Plugging this into the equation for f , we get

$$f^m + (a_1^{(0)} + \cdots + a_1^{(l_1)}) f^{m-1} + \cdots + (a_m^{(0)} + \cdots + a_m^{(l_m)}) = 0,$$

rearranging by the degrees of the homogeneous components gives that each of those must vanish, in particular $f^m + a_1^{(0)} f^{m-1} + \cdots + a_m^{(0)} = 0$, in other words f is algebraic over k . Because k is algebraically closed, $f \in k$. □

Proposition 10.16. *Let X be any variety and Y be an affine variety. Then there is a natural bijective map of sets*

$$\alpha: \text{Hom}(X, Y) \xrightarrow{\cong} \text{Hom}((A(Y), \Gamma(X, \mathcal{O}_X))).$$

Proof. (1) *Define α :* Let $\varphi: X \rightarrow Y$ be a morphism. By definition, φ maps regular functions on Y to regular functions on X , i.e., φ induces a map $\Gamma(Y, \mathcal{O}_Y) \rightarrow \Gamma(X, \mathcal{O}_X)$ which is a homomorphism of k -algebras. In particular it induces a map $A(Y) \rightarrow \Gamma(X, \mathcal{O}_X)$. This defines α .

(2) Let $h \in \text{Hom}(A(Y), \Gamma(X, \mathcal{O}_X))$ and suppose $Y \subseteq \mathbb{A}^n$ is closed. Then $A(Y) = k[X_1, \dots, X_n]/I(Y)$. Let $[X_i]$ be the image of X_i in $A(Y)$ and let $\xi_i = h([X_i]) \in \Gamma(X, \mathcal{O}_X)$. Define $\psi: X \rightarrow \mathbb{A}^n$ via

$$p \mapsto (\xi_1(p), \dots, \xi_n(p)) \in \mathbb{A}^n.$$

(3) We show $\psi(X) \subseteq Y$: We have $Y = Z(I(Y))$, it is thus enough to show that for all $p \in X$ and for all $f \in I(Y)$ it holds $f(\psi(p)) = 0$. We have

$$\begin{aligned} f(\psi(p)) &= f(\xi_1(p), \dots, \xi_n(p)) \\ &= f(h([X_1])(p), \dots, h([X_n])(p)) = h(f(X_1), \dots, f(X_n))(p) = 0 \end{aligned}$$

as f is a polynomial, $\xi_i = h([X_i])$, h is a homomorphism of k -algebras and $f \in I(Y)$. We thus have defined a map in the other direction.

(4) We show that ψ is a morphism and ψ and α are inverse to each other:

$$\begin{array}{ccc} \text{Hom}(X, Y) & \longrightarrow & \text{Hom}(A(Y), \Gamma(X, \mathcal{O}_X)) \\ \varphi & \longmapsto & \alpha(\varphi) := (f \mapsto f \circ \varphi) \\ & & \psi(h) \text{ long maps from } h. \end{array}$$

Evaluating the maps gives

$$\varphi \mapsto (f \mapsto f \circ \varphi) \mapsto (p \mapsto ([X_1] \circ \varphi)(p), \dots, [X_n \circ \varphi](p)) = \varphi$$

and

$$h \mapsto (p \mapsto (h([X_1])(p), \dots, h([X_n])(p))) \mapsto (f \mapsto h(f([X_1], \dots, f([X_n]))) = h,$$

i.e., the maps are inverse to each other. That ψ is a morphism follows from the following Lemma.

□

Remark 10.17. Consider the situation in (Proposition I.69). Let $h \in \text{Hom}(A(Y), \Gamma(X, \mathcal{O}_X))$ be injective and

$$\psi: X \longrightarrow \mathbb{A}^n, \quad p \mapsto (\xi_1(p), \dots, \xi_n(p))$$

be the inverse of α . Then $\text{cl}(\psi(X)) = Y$.

Proof. Assume, the assertion was false. We could thus find a closed subvariety $U \subseteq Y$ such that $\psi(X) \subseteq U$. Let $I = I(U)$ and $f \in I$. From $f(\psi(p)) = 0$ if and only if $h(f(\bar{x}_1, \dots, \bar{x}_n))(p) = 0$ for all points p , and the injectivity of h , we conclude that because $h(f(\dots)) = 0$, $f(\dots) = 0$ on Y for every $f \in I$. Since $U = Z(I(U))$, we must have $U = Y$. □

Lemma 10.18. *Let X be any variety and let $Y \subseteq \mathbb{A}^n$ be an affine variety. A map of sets $\psi: X \rightarrow Y$ is a morphism if and only if $X_i \circ \psi$ is a regular function on X for $1 \leq i \leq n$, where $X_1, \dots, X_n$ are the coordinate functions on $\mathbb{A}^n$.*

Proof. “ $\Rightarrow$ ”: If ψ is a morphism, then by the definition $X_i \circ \psi$ is regular.

“ $\Leftarrow$ ”: If $X_i \circ \psi$, $1 \leq i \leq n$, is regular, for every polynomial $f \in k[X_1, \dots, X_n]$, $f \circ \psi$ is also regular.

For regularity: For any open subset $U \subseteq Y$ and any regular function g on U , $g \circ \psi$ is regular. Indeed, since regular functions on open sets of Y are locally quotients of polynomials, we can write $g = h_1/h_2$ near a point p , where $h_2 \neq 0$ and get that

$$g \circ \psi = \frac{h_1 \circ \psi}{h_2 \circ \psi},$$

is regular, where $h_2 \circ \psi$ is again non-zero on U .

For continuity, we show that ψ^{-1} is closed. The closed subsets of Y are defined by zero sets of polynomials and $f \circ \psi$ is continuous. ψ^{-1} takes closed sets to closed sets. $\square$

Corollary 10.19. *Two affine varieties X and Y are isomorphic if and only if $A(X)$ and $A(Y)$ are isomorphic as k -algebras.*

Corollary 10.20. *The functor $X \mapsto A(X)$ induces an arrow-reversing equivalence of categories between the category of affine varieties over k and the category of finitely generated integral domains over k .*